

claude-windows / 45ea3a91-654d-4972-9cd3-65f35db54caf

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\subagents\workflows\wf\_2a64bc88-2a8\agent-ab17694eae9195645.jsonl`
- SHA-256: `6e90ec7bf16d3f43f8094a31df384d3e2f1c26c9fccd2b8bf8077d4be90a70ce`
- Source modified: `2026-06-10T05:44:05+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf\_2a64bc88-2a8`
- session_id: `45ea3a91-654d-4972-9cd3-65f35db54caf`

Transcript

1. user (2026-06-10T05:33:14.935Z)

CONTEXT ? two sibling repos on a Windows machine:

- INDEXER: C:/Users/avidu/Projects/badminton-highlight-indexer ? local AI badminton(->racket-sports) highlight indexer + rally detector. FastAPI + SQLite + ffmpeg + GPU CV via WSL; Gemini = paid cloud AI. docs/ tree is rich. Current branch docs/next-steps-multisport (master = main branch).

- COLLECTOR: C:/Users/avidu/Projects/sports-data-collector ? golden-label acquisition/curation/ingestion CLI (system-of-record candidate for training corpus).

Project state (June 2026): scaffolding complete; owned-model roadmap agreed (docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md) but NO platform/owned-model implementation started; next committed platform slice = async job model. Licensing guardrail: MIT/Apache-2.0/BSD only for code+data+weights; paid LLMs are inference-only, never training-data sources. ENVIRONMENT RULES: this dev environment has NO GPU/torch/cv2 ? do NOT attempt to run the video pipeline or import torch/cv2 modules. Test suites are offline/fully-mocked and SAFE to run. You are READ-ONLY: never edit/create/delete repo files; running tests and git read commands is allowed.

QUALITY BAR: cite file paths (and line numbers where possible) for every finding. Report what IS in the code/docs, not what you assume. Be skeptical and concrete. Your final structured output is consumed programmatically by an orchestrator ? return raw data, not prose for a human.

TASK: deep RELIABILITY / OPERABILITY / SECURITY audit of the indexer as a system that must one day run as a hosted paid API (see docs/PLATFORM_ARCHITECTURE.md + docs/DEFERRED_HARDENING_PLAN.md ? read both first to know what is ALREADY deferred deliberately; do not re-report deliberately-deferred items as discoveries, instead assess whether the deferral list is COMPLETE).

Deep-read with a failure-mode lens: backend/main.py (API endpoints, sync blocking, global state), backend/infrastructure/database.py (SQLite concurrency, INSERT OR REPLACE cascade, transaction discipline, WAL mode?), backend/pipeline/detectors/ (WSL seam: timeout_sec=0 hangs, staging, healthchecks, cache healing), backend/providers/gemini.py (upload lifecycle, orphaned remote files, API-key handling, error swallowing), backend/pipeline/stitcher/compiler.py (ffmpeg invocation ? command injection via filenames? temp cleanup), backend/reporting/ (never-raises swallow-all), run_*.py entrypoints.

ASSESS: (1) silent-failure inventory ? every path where an error becomes an empty result or swallowed exception, and whether the observability report actually disambiguates each one; (2) concurrency: what happens with 2 simultaneous /api/process calls (SQLite locking, shared globals, GPU contention, same video twice)? (3) security: path/command injection surfaces (video filenames into ffmpeg/WSL bash -c), API key leakage into logs/reports, FastAPI input validation; (4) data integrity: can a crash mid-run corrupt the DB or cache? is re-ingest destructive? (5) resource lifecycle: temp dirs, staged WSL copies, Gemini remote files, frame caches ? what leaks; (6) the DEFERRED_HARDENING_PLAN completeness: name concrete failure modes it does NOT list.

Severity-rate as if a paying customer's video run is on the line.

2. user (2026-06-10T05:33:21.891Z)

```

1 # Deferred Hardening Plan ? Train/Eval Guardrails, Async Jobs, Packaging
2
3 **Status:** ? Design for review ? **no code yet**. Owner-requested follow-up to the
4 "config & identity consistency" tech-debt pass. Each item below becomes its own PR
5 only after this doc is approved.
6
7 **Why these three:** they were cataloged as out-of-scope during the consistency
8 pass because each is larger/design-heavy or needs the owner's machine. They are the
9 remaining high-value hygiene items not already done (config/structure/Result/logging/
10 reports) or in flight (detector abstraction, PR #49).
11
12 **Recommended sequence:** #7 Packaging (lowest risk, unblocks clean installs/CI) ?
13 #13 Train/eval guardrails (correctness of the data flywheel) ? #16 Async jobs
14 (largest, product-facing).
15
16 ---
17
18 ## 1. Packaging modernization (debt #7)
19
20 ### Problem
21 - No `pyproject.toml` on `master`. (PR #49's description lists packaging as "done",
22   but it isn't present ? reconcile with the owner.)
23 - The root `setup.py` is a **diagnostics/dependency-checker**, not a PEP 517 build
24   script ? its name collides with setuptools semantics and will confuse `pip`/`build`.
25 - Deps live only in `requirements.txt`; `obsreport` is a commented soft dep
26   (BACKLOG) and `supervision<0.30.0` is pinned (debt #14).
27
28 ### Approach
29 - Add `pyproject.toml` (PEP 621): project metadata + runtime deps mirroring
30   `requirements.txt`, with optional-dependency groups:
31   - `[project.optional-dependencies] dev` ? `pytest`, coverage tooling.
32   - `reporting` ? `obsreport` pinned to its git tag (`v0.1.3`, per BACKLOG).
33   - `gpu` ? torch/torchvision (note: the cu124 `--extra-index-url` is **not**
34     expressible in standard PEP 621 deps; document the install line or use a
35     `[tool.uv]/pip-conf` approach rather than faking it).
36 - Console entry points: `indexer = backend.main:main` (so `indexer --server` works),
37   optionally eval drivers.
38 - **Rename the diagnostic `setup.py`** (e.g. `scripts/doctor.py`) so it no longer
39   shadows the build system; update README/`run_all_tests.py`/CI references.
40 - Choose a build backend (hatchling recommended) and confirm the `backend/` import
41   root stays valid (keep current layout; no src/ move required).
42
43 ### Risks
44 - `setup.py` rename touches docs and any muscle-memory invocations ? grep first.
45 - Torch CUDA wheels can't be pinned cleanly in PEP 621; must document the
46   `--extra-index-url` install path (the current `setup.py` logic can move into docs).
47 - CI must switch to `pip install -e .[dev]`.
48
49 ### Verification
50 - `pip install -e .[dev]` in a clean venv ? `run_all_tests.py` green.
51 - `indexer --help` resolves via the entry point.
52
53 ---
54
55 ## 2. Train/eval separation guardrails (debt #13)
56
57 ### Problem
58 Separation is **by convention only**. `backend/eval/calibrate_local.py` and
59 `calibrate_wasb.py` do honest leave-one-video-out (LOVO), but nothing *enforces*
60 that a video used to train the learned segmenter
61 (`backend/eval/training.py::build_training_set` ? `distill_local`) is absent from the
62 eval/regression set. Leakage would silently inflate the very metrics the product
63 story rests on (? in BACKLOG).
64
65 ### Approach

```

```

66 - **Split unit = match, never clip** (per `GOLDEN_SET_PHASE1_VIDEOS.md`: split by
67 match, hold ~? of each sport as eval).
68 - Add a `split: "train" | "eval"` field to the collector's `curate export`
69 manifest (cross-repo contract, ADR-002 lineage) ? or a separate eval manifest.
70 - New `backend/eval/splits.py`:
71   - `load_split(manifest) -> (train_ids, eval_ids)` keyed by **match id** (not just
72     file MD5 ? re-encoded duplicates share a match but differ in
73     `compute_video_id`, so group at match level).
74   - `assert_disjoint(train_ids, eval_ids)` ? raise on any overlap.
75 - Wire the guard into the training entrypoint (`distill_local`/`training`) and into
76   `run_regression.py` so the baseline is computed strictly on the eval split.
77
78 ### Risks
79 - Requires a collector-side manifest change (coordinate the cross-repo contract).
80 - Small corpus ? tiny held-out set ? high metric variance; the guardrail is correct
81   but the numbers will be noisy until the golden set grows.
82 - MD5-only dedup is insufficient (re-encodes evade it) ? must track match identity.
83
84 ### Verification
85 - Unit test: overlapping train/eval ids ? `assert_disjoint` raises.
86 - Integration: a deliberately-leaked manifest fails the training entrypoint.
87
88 ---
89
90 ## 3. Async job model (debt #16)
91
92 ### Problem
93 `POST /api/process` runs validate ? segment ? report **inline**, blocking the
94 request for the full duration of a long video. Unworkable as a hosted/product API.
95
96 ### Approach (minimal first slice ? single self-hosted box)
97 - `POST /api/process` returns **202 + `job_id`** ; add `GET /api/jobs/{job_id}`
98   returning `{status, progress, result, reports}`.
99 - Run the pipeline on a background worker ? start with a `ThreadPoolExecutor`
100   (segmenters already use threads internally for AI handoff, so this fits); leave a
101   real queue (RQ/Celery) for later if multi-process is needed.
102 - Persist job state in the existing SQLite DB via the migration ladder
103   (`backend/infrastructure/database.py::init_db`, bump `PRAGMA user_version` 1 ? 2,
104   add a `jobs` table). The observability report becomes the job's artifact.
105 - Update the static UI to poll `GET /api/jobs/{id}` instead of blocking.
106
107 ### Risks
108 - **SQLite thread-safety:** verify `check_same_thread`/connection-per-thread;
109   `_get_connection` already re-issues `PRAGMA foreign_keys=ON` per connection ?
110   confirm it's safe under the executor.
111 - Cancellation/timeout and crash recovery (a job in RUNNING after a restart).
112 - In-process store is not horizontally scalable ? acceptable for the current
113   single-box self-hosted target; call it out so it isn't mistaken for production-HA.
114
115 ### Verification
116 - Submit a job ? poll ? DONE with the same segments the sync path produced.
117 - Server restart mid-job leaves a recoverable/clearly-FAILED job, not a hang.
118
119 ---
120
121 ## Out of this doc
122 - **Detector/WSL abstraction (#6):** in flight (PR #49).
123 - **Unify hybrid segmenters (#7 in LOW_REGRET):** owner-deferred.
124 - **ByteTrack `supervision<0.30.0` migration + real-footage validation (#14/#15):**
125   need a GPU/WSL box and real video ? owner-machine work, tracked separately.
126

```

3. user (2026-06-10T05:33:23.465Z)

```

1 # Platform Architecture ? Multi-User "Assistant Coach"

```

```

2
3  **Status:** ? Strategy / research ? for review. No code yet.
4  **Date:** 2026-06-07 . **Owner:** avidullu
5
6  **Purpose.** Map the path from today's **single-user, local** tool to a **multi-tenant
7  "assistant coach" platform**: users upload videos, get highlights **stored and
8  browsable by history**, and choose **where their data lives** (Google Drive / cloud
9  bucket / local) and **where the compute runs** (our GPU / their VM). The organizing
10 principle is to **decouple the layers** ? identity, ingestion, storage, orchestration,
11 execution/compute, the detection+highlight pipeline, and data-serving ? behind small
12 **pluggable interfaces**, reusing the registry pattern the codebase already uses for
13 segmenters/providers/validators/sports/annotations.
14
15  **Product framing (the north star *and* the v1 line).** This is an **assistant coach**,
16 not just a clipper ? but the strategy is **phased and unambiguous about scope**:
17 - **v1 (concrete, near-term):** **rally detection ? highlight-reel compilation ? per-
user
18 history ? a coach/user correction loop that feeds the golden set** ("assistant-coach
19 *lite"). Essentially today's pipeline made multi-user ? this is the real v1
deliverable.
20 - **Mid-term:** **rich rally/shot semantic metadata** (shot type, rally outcome) and the
21 **owned video-QA model** ($5A; $6 P4?P5).
22 - **North-star vision (strictly future ? not* v1):** **gait / biomechanics, per-athlete
23 identifying templates, and deep coaching feedback under a supervising coach** ($6 P6).
24 Inspiring, but explicitly deferred ? and the part that raises the **biometric**
privacy
25 bar ($7).
26
27 The buyer is the **academy/coach** (beachhead in
28 [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md)); **power users self-serve**. Two
29 consequences ripple through the design: (1) the data model keeps an **optional coach ?
30 athlete overlay over a generic core** ($4.1) ? not baked in; (2) gait/biometrics is
31 **special-category data**, gated and deferred ($7).
32
33 > **Competitive note (validated by research):** none of Veo / Hudl / SwingVision /
34 > Spiideo offer true BYO-storage or BYO-compute ? they're vendor-cloud SaaS
35 > (SwingVision uniquely runs *edge* compute on the user's iPhone). So **"your video,
36 > your storage, your compute" is a genuine differentiator for privacy-sensitive
37 > academies ? and SwingVision proves user-side compute is commercially viable. All four
38 > share the same **teams ? athletes ? clips** data model, which we adopt ($4).
39
40 > Scope: this is a **direction-setting doc**. It does not start implementation; the
41 > first infra slice (an async job model) is the on-hold #16 in
42 > [`DEFERRED_HARDENING_PLAN.md`](DEFERRED_HARDENING_PLAN.md). Commercial constraints
come
43 > from [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) (C4 paid-LLM no-train; C12 SaaS
44 > foundations) and `archives/MONETIZATION_AUDIT.md`.
45
46 ## 1. Current state & gaps (what we're evolving from)
47
48 | Concern | Today | Gap for multi-user |
49 |---|---|---|
50 | Identity / tenancy | none | no users, orgs, auth, or isolation |
51 | Storage | local `filepath` string; output to `output_dir` | no abstraction; can't BYO
Drive/bucket; raw video must be on the box |
52 | Data model | `videos` keyed by file **MD5**, `play_segments`, `events`,
`candidate_segments` (SQLite) | no owner/athlete link; **highlights aren't a first-class
persisted entity** (compile writes a file and returns a path); no per-user history |
53 | Execution | pipeline runs **in-process, synchronously** in `/api/process` | blocks;
can't dispatch to a user's VM/GPU; no job status |
54 | Pipeline | **already pluggable** (segmenter/provider/validator/sport/annotation
registries) | reuse as-is ? this layer is the model for the rest |
55 | API | `/api/process`, `/api/query`, `/api/compile`, `/api/config`; no auth | needs
auth, per-tenant scoping, async job + history endpoints |
56

```

```

57 Grounding: `backend/infrastructure/database.py` (schema + `PRAGMA user_version`
58 migration ladder, v1), `backend/main.py` (endpoints, sync flow), the registry pattern
59 in `backend/pipeline/segmenters/base.py::SegmenterRegistry` and
60 `backend/providers/base.py`.
61
62 ## 2. Target layered architecture
63
64 Decouple into layers with a clean **control-plane vs data-plane** split ? the
65 orchestrator coordinates *metadata and jobs*; the heavy *bytes and GPU work* can live
66 elsewhere (even in the user's own cloud).
67
68 ...
69
70 CONTROL      ?????????????????????????????????????????????????????????
71 PLANE        ? Presentation / API (auth, per-tenant) ?
72              ? Identity & Tenancy (org?coach?athlete?user) ?
73              ? Orchestration / Jobs (submit?status?result) ?
74              ?????????????????????????????????????????????????????????
75              ? dispatch ? metadata
76 DATA       ?????????????????????????????????????????????????????????
77 PLANE        ? Execution / Compute ? ? Data / Serving ?
78              ? (local | BYO-VM | ? ? (per-tenant DB: ?
79              ? hosted GPU) ? ? videos, segments, ?
80              ? runs the Pipeline: ? ? highlights, jobs, ?
81              ? detect ? highlight ? ? history) ?
82              ?????????????????????????????????????????????????????????
83              ? read input / write artifacts
84              ?????????????????????????????????????????????????????????
85              ? Storage (local upload | Google Drive | S3 | ?
86              ? GCS | Azure) ? raw video + highlight outputs ?
87              ?????????????????????????????????????????????????????????
88
89 The two **new pluggable seams** are **Storage** and **Compute/Execution**; everything
90 else either exists (pipeline) or is net-new infra (identity, orchestration, serving). A
91 **model/training layer** ? the owned generative video-QA model ($5A) ? sits atop these,
92 fed by Storage (training corpus), Data (labels), and Compute (training + inference), and
93 is delivered through the existing **AI-provider registry with a paid fallback**.
94
95 ## 3. Pluggable abstractions (extend the existing registry pattern)
96
97 The codebase already proves the model: an `ABC` + a `Registry` singleton with
98 `register/get/list_available` (e.g. `SegmenterRegistry`). Add two more registries with
99 the same shape so they're plug-and-play and testable in isolation.
100
101 ### 3.1 `StorageBackend` registry
102 Replace the bare `video_path: str` that flows through the pipeline with a
103 **`StorageRef`** (a backend id + a key/URI), resolved by a backend.
104
105 ```python
106 class StorageBackend(ABC):
107     name: str
108     def open_read(self, ref: StorageRef) -> BinaryIO: ... # stream large video
109     def download_to_local(self, ref: StorageRef, dst: str): ... # for ffmpeg/cv2/WSL
110     def upload(self, local_path: str, ref: StorageRef): ... # persist
111     def signed_url(self, ref: StorageRef, ttl: int) -> str: ... # hand to LLM / browser
112     def stat(self, ref: StorageRef) -> StorageStat: ...
113     def list(self, prefix: StorageRef) -> list[StorageRef]: ...
114
115
116 - **Recommended implementation:** build the backend over **`fsspec`** (one matrix ?
117 local / S3 (`s3fs`) / GCS (`gcsfs`) / Azure (`adlfs`) + Google Drive via `gdrivefs`,
118 Dropbox via `dropboxdrivefs`), reaching for **`smart_open`** as the robust large-MP4
119 byte-streaming primitive. Storage becomes one more registry. (Treat Drive/Dropbox as
120 add-ons ? less battle-tested than `s3fs`/`gcsfs`; cover with integration tests.)

```

```

121 - **Backends:** `local` (upload dir), `gdrive`, `s3`, `gcs`, `azure`. **BYO-storage**:
122 the tenant registers their own bucket/Drive; raw video **never leaves their account**.
123 - **Auth ? never persist users' long-lived keys.** For buckets, have the user create an
124 IAM role you **AssumeRole** into, scoped to one bucket/prefix, and mint **presigned
125 URLs** so raw video moves **browser ? the user's bucket directly, never through our
126 servers**. For Drive, OAuth with the narrow **drive.file** scope; for academies, a
127 **service account with domain-wide delegation**. Store any per-tenant creds encrypted.
128 - **Why it matters for the pipeline:** ffmpeg/OpenCV/WSL need a **local file**, while
the
129 LLM provider can take a **signed URL**. The backend exposes both `download_to_local`
130 (for compute) and `signed_url` (for the provider/browser), so the pipeline stops
caring
131 *where* the bytes live.
132 - **Migration path:** introduce `StorageRef` alongside the current local path; a `local`
133 backend wraps today's behavior so nothing breaks; adapters land incrementally.
134
135 ### 3.2 `ComputeBackend` / `ExecutionBackend` registry
136 Where the pipeline actually runs for a given job.
137
138 - **`local`** ? in-process / local subprocess (today's behavior; dev + self-host).
139 - **`hosted`** ? our GPU VM pool (the SaaS default).
140 - **`byo_worker`** ? the tenant runs a **pull-based worker** (a small agent) on their
own
141 VM/GPU. **Critical security property:** the worker *pulls* jobs scoped to its tenant
142 from the control plane, pulls input from the tenant's **own** storage, runs the
143 pipeline, and writes artifacts + results back ? so **the orchestrator never holds the
144 user's cloud credentials** and needs **no inbound access** to their network. This
145 mirrors the **self-hosted CI runner** pattern (GitHub Actions runners use a 50s
146 outbound-only HTTPS long-poll). This pull-based runner is the **cheapest, most
147 credential-safe v1** for BYO-compute.
148 - For users who want the platform to **provision GPUs in their *own* cloud account on
149 demand** (an academy with an AWS account but no standing GPU box), add **SkyPilot**
150 (BYOC: everything launches in the user's accounts/VPCs; creds stay with them).
151 **Modal** is the managed fallback for users wanting zero infra (accepting that raw
152 video then flows through Modal).
153 - **?? Operational / support burden (a real adoption risk).** For the primary beachhead
?
154 Indian badminton **academies/coaches** ? asking them to *run and maintain a worker on
155 their own VM/GPU* is a meaningful ongoing load (updates, monitoring, firewall/NAT, the
156 "it stopped after our IT change" ticket). So: **default academy users to `hosted`**,
and
157 treat `byo_worker`/SkyPilot as a **power-user / enterprise / privacy-sensitive tier**,
158 not the academy default. If/when BYO ships, plan for **auto-updating, self-healing
159 agents + health heartbeats + clear worker status in the UI**, and budget the support
cost.
160 - **Middle tier to consider ? "zero-infra BYO":** a managed-but-isolated path where the
161 **storage** stays in the user's bucket (signed URLs) while compute runs on **our**
GPUs
162 in a **strictly isolated, per-tenant, ephemeral sandbox** (nothing persists our side).
163 This gives the BYO *privacy* benefit (raw video never persists on our infra)
**without**
164 the user running a worker ? a pragmatic default between full-hosted and full-BYO.
165
166 The pipeline code is unchanged; only *where it's invoked* and *how it gets bytes* vary.
167
168 ### 3.3 Reuse, don't reinvent
169 The existing `segmenter` / `provider` / `validator` / `sport` / `annotation` registries
170 are the proven precedent and stay as the **pipeline** layer. The `annotation` registry +
171 `candidate_segments` table are also the natural seam for the **coach-correction / golden
172 -set** loop in the assistant-coach phase ($6, P4).
173
174 ## 4. Multi-tenant data model & serving layer
175
176 ### 4.1 Entities ? generic core, with the coach model as an optional overlay
177 The foundational schema is deliberately **domain-neutral** so we don't overfit to the

```

```

178     coach mindset ? the coach/athlete framing is a **market-fit hypothesis that will
iterate**
179     (see C13 / PMF), not a load-bearing assumption. **Generic core:**
180     ```
181     User ??< VideoCollection ??< Video ??< Highlight ??< HighlightMetadata
182             ???< PlaySegment ??< Event
183     Video / Highlight ??< Job (processing record: status, compute backend, timings)
184     ```
185     - **Generic & re-targetable:** a `VideoCollection` groups a user's videos (a season, an
186     event, a "folder") with no assumption of a coach or academy ? a power user, a coach,
or a
187     parent are all just `User`s with collections.
188     - **Highlights are first-class** (today compile only writes a file): a `highlights`
table
189     (segment ids + output `StorageRef` + params + created_at) powers "see past videos and
190     highlights by history," and **`HighlightMetadata`** carries the **rich semantic tags**
191     (shot type, rally outcome, etc. ? see §6 metadata stage) plus titles / ratings /
notes.
192     - **`video_id` stays the content MD5** (dedup within a tenant), scoped by `tenant_id`.
193     - **Coach overlay (optional, additive):** for the academy/coach GTM, layer
194     `Org/Academy ? Coach ? Athlete` *on top* (e.g. a `Membership`/role table + an
`athlete_id`
195     tag on collections/videos) **without** baking it into the core tables ? so the schema
196     stays usable for solo power users and the coach model can evolve, or be dropped, as
PMF
197     dictates.
198     - **History/serving API:** list collections/videos/highlights per user (and, with the
199     overlay, by athlete/coach/date); **query highlights by `HighlightMetadata`** (e.g.
shot
200     type / outcome); re-stitch from stored segments without reprocessing.
201
202     ### 4.2 Tenancy isolation ? options & recommendation
203     | Model | Isolation | Ops cost | Best for |
204     |---|---|---|---|
205     | **Shared DB + `tenant_id`** (row-level, ideally Postgres **RLS**) | logical | low |
**Hosted SaaS, many small tenants** ? **recommended default** |
206     | Schema-per-tenant | medium | medium | tens/hundreds of mid tenants |
207     | **DB-per-tenant** (incl. **SQLite-per-tenant**) | strong/physical | higher at scale |
**self-hosted single-academy**, or high-isolation enterprise |
208
209     **Recommendation:** for the hosted product, **migrate SQLite ? Postgres with shared-DB +
210     **`tenant_id` + Row-Level Security**; keep **SQLite-per-tenant** as the **self-hosted /
211     **single-academy** deployment (it reuses today's code almost verbatim). Either way,
212     **abstract DB access behind a repository layer now** so the engine is swappable, and
213     graduate the `PRAGMA user_version` migration discipline to **Alembic** under Postgres.
214
215     ## 5. Orchestration / async jobs
216
217     This is the serving layer that makes multi-user real. It is the **multi-user superset of
218     [DEFERRED_HARDENING_PLAN.md](DEFERRED_HARDENING_PLAN.md) #16** (async job model) ?
build
219     that first, single-tenant, then generalize.
220
221     - **Job record** (`jobs` table): `id, tenant_id, video_id, type (index|compile|analyze),
222     status (queued|running|done|failed), progress, compute_backend, result_ref, error,
223     timestamps`. `/api/process` returns **202 + job_id**; `/api/jobs/{id}` polls.
224     - **Idempotency** keyed on `(tenant_id, video_id, pipeline_version)` ? reuse the
225     content-hash discipline already in `compute_video_id`.
226     - **Dispatch:** the orchestrator enqueues; a worker (hosted or `byo_worker`) claims jobs
227     for its tenant, resolves input via the tenant's `StorageBackend`, runs the pipeline,
228     writes artifacts + DB rows. **Observability reports** (already per-video) become the
229     job artifact surfaced in history.
230     - **Queue/worker backend ? a decision to prototype in #16, *not* a default here.** The
231     **requirements** are firm ? **idempotency**, **resumability** (a 40-min GPU job mustn't
232     restart on a worker blip), **progress states** ? which rule out fire-and-forget

```

Celery/RQ

```
233     for the long jobs. The *choice* should be made by prototyping in the
234     [DEFERRED_HARDENING_PLAN.md](DEFERRED_HARDENING_PLAN.md) #16 slice, weighing:
235     - **Start simplest:** *Arq* (async-native, natural FastAPI fit) or even a **plain
236       Postgres-backed queue + worker** ? likely sufficient for the *first* single-tenant
237       async slice, with the lowest operational complexity and the easiest debugging.
238     - **Durable-execution upgrade if/when needed:** *Hatchet* (Postgres-backed, per-step
239       retries/timeouts/concurrency; pairs neatly with the pull-based worker) ? attractive
240       because it adds **no new datastore** once we're on Postgres, but it *is* a new
241       dependency + operational/debugging surface; adopt only when long-job resumability/
242       observability demands it.
243     - **Heaviest:** *Temporal* ? most mature durable workflows, operationally heavy;
244       reserve for genuinely complex multi-stage pipelines at scale.
245
246     Record the decision criteria in #16: operational complexity, maturity, **debugging
247     experience for long GPU jobs**, and whether durability is worth a new dependency.
248     **Bias: start with the simplest thing that gives retries + status; graduate only on
249     evidence.**
250
251     ## 5A. Owned generative model (the video-QA assistant) & the data flywheel
252
253     The north-star capability: an **owned, self-hostable generative video model** ?
254     fine-tuned on sports video (including **consented, user-contributed** footage) ? that
255     **answers questions about rallies, shots, and technique**. This is the conversational
256     brain of the "assistant coach." Crucially, it does **not** replace the known stack; it
257     is introduced **behind the same abstraction, with the paid API as an always-ready
258     fallback**.
259
260     > **The moat is *not* the model.** Anyone can download Qwen-VL/Gemma. The defensible
261     > assets are *(a) the consented, badminton-specific video?QA dataset* and *(b) the
262     > eval harness**. Build and message *those* ? not "we own an LLM."
263
264     > **? Committed guardrail (legal, non-negotiable).** We respect the licensing findings
265     in
266     > full. **Paid frontier models (Gemini / OpenAI / Anthropic) are used for *inference /
267     > serving / fallback* only ? never as a *training* data source.** Harvesting their
268     outputs
269     > to train our owned model would breach their "no competing model" terms (a
270     terms/copyright
271     > violation), so it is prohibited in both engineering and legal process. **Training
272     teachers
273     > are open, Apache-2.0 models only** (Qwen-VL / InternVL / Gemma 4). Detail in the
274     > distillation constraint below.
275
276     ### Core principle: AI is a pluggable provider with a paid fallback
277     The codebase already abstracts the cloud AI behind `provider_registry` /
278     `AIVideoProvider` (`backend/providers/`). Make that a **first-class platform
279     principle**:
280     - Every generative capability routes through the **provider registry**.
281     - The **owned model** plugs in as a new provider (`owned_vlm`); **Gemini (paid, via
282       Vertex AI)** stays registered as the **fallback** (failover on error/low-confidence,
283       or as the quality ceiling/teacher reference).
284     - This lets us **ship today on the known paid stack** and **swap in the owned model when
285       its quality clears a measured bar** ? with zero architectural churn and the paid path
286       always one config flip away. (Mirrors the COMMERCIALIZATION C4 A/B/C strategy: the
287       owned model is the long-run "(A) eliminate the per-call dependency"; paid is "(C)".)
288
289     ### The model / training layer (new sub-layer, fed by the scaffolding)
290     ```
291     Data curation      Training      Eval      Serving
292     (consented user vid (fine-tune an open, (extend the existing (self-host inference
293     + golden set +      commercially-usable  eval harness to QA  as a provider /
294     rally/segment/event base VLM via      quality + the      ComputeBackend;
295     annotations ?       LoRA/QLoRA)      regression gate)    paid fallback)
296     video-QA pairs)
```

```

292     ``
293     This layer **reuses every other layer**: **Storage** holds the training corpus;
294     **Data** supplies labels (the `annotation` registry + `candidate_segments` +
295     golden set are already the labeling seam); **Compute/Orchestration** runs training and
296     inference jobs (even **BYO-compute** can serve inference on the tenant's GPU); the
297     **eval harness** already in `backend/eval/` extends to grade QA quality. **In other
298     words, the rock-solid scaffolding is exactly what makes the model flywheel possible ** ?
299     which is why scaffolding comes first (see S6).
300
301     ### Base model & training (Apache-2.0 only ? licensing is decisive)
302     **Recommended near-term base: `Qwen2.5-VL-7B-Instruct`** ? Apache-2.0, native long-video
303     (dynamic-FPS sampling, temporal mRoPE), mature LoRA tooling. Upgrade path
304     **Qwen3-VL-4B/8B (Apache-2.0)**; **Gemma 4 multimodal (Apache-2.0, 2026)** is a clean
305     Google-stack alternative to evaluate.
306     - ?? **Lock to Apache-2.0-clean bases** so you can fine-tune, **keep weights private,
307     gate behind subscriptions, and never owe open-sourcing or branding**. (Note Qwen2.5-VL
308     is Apache **only at 7B/32B** ? 3B/72B use the custom Qwen license.)
309     - ? **Reject for the flagship:** **Gemma 3** ? custom terms with a **viral derivative
310     clause** (anything fine-tuned on it, *or even trained on its synthetic outputs*, stays
311     permanently bound to Google's *updatable* Prohibited-Use Policy); **Gemma 4's switch
312     to
313     Apache-2.0 removes this**. **Llama 3.2 Vision** ? custom license forces a "Llama-?"
314     model name + "Built with Llama" badge and a 700M-MAU ceiling. **`gpt-oss`** ?
315     Apache-2.0
316     but **text-only**, so usable only as an optional text-reasoning layer, never the video
317     encoder.
318     - **Training method:** **QLoRA** (? r=64, ?=128) on the 7B ? fits one prosumer GPU and
319     avoids catastrophic forgetting of general video skills. Build the dataset by
320     converting
321     the existing rally/shot/segment annotations + golden set into templated **clip?QA
322     pairs**, expand synthetically with **open-model teachers only** (see constraint
323     below),
324     and hold out a **golden QA eval** that gates every fine-tune. A few hundred gold pairs
325     ?
326     **5k?20k** curated is a realistic band for this narrow domain.
327     - **Serving economics:** a 7B VLM ? 14 GB VRAM (FP16) / ~3.5 GB (INT4); a ?16 GB GPU is
328     a
329     comfortable serve target (vLLM for throughput). Self-hosting wins **only** once you
330     have
331     sustained video-QA volume **+** a defensible consented dataset **+** real per-call API
332     cost pain ? below that, the paid API is cheaper than a standing GPU and the second
333     stack
334     is overhead. Hence: **paid-primary now ? owned-primary / paid-fallback later.**
335
336     Base-model licensing + training-data policy are tracked as **COMMERCIALIZATION C14**.
337
338     ### Training backend ? managed service vs self-managed (the fine-tune "forge")
339     *Where* you run the LoRA/RL fine-tune. Weigh on **video support, weight
340     export/ownership,
341     smallest servable model, RL support, no-train DPA, and infra burden**.
342
343     **Decisive finding (researched 2026-06):** managed fine-tuning services are
344     **image-based** (JSONL + base64 images) ? for video you pre-sample frames and feed them
345     as image sequences. **Only the self-managed `ms-swift` explicitly advertises *native*
346     video (+audio) training**. So for true video-QA, a self-managed framework is the
347     strongest technical fit ? and it also lets you fine-tune the **small 7B** (cheap to
348     serve) and keep full ownership. Managed services trade some of that for zero infra.
349
350     | Option | Type | Video / VLM support | Smallest VL | RL | Export weights | Best for |
351     |---|---|---|---|---|---|---|
352     | **ms-swift** (ModelScope) | Self-managed (rented/own GPU) | **Native video+audio**;
353     Qwen3-VL/Omni, 300+ MLLMs | any incl **7B** | ? DPO/GRPO | ? full ownership | **? video-QA;
354     slots into BYO-compute (§3.2)** |
355     | **LLaMA-Factory** | Self-managed | Qwen3-VL + 100+ VLMs; video datasets | any incl 7B
356     | ? | ? | popular, well-documented |

```

```

345 | **Unslloth** | Self-managed (1-GPU optimized) | Qwen3-VL LoRA (vision+lang; frames) |
any incl 7B | ? GRPO | ? | cheapest first runs |
346 | **Tinker** (Thinking Machines) | Managed API (low-level) | Qwen3-VL **30B-A3B+ only**;
image input (video **unverified**) | 30B-A3B MoE | ? PP0/GRPO, custom loss | ? adapters + ckpt;
no-train ToS | zero-infra RL/research at MoE scale |
347 | **Together AI** | Managed (tune + serve) | Qwen3-VL / Gemma-3 / Llama-4; image data |
verify | partial | ? adapters downloadable | managed Qwen3-VL + serving |
348 | **Fireworks AI** | Managed (tune + serve) | Full Qwen2.5-VL (3B?72B SFT; LoRA@32B +
LoRA-upload) | **7B** | partial | ?? verify export (lock-in risk) | managed *small*-VL + serving
|
349 | **Predibase** (Rubrik) | Managed *** VPC self-host** | VLM instruct-tune; **LoRAX**
multi-adapter | verify | ? (VPC) | ? **RFT/GRPO** flagship | RL "creativity" + multi-adapter +
privacy |
350 | ~~OpenAI GPT-4o / Gemini tuning~~ | Closed | ? | ? | ? | ? no export | **non-fit ?
violates "own the model"** |
351
352 *(Enterprise clouds ? Databricks Mosaic / SageMaker / Vertex on open models ? give full
353 control but are only worth it if you're already on that stack.)*
354
355 **Recommendation:**
356 - **Early iteration / true video-QA ?** self-managed **`ms-swift`** (or LLaMA-Factory /
357 Unslloth) on a **rented GPU** (RunPod / Lambda / Modal / SkyPilot-to-your-cloud):
native
358 video, any size incl. **7B** (cheap to serve), full ownership, and it drops into the
359 **ComputeBackend / BYO-compute** seam ($3.2). **Likely a better fit than Tinker** for
360 the video requirement *and* serving economics.
361 - **Managed, no GPU ops ?** **Together AI** (Qwen3-VL + downloadable adapters + serving)
362 or **Fireworks** (full Qwen2.5-VL incl. 7B + serving) ? confirm frame/video support
and
363 adapter export.
364 - **If RL "coaching creativity" + per-sport multi-adapter is central ?** **Predibase**
365 (RFT + LoRAX, VPC self-host for the biometric/privacy posture) or **Tinker** (lowest-
366 level RL primitives).
367 - **Tinker** remains a strong zero-infra RL-research option with clean weight export ?
just
368 not the cheapest path to a *small video* model (its VL floor is 30B-A3B).
369 - **Portability is the hedge:** you own the **data + recipe + exported weights**, so the
370 backend is swappable ? same ethos as the provider/registry pattern. Don't get locked
in.
371
372 **Verify before committing (any backend):** (1) **native video / frame-sequence**
373 ingestion + max frames per example; (2) **weight export** + a **commercial DPA**
covering
374 consented, minors-excluded video (biometric, $7); (3) the **inference plan** for the
size
375 you train (a self-managed **7B** is far cheaper to serve than a 30B-A3B MoE).
376
377 ### Training harness ? make it agent-runnable (nanochat-style operability)
378 The goal: **an agent (or CI) can run a fine-tune on a fresh batch of golden-labeled
video
379 and get a trustworthy ship / no-ship verdict, unattended.** Karpthy's **nanochat** is
the
380 reference for that *operability* ? one MIT repo, one `speedrun.sh` that runs
381 tokenizer?train?eval?serve, a single complexity dial, and an automatic **report-card**
382 score. It is **not** a usable engine here (text-only, from-scratch GPT-2-class), but its
383 five properties are exactly what make training agent-runnable, and we copy them:
384 ** (1) one command, end-to-end; (2) one complexity dial (sane auto-derived
defaults);
385 ** (3) an automatic **report card** ? a deterministic pass/fail vs a target (the crux);
386 ** (4) reproducible + deterministic + cheap; (5) minimal/readable.
387
388 **Engine (don't reinvent):** **`ms-swift`** already delivers this for VLMs ? CLI-driven,
389 **native video**, Qwen3-VL support, an eval backend (EvalScope), and reproducibility via
390 `args.json` + `full_determinism`. (LLaMA-Factory is the close second; HF **`nanoVLM`** ?
391 ~750 lines ? is the minimal reference to *learn* VLM training, not the production path.)

```

```

392
393     **The harness = thin glue over components we already have** ? one declarative,
394     agent-invokable job (call it `train_speedrun`):
395     1. **Build dataset** from golden labels ? clip?QA pairs (reuse the `annotation` registry
+
396         `candidate_segments` + golden set ? the labeling seam in `backend/`).
397     2. **Fine-tune** via ms-swift on a **ComputeBackend** GPU ($3.2), run as an
orchestration
398         job ($5 / DEFERRED #16): one config, idempotent, logged.
399     3. **Auto-eval + report card** ? extend the existing `backend/eval/` regression harness
to
400         **QA quality**; emit a `report.md` pass/fail vs the **held-out golden QA eval**.
401     4. **Register** the result as the `owned_vlm` provider (Gemini paid fallback stays)
**only
402         if it beats the bar**.
403
404     So: **golden labels** ? ms-swift train (on a compute backend) ? eval gate ? register
405     provider**, behind one command an agent or CI runs. nanochat is the inspiration for the
406     wrapper; ms-swift is the engine; the **golden QA eval is the report card**.
407
408     > **Prerequisite (the crux):** the harness is only as trustworthy as the **golden QA
eval
409     > set**. nanochat's speedrun is meaningful *because* its eval (CORE score) is automatic
and
410     > trustworthy; ours is the golden QA eval ? **build that first**, or an agent-run loop
411     > optimizes blind. (Same "moat = dataset + eval", scaffolding-first principle.)
412
413     ### ?? Teacher / distillation constraint (the single biggest legal landmine)
414     You **cannot** train the owned model on outputs from the paid frontier APIs ? a general
415     video-QA assistant is squarely a **competing model**: **Gemini** ("may not use the
416     Services to develop models that compete"), **OpenAI** ("may not use Output to develop
417     models that compete"), **Anthropic** (bars using Outputs to train competing models). Two
418     safe lanes: **(1)** generate synthetic QA with **open-model teachers only** (Qwen-VL /
419     InternVL / Gemma 4 ? Apache-clean); **(2)** keep the paid API strictly as a **runtime
420     fallback** ? serving answers is fine, *harvesting its outputs to train the model that
421     replaces it is not*. Write a hard ***"no training on competitor-API outputs"*** rule into
422     eng + legal process. (This makes the ROADMAP's "Gemini as offline teacher only" precise:
423     the teacher is *open models*, not the paid APIs.)
424
425     ### Data flywheel, personalization & consent tiers
426     Two complementary loops, each with its **own** opt-in (training consent is separate from
427     processing consent ? $7):
428
429     - **Global model (shared ? raises the floor).** Train the base owned model on an
430     **explicitly-contributed, licensed** corpus ? *not* on whatever sits in a tenant's
bucket
431     (preserves "your video, your storage"). Flywheel: *more consented, labeled matches ? a
432     better shared model ? better highlights/answers ? more reason to contribute.* Higher
433     consent bar (pooled data, biometric/minor care ? $7); **minors excluded by default**.
434
435     - **Per-user adapters (personalized ? raises the ceiling) ? owner hunch, under
exploration.**
436     A small **per-user (or per-tenant) LoRA adapter, fine-tuned *only on that user's own
437     consented videos*** on top of the shared base, served via **multi-LoRA** (LoRAX /
438     Together / Predibase). Why it's compelling:
439     - **Precision:** adapts to *that user's* court, camera angle, lighting, and players'
440     styles ? materially better detection/highlights for them.
441     - **Cleanest consent story:** **improve *your* detection using *your* data** is
far
442     lower-friction and lower-risk than contributing to a global model ? the data
improves
443     **only the user's own adapter**, with **no cross-user mixing**, sidestepping much of
the
444     pooled-corpus biometric/minor entanglement. Reinforces the differentiator: *your
video,
```

```

445     your storage, your compute ? and now **your model**. With BYO-compute the adapter
can be
446     trained/served on the user's own GPU and **never leave their boundary**; **deletion
=
447     drop the adapter** (clean erasure).
448 - **Open questions (still a hunch)** cold-start (need enough of the user's labeled
data
449     before a personal adapter beats the base ? fall back to the shared model meanwhile);
a
450     **per-user held-out eval** to prove the adapter helps (don't ship a regression);
base +
451     optional global-adapter + per-user-adapter stacking; per-user training cost at
scale.
452     Worth a spike once the harness + golden eval exist.
453
454     **Two consent tiers, then:** **a)** "improve **my own** detection with **my** data"
455     (per-user, low friction) and **b)** "contribute to the **shared** model" (global,
higher
456     bar). **Default for both is off.**
457
458     ## 6. Phased roadmap
459
460     | Phase | Deliverable | Build / reuse |
461     |---|---|---|
462     | **P0** | **Async job model, single-tenant** (DEFERRED #16) | `jobs` table via
`user_version` ladder; worker; 202+status endpoints |
463     | **P1** | **Identity + multi-tenant data model + upload + highlight persistence +
history** | auth (build vs Auth0/Clerk/Supabase ? decide); Postgres + `tenant_id`/RLS;
`highlights` table; repository layer; history API + UI |
464     | **P2** | **`StorageBackend` registry + BYO-storage** | new ABC+registry;
`local`>`gdrive`>`s3`>`gcs`; replace `video_path` with `StorageRef`; encrypted per-tenant creds
|
465     | **P3** | **`ComputeBackend` registry + BYO-compute** | `local`>`hosted`>`byo_worker`;
pull-based worker agent; control/data-plane split |
466     | **P4** | **Rich rally/shot semantic metadata & tagging** (the next value stage after
solid rally detection + highlights) | shot type (smash / backhand drive / net), rally
**outcome** (winner / forced / **unforced error**), context (e.g. **open court but error**) ?
query-API filters + `HighlightMetadata` + a **human correction/review loop** feeding the golden
set via the `annotation` registry + `candidate_segments` |
467     | **P5** | **Owned generative video-QA model** ($5A) | global flywheel **+ per-user
adapters**; P4 metadata is training signal ? fine-tune an open VLM (Qwen-VL) via LoRA ? serve as
`owned_vlm` **provider with Gemini paid fallback**; extend the eval harness to QA quality |
468     | **P6** ***(visionary)* | **Gait / biomechanics under a supervising coach** |
deliberately **back-burner**; technique feedback; highest Art. 9 bar ? gate behind explicit
biometric + parental consent ($7) |
469
470     P0?P1 deliver the headline ask (per-user upload + stored highlights + history). P2?P3
471     deliver the "plug-and-play storage/compute" ask. **P4 (rich semantic metadata) is the
472     immediate next value stage once rally detection + highlights are solid** ? e.g.
**rallies
473     that ended in a smash or backhand drive," **rallies where the court was open but the
474     player made an unforced error.** **P5** owns the model; **P6 (gait) is the visionary
475     horizon, intentionally deferred.**
476     **Sequencing principle (owner directive):** make the scaffolding (P0?P4) **rock-solid
477     first**, keep the **known paid stack as an always-ready fallback**, and only then shift
478     focus to **owned-model quality** (P5). The provider registry keeps the paid path one
479     config flip away.
480
481     ## 7. Privacy & legal (gait raises the bar ? design for it now)
482
483     - **The load-bearing nuance:** biometric data is special-category **only when used to
484     *uniquely identify* a person** (GDPR Art. 9). **Highlight/shot detection on players a
485     coach already knows is general analytics ? generally *not* Art. 9.** But
486     **gait/biomechanics that builds a per-athlete identifying template tips into Art. 9**
487     and needs **explicit, granular consent** referencing biometric data. Engineer gait

```

```

488     with that boundary (consent gate, purpose limitation, no cross-account
identification);
489     decide whether it runs **locally/on-tenant only** (strong posture) vs our cloud. (EDPB
490     Guidelines 3/2019 on video devices apply.)
491     - **Minors (e.g. academies).** Parental/guardian consent and safeguarding are first-
order,
492     not an afterthought. Model consent explicitly per relationship (user?subject,
coach?athlete
493     where the overlay applies, parent?minor).
494     - **BYO-storage is the central privacy/liability lever:** if raw video stays in the
495     tenant's own bucket/Drive and we process via signed URL or their compute, we minimize
496     what we hold ? shrinking breach surface, residency, and erasure obligations.
497     - **Paid-LLM no-train (C4):** use **Gemini via Vertex AI** (contractual no-training),
498     **not the consumer/free tier**; set short/zero retention, document it in the DPA, and
499     surface "your footage is never used to train AI" as a trust feature for academies. For
500     the gait phase, prefer keeping biometric inference off third-party APIs entirely.
501     - **Training consent is a distinct, higher bar (for the owned model, §5A).** Training on
502     user video needs a **separate, explicit, opt-in training licence** ? not bundled with
503     processing consent ? and it **compounds** the biometric (Art. 9) and minor-consent
504     issues above. Train **only on explicitly contributed/licensed video**; never train on
505     what merely passes through a tenant's bucket. **Exclude minors' video from the
training
506     pool by default**, keep the training corpus a **separate, firewalled, opt-in sub-
pool**,
507     and log **per-clip consent/provenance** for revocation + audit. Incentives must not
make
508     consent non-free (the product stays fully usable without contributing). And the
509     **teacher-distillation rule** (§5A): never train on paid-LLM outputs. **Two consent
tiers
510     (§5A):** (a)* "improve **my own** detection with **my** data" ? a per-user adapter
trained
511     only on the user's own video (lowest friction; no cross-user mixing; deletion = drop
the
512     adapter); (b)* "contribute to the **shared** model" ? pooled corpus, higher bar.
Default
513     for both is **off**.
514     - **Right to erasure / retention:** per-tenant, per-athlete delete must cascade across
DB +
515     storage artifacts (the `ON DELETE CASCADE` discipline already in `database.py` extends
here).
516
517     ## 8. Key decisions for the owner (the forks)
518
519     1. **Hosted-first vs BYO-first.** Recommended: **hosted control-plane first** (P0?P1)
with
520     **BYO-storage soon after** (P2); **BYO-compute later** (P3) as a power-
user/enterprise
521     option ? it's the highest-complexity, highest-security item.
522     2. **Database now or later.** Move to **Postgres at P1** for the hosted path, or stay
523     SQLite-per-tenant if self-hosted-single-academy is the first GTM. (Abstract DB access
524     regardless.)
525     3. **Auth: build vs buy** (Auth0 / Clerk / Supabase Auth / Cognito).
526     4. **Orchestration tech** (see research recommendation) ? bias to the simplest that
gives
527     retries + status.
528     5. **Gait processing location** ? local/on-tenant (privacy-max) vs cloud (capability-
max).
529     6. **How far to push BYO-compute** given its security/ops complexity.
530     7. **Owned-model base + licensing (§5A).** Which commercial-OK video VLM to build on
531     (Qwen-VL Apache-2.0 vs Gemma 3 custom license vs ?) and confirm a gated commercial
532     derivative is permitted ? see COMMERCIALIZATION C14.
533     8. **Training-data sourcing & consent.** Opt-in contribution terms, incentives, and the
534     teacher-distillation constraint (whether paid-LLM outputs may seed the corpus).
535     9. **When to start the owned model** ? strictly after the scaffolding + flywheel are
536     solid (the explicit directive), with the paid provider as the standing fallback.

```

537

538 ****Risks:**** scope explosion (sequence strictly P0?P5); multi-tenant data-leak bugs (RLS +
539 tests); BYO-compute security; ****biometric/minor liability**** (get this right before any
540 gait feature ships); ****training on user video without proper opt-in/licence****, and
541 ****base-model / teacher-distillation licence violations****; cost of hosted GPU (training
542 ***and* inference**); ****premature model-building**** before data/eval are ready (mitigated by
543 keeping the paid fallback and gating P5 behind the flywheel).

544

545 **## 9. Sources**

546

547 External best-practices research (2026-06-07) backing the recommendations above. Each
548 underlying claim in the research was tagged fact-vs-synthesis; figures are as-of and
549 should be re-validated before committing budget.

550

551 ****Storage abstraction:**** [fsspec](https://fsspec.com/) ·
[cloudpathlib](https://github.com/drivendataorg/cloudpathlib) ·
[smart_open](https://github.com/piskvorky/smart_open) · [Apache
Libcloud](https://libcloud.apache.org/) · [gdrive-fsspec](https://github.com/fsspec/gdrive-
fsspec) · [dropboxdrivefs](https://github.com/fsspec/dropboxdrivefs) · [AWS S3 presigned
URLs](https://docs.aws.amazon.com/AmazonS3/latest/userguide/using-presigned-url.html) · [token-
based S3 with STS](https://oneuptime.com/blog/post/2026-02-12-token-based-access-s3-sts/view) ·
[signed URLs across clouds](https://medium.com/@shaikhsharwan49/signed-urls-in-aws-gcp-and-azure-
a-beginner-friendly-guide-e2e549425865)

552

553 ****Multi-tenancy:**** [Bytebase: multi-tenant DB
patterns](https://www.bytebase.com/blog/multi-tenant-database-architecture-patterns-explained/) ·
[Hunchbite: RLS vs schema-per-tenant](https://hunchbite.com/guides/multi-tenant-saas-
architecture) · [PlanetScale: tenancy in Postgres](https://planetscale.com/blog/approaches-to-
tenancy-in-postgres) · [DEV: DB-per-tenant vs shared schema](https://dev.to/young_gao/multi-
tenant-architecture-database-per-tenant-vs-shared-schema-1n2e)

554

555 ****BYO-compute / orchestration:**** [Pinecone BYOC](https://www.pinecone.io/blog/byoc/) ·
[Nuon: control/data-plane](https://nuon.co/blog/byoc-control-plane-data-plane-architectures) ·
[GitHub self-hosted runners (ARC)](https://docs.github.com/en/actions/hosting-your-own-
runners/managing-self-hosted-runners-with-actions-runner-controller/about-actions-runner-
controller) · [SkyPilot](https://github.com/skypilot-org/skypilot) · [SkyPilot AI control
plane](https://blog.skypilot.co/ai-job-orchestration-pt2-ai-control-plane/) · [Modal (AWS
Marketplace)](https://aws.amazon.com/marketplace/pp/prodview-j727623xqhh2k) · [Northflank:
Anyscale alternatives](https://northflank.com/blog/anyscale-alternatives-for-ai-ml-model-
deployment) · [DevPro: Python task queues
2025](https://devproportal.com/languages/python/python-background-tasks-celery-rq-dramatiq-
comparison-2025/) · [Judoscale: choosing a task queue](https://judoscale.com/blog/choose-python-
task-queue) · [Hatchet vs Temporal](https://hatchet.run/versus/hatchet-vs-temporal) ·
[Hatchet](https://github.com/hatchet-dev/hatchet)

556

557 ****Privacy / legal:**** [GDPR Art. 9](https://gdpr-info.eu/art-9-gdpr/) · [GDPR Local:
biometric compliance](https://gdprlocal.com/biometric-data-gdpr-compliance-made-simple/) ·
[Legiscope: Art. 9](https://www.legiscope.com/blog/gdpr-article-9-special-categories.html) ·
[EDPB Guidelines 3/2019 (video)](https://www.edpb.europa.eu/sites/default/files/files/file1/edpb_
_guidelines_201903_video_devices_en_0.pdf) · [Fox Rothschild: EDPB video
guidance](https://www.foxrothschild.com/publications/video-surveillance-under-gdpr-edpb-issues-
final-guidance) · [ICO: children & UK GDPR](https://ico.org.uk/for-organisations/uk-gdpr-
guidance-and-resources/childrens-information/children-and-the-uk-gdpr-old/what-are-the-rules-
about-an-iss-and-consent/) · [EDPB Guidelines 05/2020 (consent)](https://www.edpb.europa.eu/site
s/default/files/files/file1/edpb_guidelines_202005_consent_en.pdf) · [AI data-retention
compared](https://ax-sentinel.com/blog/ai-data-retention-policies-compared) · [Xenoss:
enterprise LLM platforms](https://xenoss.io/blog/openai-vs-anthropic-vs-google-gemini-
enterprise-llm-platform-guide)

558

559 ****Comparable products:**** [Veo](https://www.veo.com/) · [Veo Help
Center](https://support.veo.com/hc/en-us/articles/4459159526929-Overview-of-cam-veo-co-Record-
manage-and-upload-footage-with-your-Veo-Cam-1) · [Latterly: Hudl
competitors](https://www.latterly.org/hudl-competitors/) · [Apple: Behind the Design ?
SwingVision](https://developer.apple.com/news/?id=0pg4dthn) · [SwingVision
FAQ](https://swing.vision/faq) · [Sportsvisio: coach video-analysis

comparison](https://www.sportsvisio.com/stories/video-analysis-software-for-coaches-complete-comparison)

560

561 **Training backends & harness (§5A):** [karpathy/nanochat (harness reference)](https://github.com/karpathy/nanochat) · [HF nanoVLM (minimal VLM)](https://github.com/huggingface/nanoVLM) · [nanoVLM blog](https://huggingface.co/blog/nanovlm) · [Tinker (GA + vision)](https://thinkingmachines.ai/blog/tinker-general-availability/) · [Tinker model lineup](https://tinker-docs.thinkingmachines.ai/model-lineup) · [Tinker ToS (weights/no-train)](https://thinkingmachines.ai/legal/tos/) · [Fireworks VLM fine-tuning](https://fireworks.ai/blog/vlm-tuning) · [Together AI fine-tuning](https://www.together.ai/fine-tuning) · [Predibase: fine-tune & serve VLMs](https://predibase.com/blog/how-to-fine-tune-and-serve-vlms-in-predibase) · [Predibase RFT/GRPO](https://docs.predibase.com/user-guide/fine-tuning/grpo) · [ms-swift (native video/audio)](https://github.com/modelscope/ms-swift) · [LLaMA-Factory (VLMs)](https://github.com/hiyouga/LlamaFactory) · [Unsloth (Qwen3-VL)](https://qwen.readthedocs.io/en/latest/training/unsloth.html) · [Fine-tuning tools 2026 (LoRA/QLoRA)](https://www.index.dev/blog/top-ai-fine-tuning-tools-lora-vs-qlora-vs-full)

562

563 **Owned model & licensing (§5A):** [Qwen2.5-VL](https://qwenlm.github.io/blog/qwen2.5-vl/) · [Qwen2.5 license split](https://qwenlm.github.io/blog/qwen2.5/) · [Qwen3-VL](https://github.com/QwenLM/Qwen3-VL) · [Gemma 3 (multimodal)](https://huggingface.co/blog/gemma3) · [Gemma Terms of Use](https://ai.google.dev/gemma/terms) · [Gemma license-risk analysis](https://wcr.legal/google-gemma-license-risks/) · [Gemma 4 / Apache-2.0](https://www.mindstudio.ai/blog/what-is-gemma-4-apache-2-license-commercial-ai-deployment) · [Llama 3.2 license (700M MAU + naming)](https://www.llama.com/llama3_2/license/) · [gpt-oss (text-only, Apache-2.0)](https://openai.com/index/introducing-gpt-oss/) · [Gemini API terms (compete clause)](https://ai.google.dev/gemini-api/terms) · [OpenAI terms](https://openai.com/policies/row-terms-of-use/) · [Anthropic: outputs & training](https://support.claude.com/en/articles/12326764-can-i-use-my-outputs-to-train-an-ai-model) · [QLoRA practical](https://apxml.com/courses/fine-tuning-adapting-large-language-models/chapter-4-parameter-efficient-fine-tuning/practice-qlora-fine-tuning) · [VLM LoRA cookbook](https://huggingface.co/learn/cookbook/en/fine_tuning_vlmtrl) · [GDPR Art. 9](https://gdpr-info.eu/art-9-gdpr/) · [athletes & biometric data](https://www.architectureandgovernance.com/applications-technology/biometric-data-and-athletes-privacy-law-and-compliance-implications/) · [federated learning + DP](https://research.google/blog/federated-learning-with-formal-differential-privacy-guarantees/) · [self-hosted LLM VRAM/cost](https://www.gpu-mart.com/guides/self-hosted-llm) · [Gemini API pricing](https://ai.google.dev/gemini-api/docs/pricing)

564

565 ## See also

566 - [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) ? C12 (SaaS foundations), C4 (paid-LLM), economics.

567 - [`DEFERRED\_HARDENING\_PLAN.md`](DEFERRED_HARDENING_PLAN.md) ? #16 async job model (the P0 infra slice), #7 packaging.

568 - [`PMF\_MARKET\_RESEARCH.md`](PMF_MARKET_RESEARCH.md) ? academy/coach beachhead (who this serves).

569 - [`CODE\_MAP.md`](CODE_MAP.md) ? current architecture + the registry extension points.

570 - [`ROADMAP.md`](ROADMAP.md) ? offline self-improving rally detection (the pipeline's own trajectory).

571

4. user (2026-06-10T05:33:29.416Z)

```

1      import os
2      import sys
3      import argparse
4      import json
5      import logging
6      import uvicorn
7      from dotenv import load_dotenv
8
9      load_dotenv()
10

```

```

11     # Centralized logging setup for the indexer (hot paths use logger, CLI output uses print
for direct user-facing messages).
12     logging.basicConfig(
13         level=logging.INFO,
14         format='%(asctime)s [%(levelname)s] %(name)s: %(message)s',
15         datefmt='%H:%M:%S'
16     )
17     logger = logging.getLogger(__name__)
18     from fastapi import FastAPI, HTTPException
19     from fastapi.middleware.cors import CORSMiddleware
20     from pydantic import BaseModel
21     from typing import List, Dict, Any, Optional
22
23     from fastapi.staticfiles import StaticFiles
24     from fastapi.responses import HTMLResponse
25     from backend.pipeline.validators.base import registry
26     from backend.infrastructure.database import Database
27     from backend.pipeline.segmenters.base import segmenter_registry, BasePlaySegmenter
28     from backend.pipeline.stitcher.compiler import VideoCompiler
29     from backend.utils.hashing import compute_video_id # canonical file-identity hashing
30     from backend.config import load_config as load_app_config, AppConfig # new typed config
31     from backend.results import Result, Failure # standardized error/result contract
32     from backend.reporting import emit_indexer_report
33
34     app = FastAPI(title="Sports Highlight Indexer API")
35
36     # Enable CORS for local web interface
37     app.add_middleware(
38         CORSMiddleware,
39         allow_origins=["*"],
40         allow_credentials=True,
41         allow_methods=["*"],
42         allow_headers=["*"],
43     )
44
45     # Serves static frontend files directly (now under api/ per approved structure)
46     os.makedirs("backend/api/static", exist_ok=True)
47     app.mount("/static", StaticFiles(directory="backend/api/static"), name="static")
48
49     @app.get("/", response_class=HTMLResponse)
50     def serve_index():
51         index_path = "backend/api/static/index.html"
52         if os.path.exists(index_path):
53             with open(index_path, "r", encoding="utf-8") as f:
54                 return f.read()
55         return "<h3>Sports Highlight Indexer Server is Running. Please create index.html in
backend/api/static/</h3>"
56
57     # Global variables initialized at startup
58     db_instance: Optional[Database] = None
59     global_config: Optional[AppConfig] = None # now typed (Pydantic model)
60
61     # Legacy thin wrapper for any remaining direct calls during transition.
62     # New code should import load_app_config / AppConfig from backend.config directly.
63     def load_config(config_path: str = "config.json") -> Dict[str, Any]:
64         cfg = load_app_config(config_path)
65         return cfg.model_dump(mode="json")
66
67     # --- API Models ---
68     class ProcessRequest(BaseModel):
69         video_path: str
70         player_pool: Optional[List[str]] = None
71         max_frames: Optional[int] = None
72         segmenter: Optional[str] = None
73

```

```

74     class QueryRequest(BaseModel):
75         video_id: str
76         server: Optional[str] = None
77         receiver: Optional[str] = None
78         duration_min: Optional[float] = None
79         event_type: Optional[str] = None
80
81     class CompileRequest(BaseModel):
82         video_id: str
83         segment_ids: List[int]
84         output_filename: str
85
86     # --- API Endpoints ---
87     @app.get("/api/config")
88     def get_config():
89         return global_config
90
91     @app.post("/api/process")
92     def process_video(req: ProcessRequest):
93         if not os.path.exists(req.video_path):
94             raise HTTPException(status_code=404, detail=f"Video file not found at
95 '{req.video_path}'")
96
97         video_id = compute_video_id(req.video_path)
98         was_reingest = db_instance.get_video(video_id) is not None
99
100        # 1. Run pluggable validators (with-context variant surfaces ffprobe_meta for the
report)
101        logger.info(f"Running validations for {req.video_path}...")
102        val_results, val_context = registry.run_all_with_context(req.video_path,
global_config)
103        failures = [r for r in val_results if not r.passed]
104
105        # typed AppConfig: attribute access for the default segmenter (with safe fallback)
106        default_seg = getattr(getattr(global_config, "indexing", None), "default_segmenter",
"motion")
107        seg_name = req.segmenter or default_seg
108        segmenter_actual = None
109        segmentation_ran = False
110        response: Optional[Dict[str, Any]] = None
111        try:
112            if failures:
113                db_instance.add_video(video_id, req.video_path, "failed", [r.dict() for r in
val_results])
114                response = {
115                    "video_id": video_id,
116                    "status": "failed",
117                    "message": "Video failed validation checks.",
118                    "results": [r.dict() for r in val_results],
119                }
120            return response
121
122        # 2. Run segmenter & indexer
123        print(f"[API] Video passed checks. Segmenting and indexing...")
124        db_instance.add_video(video_id, req.video_path, "processed", [r.dict() for r in
val_results])
125
126        segmenter_cls = segmenter_registry.get(seg_name)
127        if not segmenter_cls:
128            available = list(segmenter_registry.list_available().keys())
129            raise HTTPException(
130                status_code=400,
131                detail=f"Segmenter '{seg_name}' not found. Available segmenters:
{available}"

```

```

132
133     print(f"[API] Using segmenter: {seg_name}")
134     segmenter_actual = seg_name
135     segmenter = segmenter_cls(db_instance, global_config)
136     result = segmenter.process_video(video_id, req.video_path, req.player_pool,
req.max_frames)
137     segmentation_ran = True
138
139     if isinstance(result, Failure):
140         print(f"[API] Segmenter failed: {result.message}")
141         response = {
142             "video_id": video_id,
143             "status": "failed",
144             "message": f"Segmenter '{seg_name}' failed: {result.message}",
145             "results": [r.dict() for r in val_results],
146             "play_segments": [],
147         }
148         return response
149
150     segments = result.value if hasattr(result, "value") else result
151
152     response = {
153         "video_id": video_id,
154         "status": "success",
155         "message": f"Successfully indexed {len(segments)} play segments using
'{seg_name}' segmenter.",
156         "results": [r.dict() for r in val_results],
157         "play_segments": segments,
158     }
159     return response
160 finally:
161     # Always emit the per-video observability report (next to the video, or to
162     # report.output_dir). Never raises. Surface the paths in the response.
163     paths = emit_indexer_report(
164         db=db_instance, video_id=video_id, video_path=req.video_path,
config=global_config,
165         val_results=val_results, val_context=val_context,
166         segmenter_requested=req.segmenter, segmenter_actual=segmenter_actual,
167         was_reingest=was_reingest, segmentation_ran=segmentation_ran,
168     )
169     if paths and isinstance(response, dict):
170         response["reports"] = paths
171
172 @app.post("/api/query")
173 def query_play_segments(req: QueryRequest):
174     filters = {
175         "server": req.server,
176         "receiver": req.receiver,
177         "duration_min": req.duration_min,
178         "event_type": req.event_type
179     }
180     segments = db_instance.query_play_segments(req.video_id, filters)
181     return {"play_segments": segments}
182
183 @app.post("/api/compile")
184 def compile_highlights(req: CompileRequest):
185     video = db_instance.get_video(req.video_id)
186     if not video:
187         raise HTTPException(status_code=404, detail="Indexed video record not found.")
188
189     # Retrieve matching play segments from db
190     all_segments = db_instance.query_play_segments(req.video_id, {})
191     matched_segments = [s for s in all_segments if s["id"] in req.segment_ids]
192
193     if not matched_segments:

```

```

194         raise HTTPException(status_code=400, detail="No matching play segments found to
stitch.")
195
196     # Extract start/end time pairs
197     time_pairs = [(s["start_time"], s["end_time"]) for s in matched_segments]
198
199     # Run compiler
200     output_dir = getattr(getattr(global_config, "output", None), "output_dir", "output")
or "output"
201     compiler = VideoCompiler(output_dir)
202     try:
203         out_path = compiler.compile_highlights(video["filepath"], time_pairs,
req.output_filename)
204         return {"status": "success", "filepath": out_path}
205     except Exception as e:
206         raise HTTPException(status_code=500, detail=f"Highlight stitching failed:
{str(e)}")
207
208     # --- CLI Debug Utility & Orchestration ---
209     def run_cli_diagnose_clip(video_path: str, timestamp: float):
210         """CLI debugging utility to examine segment properties around a timestamp."""
211         print("-" * 60)
212         print(f"DEBUGGING VIDEO CLIP: {video_path} at {timestamp}s")
213         print("-" * 60)
214
215         cfg = load_config() # legacy wrapper still works, returns dict for now
216
217         # 1. Validation check diagnostics
218         print("[DIAGNOSTIC] Running validation framework...")
219         results = registry.run_all(video_path, cfg)
220         for r in results:
221             status_symbol = "[PASS]" if r.passed else "[FAIL]"
222             print(f" {status_symbol} {r.validator_name}: {r.message}")
223             if r.details:
224                 print(f"      Details: {r.details}")
225
226         # 2. Local segment analysis around the timestamp (sampling +/- 5 seconds)
227         print("-" * 60)
228         print("[DIAGNOSTIC] Analyzing local motion around the timestamp...")
229         import cv2
230         cap = cv2.VideoCapture(video_path)
231         if not cap.isOpened():
232             print("[FAIL] OpenCV could not open video.")
233             return
234
235         fps = cap.get(cv2.CAP_PROP_FPS)
236         total_frames = cap.get(cv2.CAP_PROP_FRAME_COUNT)
237
238         start_frame = max(0, int((timestamp - 3.0) * fps))
239         end_frame = min(int(total_frames), int((timestamp + 3.0) * fps))
240
241         # Measure frame-by-frame changes in sample region
242         cap.set(cv2.CAP_PROP_POS_FRAMES, start_frame)
243         prev_gray = None
244         motions = []
245
246         for f in range(start_frame, end_frame):
247             ret, frame = cap.read()
248             if not ret:
249                 break
250             gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
251             gray = cv2.GaussianBlur(gray, (21, 21), 0)
252
253             if prev_gray is not None:
254                 diff = cv2.absdiff(prev_gray, gray)

```

```

255         _, thresh = cv2.threshold(diff, 25, 255, cv2.THRESH_BINARY)
256         motion_ratio = cv2.countNonZero(thresh) / (thresh.shape[0] *
thresh.shape[1])
257         motions.append(motion_ratio)
258         prev_gray = gray
259
260     cap.release()
261
262     if motions:
263         avg_motion = sum(motions) / len(motions)
264         # Support both legacy dict (from wrapper) and future direct model
265         if hasattr(cfg, "indexing"):
266             threshold = getattr(cfg.indexing, "motion_threshold", 0.08)
267         else:
268             threshold = cfg.get("indexing", {}).get("motion_threshold", 0.08) # type:
ignore[attr-defined]
269         print(f" Sample Frame Count: {len(motions)}")
270         print(f" Average motion index around timestamp: {round(avg_motion, 4)}
(threshold: {threshold})")
271         if avg_motion > threshold:
272             print(" Result: [ACTIVE PLAY] High motion detected. Segment classifies as
active play.")
273         else:
274             print(" Result: [DEAD TIME] Idle movement. Segment classified as dead
time.")
275     else:
276         print(" [ERROR] No visual frames were extracted in the sample range.")
277
278     print("=" * 60)
279
280     def main():
281         parser = argparse.ArgumentParser(description="Sports Highlight Indexer
Orchestrator")
282         parser.add_argument("--video-path", help="Local path to the MP4 sports video file")
283         parser.add_argument("--output-dir", default="output", help="Directory where
processed clips/highlights are saved")
284         parser.add_argument("--setup", action="store_true", help="Launch dependency checker
and setup utility")
285         parser.add_argument("--debug-clip", type=float, help="Timestamp (in seconds) of a
video clip to run detailed diagnostics on")
286         parser.add_argument("--server", action="store_true", help="Run the FastAPI local API
server")
287         parser.add_argument("--port", type=int, default=8000, help="Port to run the FastAPI
server on")
288         parser.add_argument("--max-frames", type=int, help="Limit processing to the first N
sub-sampled frames for debugging")
289
290         available_segmenters = list(segmenter_registry.list_available().keys())
291         parser.add_argument("--segmenter", choices=available_segmenters, help=f"Choose rally
detection segmenter. Available: {available_segmenters}")
292         parser.add_argument("--trim-start", type=float, help="Start time in seconds for the
debug trim segment feature")
293         parser.add_argument("--trim-end", type=float, help="End time in seconds for the
debug trim segment feature")
294         parser.add_argument("--trim-output", help="Output filename for the debug trimmed
segment (saves in output-dir unless absolute path)")
295
296         args = parser.parse_args()
297
298         if args.setup:
299             # Invoke setup diagnostics
300             import subprocess
301             print("[INFO] Invoking setup.py...")
302             subprocess.run([sys.executable, "setup.py"])
303             return

```

```

304
305     if args.trim_start is not None and args.trim_end is not None:
306         if not args.video_path:
307             print("[ERROR] Please provide --video-path to extract a segment.")
308             return
309
310         # Determine output path
311         out_filename = args.trim_output or "trimmed_segment.mp4"
312         if os.path.isabs(out_filename):
313             out_path = out_filename
314             out_dir = os.path.dirname(out_path)
315             out_name = os.path.basename(out_path)
316         else:
317             out_dir = args.output_dir
318             out_path = os.path.join(out_dir, out_filename)
319             out_name = out_filename
320
321         os.makedirs(out_dir, exist_ok=True)
322         print(f"[CLI] Extracting debug segment from {args.trim_start}s to
{args.trim_end}s from {args.video_path}...")
323
324         compiler = VideoCompiler(out_dir)
325         try:
326             res_path = compiler.compile_highlights(args.video_path, [(args.trim_start,
args.trim_end)], out_name)
327             print(f"[SUCCESS] Trimmed segment saved to: {res_path}")
328         except Exception as e:
329             print(f"[ERROR] Failed to compile trimmed segment: {e}")
330         return
331
332     if args.debug_clip is not None:
333         if not args.video_path:
334             print("[ERROR] Please provide --video-path to debug a specific clip.")
335             return
336         run_cli_diagnose_clip(args.video_path, args.debug_clip)
337         return
338
339     # Initialize Global Config and DB
340     global global_config, db_instance
341     global_config = load_app_config() # returns typed AppConfig
342
343     # The CLI --output-dir overrides the configured output directory. It now lives
344     # on the typed model (OutputConfig.output_dir), so every consumer ? including
345     # /api/compile ? reads the same value instead of a write-only runtime hack.
346     global_config.output.output_dir = args.output_dir
347     os.makedirs(global_config.output.output_dir, exist_ok=True)
348
349     db_path = os.path.join(global_config.output.output_dir,
global_config.output.db_name)
350     db_instance = Database(db_path)
351
352     # CLI processing mode (no web UI needed)
353     if args.video_path and not args.server:
354         print(f"[CLI] Processing video file: {args.video_path}")
355         if not os.path.exists(args.video_path):
356             print(f"[FAIL] Video file not found: {args.video_path}")
357             return
358
359         video_id = compute_video_id(args.video_path)
360         was_reingest = db_instance.get_video(video_id) is not None
361
362         # Validate (with-context variant surfaces ffmpeg_meta for the report)
363         print(f"[CLI] Validating...")
364         val_results, val_context = registry.run_all_with_context(args.video_path,
global_config)

```

```

365         failures = [r for r in val_results if not r.passed]
366
367         # Save video status
368         status = "failed" if failures else "processed"
369         db_instance.add_video(video_id, args.video_path, status, [r.dict() for r in
val_results])
370
371         seg_name = args.segmenter or getattr(getattr(global_config, "indexing", None),
"default_segmenter", "motion")
372         segmenter_actual = None
373         segmentation_ran = False
374         try:
375             print("\n" + "="*40)
376             print("VALIDATION SUMMARY")
377             print("="*40)
378             for r in val_results:
379                 status_symbol = "[PASS]" if r.passed else "[FAIL]"
380                 print(f"{status_symbol} {r.validator_name}: {r.message}")
381             print("="*40 + "\n")
382
383             if failures:
384                 print("[FAIL] Video failed checks. Indexing halted.")
385                 return
386
387             # Index
388             segmenter_cls = segmenter_registry.get(seg_name)
389             if not segmenter_cls:
390                 print(f"[FAIL] Segmenter '{seg_name}' not found.")
391                 return
392
393             print(f"[CLI] Indexing play segments using segmenter: {seg_name}...")
394             segmenter_actual = seg_name
395             segmenter = segmenter_cls(db_instance, global_config)
396             result = segmenter.process_video(video_id, args.video_path,
max_frames=args.max_frames)
397             segmentation_ran = True
398
399             if isinstance(result, Failure):
400                 print(f"[FAIL] Segmenter '{seg_name}' failed: {result.message}")
401                 print(f"[FAIL] Indexing halted for video.")
402                 segments = []
403             else:
404                 segments = result.value if hasattr(result, "value") else result
405                 print(f"[SUCCESS] Process complete. Indexed {len(segments)} play
segments inside database: {db_path}")
406             finally:
407                 # Always emit the per-video observability report. Never raises.
408                 paths = emit_indexer_report(
409                     db=db_instance, video_id=video_id, video_path=args.video_path,
config=global_config,
410                     val_results=val_results, val_context=val_context,
411                     segmenter_requested=args.segmenter, segmenter_actual=segmenter_actual,
412                     was_reingest=was_reingest, segmentation_ran=segmentation_ran,
413                 )
414             if paths:
415                 print(f"[REPORT] Technical report : {paths.get('technical_md')}")
416                 print(f"[REPORT] Customer summary : {paths.get('customer_md')}")
417                 print(f"[REPORT] Machine JSON      : {paths.get('json')}")
418             return
419
420         # Start Local Server Mode
421         if args.server or not args.video_path:
422             print(f"[SERVER] Starting local FastAPI server on
http://localhost:{args.port}...")
423             uvicorn.run(app, host="127.0.0.1", port=args.port)

```

```
424
425     if __name__ == "__main__":
426         main()
427
```

5. user (2026-06-10T05:33:29.894Z)

```
1     import sqlite3
2     import json
3     import os
4     import logging
5     from typing import List, Dict, Any, Optional
6
7     logger = logging.getLogger(__name__)
8
9     class Database:
10         def __init__(self, db_path: str):
11             self.db_path = db_path
12             self._init_db()
13
14         def _get_connection(self):
15             conn = sqlite3.connect(self.db_path)
16             conn.row_factory = sqlite3.Row
17             # Enforce ON DELETE CASCADE / SET NULL ? SQLite leaves FKs off per-connection.
18             conn.execute("PRAGMA foreign_keys = ON")
19             return conn
20
21         def _init_db(self):
22             """Initializes tables for videos, play_segments, and events."""
23             with self._get_connection() as conn:
24                 cursor = conn.cursor()
25
26                 # Videos Table
27                 cursor.execute("""
28                     CREATE TABLE IF NOT EXISTS videos (
29                         id TEXT PRIMARY KEY,
30                         filepath TEXT NOT NULL,
31                         processed_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
32                         status TEXT NOT NULL,
33                         validation_results TEXT
34                     )
35                 """)
36
37                 # Play Segments Table (Extensible metadata JSON)
38                 cursor.execute("""
39                     CREATE TABLE IF NOT EXISTS play_segments (
40                         id INTEGER PRIMARY KEY AUTOINCREMENT,
41                         video_id TEXT NOT NULL,
42                         start_time REAL NOT NULL,
43                         end_time REAL NOT NULL,
44                         duration REAL NOT NULL,
45                         server TEXT,
46                         receiver TEXT,
47                         metadata TEXT,
48                         FOREIGN KEY (video_id) REFERENCES videos(id) ON DELETE CASCADE
49                     )
50                 """)
51
52                 # Events Table
53                 cursor.execute("""
54                     CREATE TABLE IF NOT EXISTS events (
55                         id INTEGER PRIMARY KEY AUTOINCREMENT,
56                         segment_id INTEGER NOT NULL,
57                         timestamp REAL NOT NULL,
58                         type TEXT NOT NULL,
```

```

59         player TEXT,
60         details TEXT,
61         FOREIGN KEY (segment_id) REFERENCES play_segments(id) ON DELETE
CASCADE
62     )
63     """
64
65     # Versioned migrations live here. PRAGMA user_version is the schema
66     # contract ? bump it for every change and add an ordered `if version < N`
67     # block below. Tests assert the expected version, so accidental drift fails
68     CI.
69
70     version = cursor.execute("PRAGMA user_version").fetchone()[0]
71
72     if version < 1:
73         # v1: raw candidate detections (pre-confirmation), detector-agnostic.
74         # Common fields are columns; detector-specific metrics go in `stats`
75         JSON,
76         # so a new segmenter reuses this table by setting its own
77         `detector`/`stats`.
78         cursor.execute("""
79             CREATE TABLE IF NOT EXISTS candidate_segments (
80                 id INTEGER PRIMARY KEY AUTOINCREMENT,
81                 video_id TEXT NOT NULL,
82                 detector TEXT NOT NULL,
83                 chunk_index INTEGER,
84                 start_time REAL NOT NULL,
85                 end_time REAL NOT NULL,
86                 duration REAL NOT NULL,
87                 confidence REAL,
88                 stats TEXT,
89                 detection_params TEXT,
90                 confirmed_segment_id INTEGER,
91                 human_label TEXT,
92                 notes TEXT,
93                 created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
94                 FOREIGN KEY (video_id) REFERENCES videos(id) ON DELETE CASCADE,
95                 FOREIGN KEY (confirmed_segment_id) REFERENCES play_segments(id)
96             ON DELETE SET NULL
97             )
98         """)
99         cursor.execute("""
100             CREATE INDEX IF NOT EXISTS idx_candidates_video_detector
101             ON candidate_segments(video_id, detector)
102         """)
103         cursor.execute("PRAGMA user_version = 1")
104         # future: if version < 2: cursor.execute("ALTER TABLE ..."); PRAGMA
105         user_version = 2
106
107         conn.commit()
108
109     def add_video(self, video_id: str, filepath: str, status: str, validation_results:
110     List[Dict[str, Any]]) -> bool:
111         try:
112             with self._get_connection() as conn:
113                 conn.cursor().execute(
114                     "INSERT OR REPLACE INTO videos (id, filepath, status,
115                     validation_results) VALUES (?, ?, ?, ?)",
116                     (video_id, filepath, status, json.dumps(validation_results))
117                 )
118                 conn.commit()
119             return True
120         except Exception as e:
121             logger.error(f"Failed to add video: {e}")
122             return False

```

```

116         def add_play_segment(self, video_id: str, start_time: float, end_time: float,
server: Optional[str] = None, receiver: Optional[str] = None, metadata: Optional[Dict[str, Any]]
= None) -> Optional[int]:
117             try:
118                 with self._get_connection() as conn:
119                     cursor = conn.cursor()
120                     cursor.execute(
121                         "INSERT INTO play_segments (video_id, start_time, end_time,
duration, server, receiver, metadata) VALUES (?, ?, ?, ?, ?, ?, ?)",
122                         (video_id, start_time, end_time, end_time - start_time, server,
receiver, json.dumps(metadata or {})))
123                     )
124                     conn.commit()
125                     return cursor.lastrowid
126             except Exception as e:
127                 logger.error(f"Failed to add play segment: {e}")
128                 return None
129
130         def add_event(self, segment_id: int, timestamp: float, event_type: str, player:
Optional[str] = None, details: Optional[Dict[str, Any]] = None) -> bool:
131             try:
132                 with self._get_connection() as conn:
133                     conn.cursor().execute(
134                         "INSERT INTO events (segment_id, timestamp, type, player, details)
VALUES (?, ?, ?, ?, ?)",
135                         (segment_id, timestamp, event_type, player, json.dumps(details or
{})))
136                     )
137                     conn.commit()
138                     return True
139             except Exception as e:
140                 logger.error(f"Failed to add event: {e}")
141                 return False
142
143         def add_candidate_segment(self, video_id: str, detector: str, start_time: float,
end_time: float,
144                                     chunk_index: Optional[int] = None, confidence:
Optional[float] = None,
145                                     stats: Optional[Dict[str, Any]] = None,
146                                     detection_params: Optional[Dict[str, Any]] = None) ->
Optional[int]:
147             """Persist one raw (pre-confirmation) detection. `stats`/`detection_params` are
148             detector-specific dicts stored as JSON. Returns the new candidate id, or
None."""
149             try:
150                 with self._get_connection() as conn:
151                     cursor = conn.cursor()
152                     cursor.execute(
153                         """INSERT INTO candidate_segments
154                         (video_id, detector, chunk_index, start_time, end_time, duration,
155                         confidence, stats, detection_params)
156                         VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?)""",
157                         (video_id, detector, chunk_index, start_time, end_time, end_time -
start_time,
158                         confidence, json.dumps(stats or {}), json.dumps(detection_params or
{})))
159                     )
160                     conn.commit()
161                     return cursor.lastrowid
162             except Exception as e:
163                 logger.error(f"Failed to add candidate segment: {e}")
164                 return None
165
166         def link_candidate_to_segment(self, candidate_id: int, segment_id: int) -> bool:
167             """Record that a candidate detection was confirmed as a given play_segment."""

```

```

168         try:
169             with self._get_connection() as conn:
170                 conn.cursor().execute(
171                     "UPDATE candidate_segments SET confirmed_segment_id = ? WHERE id =
172                     ?",
173                     (segment_id, candidate_id)
174                 )
175                 conn.commit()
176             return True
177         except Exception as e:
178             logger.error(f"Failed to link candidate {candidate_id} to segment
179             {segment_id}: {e}")
180             return False
181
182     def query_candidate_segments(self, video_id: str, detector: Optional[str] = None,
183                                only_unlabeled: bool = False) -> List[Dict[str, Any]]:
184         """Fetch candidate detections for the annotation / fine-tuning workflow.
185         `stats` and `detection_params` are deserialized from JSON."""
186         query = "SELECT * FROM candidate_segments WHERE video_id = ?"
187         params: List[Any] = [video_id]
188
189         if detector:
190             query += " AND detector = ?"
191             params.append(detector)
192         if only_unlabeled:
193             query += " AND human_label IS NULL"
194
195         query += " ORDER BY start_time"
196
197         candidates = []
198         with self._get_connection() as conn:
199             rows = conn.cursor().execute(query, params).fetchall()
200             for row in rows:
201                 d = dict(row)
202                 d["stats"] = json.loads(d["stats"] or "{}")
203                 d["detection_params"] = json.loads(d["detection_params"] or "{}")
204                 candidates.append(d)
205             return candidates
206
207     def get_video(self, video_id: str) -> Optional[Dict[str, Any]]:
208         with self._get_connection() as conn:
209             row = conn.cursor().execute("SELECT * FROM videos WHERE id = ?",
210             (video_id,)).fetchone()
211             if row:
212                 d = dict(row)
213                 d["validation_results"] = json.loads(d["validation_results"] or "[]")
214                 return d
215             return None
216
217     def query_play_segments(self, video_id: str, filters: Dict[str, Any]) ->
218     List[Dict[str, Any]]:
219         """
220         Dynamically queries play segments based on filters:
221         - duration_min: float
222         - server: string
223         - receiver: string
224         - event_type: string (e.g. smash, foul)
225         """
226         query = "SELECT r.* FROM play_segments r WHERE r.video_id = ?"
227         params = [video_id]
228
229         if filters.get("duration_min"):
230             query += " AND r.duration >= ?"
231             params.append(filters["duration_min"])

```

```

229         if filters.get("server"):
230             query += " AND r.server = ?"
231             params.append(filters["server"])
232
233         if filters.get("receiver"):
234             query += " AND r.receiver = ?"
235             params.append(filters["receiver"])
236
237         if filters.get("event_type"):
238             # Check if there is an event of a specific type inside this segment
239             query += " AND EXISTS (SELECT 1 FROM events e WHERE e.segment_id = r.id AND
e.type = ?)"
240             params.append(filters["event_type"])
241
242         segments_list = []
243         with self._get_connection() as conn:
244             cursor = conn.cursor()
245             rows = cursor.execute(query, params).fetchall()
246
247             for row in rows:
248                 r_dict = dict(row)
249                 r_id = r_dict["id"]
250                 r_dict["metadata"] = json.loads(r_dict["metadata"] or "{}")
251
252                 # Fetch associated events
253                 event_rows = cursor.execute("SELECT * FROM events WHERE segment_id = ?",
(r_id,)).fetchall()
254                 r_dict["events"] = []
255                 for er in event_rows:
256                     e_dict = dict(er)
257                     e_dict["details"] = json.loads(e_dict["details"] or "{}")
258                     r_dict["events"].append(e_dict)
259
260                 segments_list.append(r_dict)
261
262         return segments_list
263
264     def clear_video_data(self, video_id: str):
265         with self._get_connection() as conn:
266             conn.cursor().execute("DELETE FROM videos WHERE id = ?", (video_id,))
267             conn.commit()
268

```

6. assistant (2026-06-10T05:33:56.646Z)

Now the detector/WSL seam, Gemini provider, compiler, and reporting.

7. user (2026-06-10T05:33:57.700Z)

No files found

8. user (2026-06-10T05:33:58.621Z)

No files found

9. user (2026-06-10T05:33:59.087Z)

```

C:\Users\avidu\Projects\badminton-highlight-indexer\run_all_tests.py
C:\Users\avidu\Projects\badminton-highlight-indexer\run_eval.py
C:\Users\avidu\Projects\badminton-highlight-indexer\run_phase3_eval.py
C:\Users\avidu\Projects\badminton-highlight-indexer\run_process_and_stitch.py
C:\Users\avidu\Projects\badminton-highlight-indexer\run_regression.py

```

10. user (2026-06-10T05:33:59.848Z)

```

1     import os
2     import subprocess
3     import tempfile
4     import logging
5     from typing import List, Tuple, Union
6
7     logger = logging.getLogger(__name__)
8
9     class VideoCompiler:
10         def __init__(self, output_dir: str, end_padding: float = 1.0, begin_padding: float =
11 0.5):
12             self.output_dir = output_dir
13             self.end_padding = end_padding
14             self.begin_padding = begin_padding
15             os.makedirs(output_dir, exist_ok=True)
16
17         @staticmethod
18         def _parse_time(val: Union[int, float, str]) -> float:
19             """Normalizes a timestamp value to float seconds.
20             Handles: float/int (pass-through), plain numeric strings ("12.4"),
21             and colon-separated strings ("MM:SS", "HH:MM:SS", "H:MM:SS.ms").
22             This mirrors the parse_time logic used in gemini.py to ensure
23             the stitcher can always consume whatever format the indexer produced."""
24             if isinstance(val, (int, float)):
25                 return float(val)
26             if isinstance(val, str):
27                 val = val.strip()
28                 if ":" in val:
29                     # Handle MM:SS, HH:MM:SS, etc.
30                     parts = val.split(":")
31                     parts.reverse()
32                     total = 0.0
33                     for i, p in enumerate(parts):
34                         try:
35                             total += float(p) * (60 ** i)
36                         except ValueError:
37                             pass
38                     return total
39                 try:
40                     return float(val)
41                 except ValueError:
42                     logger.warning(f"Could not parse timestamp value '{val}' as a number.
43 Defaulting to 0.0.")
44                     return 0.0
45                 logger.warning(f"Unexpected timestamp type {type(val).__name__} for value
46 '{val}'. Defaulting to 0.0.")
47                 return 0.0
48
49         def _get_video_duration(self, video_path: str) -> float:
50             """Probes the video file to get its total duration in seconds.
51             Returns 0.0 if it cannot be determined."""
52             try:
53                 result = subprocess.run(
54                     [
55                         "ffprobe", "-v", "error",
56                         "-show_entries", "format=duration",
57                         "-of", "default=noprint_wrappers=1:nokey=1",
58                         video_path
59                     ],
60                     capture_output=True, text=True, check=True
61                 )
62                 return float(result.stdout.strip())
63             except Exception as e:
64                 logger.warning(f"Could not determine video duration via ffprobe: {e}")
65                 return 0.0

```

```

63
64     def compile_highlights(self, raw_video_path: str, segments: List[Tuple[float,
float]], output_filename: str) -> str:
65         """
66         Extracts selected segments from a raw video and stitches them together
67         into a seamless highlights file, applying end_padding to prevent abrupt endings.
68         """
69         if not segments:
70             raise ValueError("No highlight segments provided to compile.")
71
72         if not os.path.exists(raw_video_path):
73             raise FileNotFoundError(f"[COMPILER ERROR] Source video file not found:
{raw_video_path}")
74
75         output_path = os.path.join(self.output_dir, output_filename)
76
77         # Probe video duration so we can detect out-of-range segments
78         video_duration = self._get_video_duration(raw_video_path)
79         if video_duration > 0:
80             logger.info(f"Source video duration: {video_duration:.2f}s")
81         else:
82             logger.warning("Video duration unknown. Cannot validate segment boundaries
against video length.")
83
84         # We will create temporary clips and stitch them using FFmpeg concat demuxer
85         temp_clips = []
86         skipped_segments = []
87
88         # Create a temp directory within output_dir
89         with tempfile.TemporaryDirectory(dir=self.output_dir) as tmp_dir:
90             logger.info(f"Trimming {len(segments)} segments
(begin_padding={self.begin_padding}s, end_padding={self.end_padding}s)...")
91             for idx, (raw_start, raw_end) in enumerate(segments):
92                 # Normalize timestamps to float seconds ? handles float, int,
93                 # plain numeric strings, and colon-separated formats like "MM:SS"
94                 start = self._parse_time(raw_start)
95                 end = self._parse_time(raw_end)
96                 segment_label = f"Segment #{idx+1}/{len(segments)} [{start:.2f}s -
{end:.2f}s]"
97
98                 # Validate the segment times
99                 if start < 0:
100                     logger.warning({segment_label}: start_time is negative
({start:.2f}s). Clamping to 0.)
101                     start = 0.0
102
103                 if start >= end:
104                     logger.error({segment_label}: start_time ({start:.2f}s) >= end_time
({end:.2f}s). Skipping invalid segment.)
105                     skipped_segments.append((idx, start, end, "start >= end"))
106                     continue
107
108                 # Check if segment extends beyond video duration
109                 if video_duration > 0:
110                     if start >= video_duration:
111                         logger.error(f"{segment_label}: start_time ({start:.2f}s) is
beyond the video duration ({video_duration:.2f}s). "
112                                     f"This segment cannot be extracted ? the video ran
out. Skipping.")
113                         skipped_segments.append((idx, start, end, "start beyond video
end"))
114                         continue
115
116                 if end > video_duration:
117                     original_end = end

```

```

118             end = video_duration
119             logger.warning(f"{segment_label}: end_time ({original_end:.2f}s)
exceeds video duration ({video_duration:.2f}s). "
120                         f"Clamping end_time to {video_duration:.2f}s. The
segment may be truncated.")
121
122             clip_path = os.path.join(tmp_dir, f"clip_{idx}.mp4")
123
124             # Precise sub-second trim using fast re-encoding to preserve stream
headers
125             padded_start = max(0.0, start - self.begin_padding)
126             padded_end = end + self.end_padding
127
128             # Clamp padded_end to video duration if known
129             if video_duration > 0 and padded_end > video_duration:
130                 padded_end = video_duration
131                 logger.info(f"{segment_label}: Padded end ({end +
self.end_padding:.2f}s) exceeds video duration. "
132                         f"Clamping to {video_duration:.2f}s (end padding
partially applied).")
133
134             trim_duration = padded_end - padded_start
135             logger.info(f"Trimming {segment_label} -> [{padded_start:.3f}s -
{padded_end:.3f}s] (duration: {trim_duration:.2f}s)")
136
137             cmd = [
138                 "ffmpeg", "-y",
139                 "-ss", str(round(padded_start, 3)),
140                 "-to", str(round(padded_end, 3)),
141                 "-i", raw_video_path,
142                 "-c:v", "libx264",
143                 "-preset", "superfast",
144                 "-crf", "22",
145                 "-c:a", "aac",
146                 "-b:a", "128k",
147                 clip_path
148             ]
149
150             try:
151                 result = subprocess.run(cmd, capture_output=True, check=True)
152                 # Verify the output clip was actually created and has content
153                 if os.path.exists(clip_path) and os.path.getsize(clip_path) > 0:
154                     temp_clips.append(clip_path)
155                 else:
156                     logger.error({segment_label}: FFmpeg exited successfully but
output clip is missing or empty. Skipping.)
157                     skipped_segments.append((idx, start, end, "empty output clip"))
158             except subprocess.CalledProcessError as e:
159                 stderr_msg = e.stderr.decode(errors='replace').strip()
160                 logger.error({segment_label}: FFmpeg trim failed.)
161                 print(f"  Command: {' '.join(cmd)}")
162                 print(f"  FFmpeg stderr: {stderr_msg[-500:]}") # Last 500 chars to
avoid flooding
163                 skipped_segments.append((idx, start, end, f"ffmpeg error:
{stderr_msg[-200:]}"))
164                 continue
165
166             # Report summary of skipped segments
167             if skipped_segments:
168                 logger.info(f"\nSUMMARY] {len(skipped_segments)} out of {len(segments)}
segments were SKIPPED:")
169                 for seg_idx, seg_start, seg_end, reason in skipped_segments:
170                     logger.info(f"  - Segment #{seg_idx+1} [{seg_start:.2f}s -
{seg_end:.2f}s]: {reason}")
171                 logger.info("")

```

```

172
173         if not temp_clips:
174             raise RuntimeError(
175                 f"[COMPILER ERROR] Failed to generate ANY valid highlight clips. "
176                 f"All {len(segments)} segments were skipped. "
177                 f"Check the errors above for details. "
178                 f"Common causes: video file is corrupted, segments are out of range,
or ffmpeg is not installed."
179             )
180
181         logger.info(Successfully trimmed {len(temp_clips)}/{len(segments)} clips.
Preparing to stitch...)
182
183         # Create a list file for FFmpeg concat demuxer
184         list_file_path = os.path.join(tmp_dir, "clips.txt")
185         with open(list_file_path, "w") as f:
186             for clip in temp_clips:
187                 # Escape single quotes in path
188                 safe_path = clip.replace("'", "'\\'")
189                 f.write(f"file '{safe_path}'\n")
190
191         logger.info(Stitching clips together...)
192         # Stitch them together without re-encoding (since they were encoded with
identical settings)
193         concat_cmd = [
194             "ffmpeg", "-y",
195             "-f", "concat",
196             "-safe", "0",
197             "-i", list_file_path,
198             "-c", "copy",
199             output_path
200         ]
201
202         try:
203             result = subprocess.run(concat_cmd, capture_output=True, check=True)
204
205             # Final verification
206             if not os.path.exists(output_path) or os.path.getsize(output_path) == 0:
207                 raise RuntimeError(
208                     f"[COMPILER ERROR] FFmpeg concatenation appeared to succeed, "
209                     f"but output file is missing or empty: {output_path}"
210                 )
211
212             output_size_mb = os.path.getsize(output_path) / (1024 * 1024)
213             logger.info(Dynamic highlight reel saved: {output_path}
({output_size_mb:.1f} MB))
214             return output_path
215         except subprocess.CalledProcessError as e:
216             stderr_msg = e.stderr.decode(errors='replace').strip()
217             raise RuntimeError(
218                 f"[COMPILER ERROR] FFmpeg concatenation failed.\n"
219                 f"Command: {' '.join(concat_cmd)}\n"
220                 f"FFmpeg stderr: {stderr_msg[-500:]}")
221
222

```

11. user (2026-06-10T05:34:15.302Z)

```

100:         logger.warning({segment_label}: start_time is negative ({start:.2f}s).
Clamping to 0.)
104:         logger.error({segment_label}: start_time ({start:.2f}s) >= end_time
({end:.2f}s). Skipping invalid segment.)
156:         logger.error({segment_label}: FFmpeg exited successfully but output
clip is missing or empty. Skipping.)
160:         logger.error({segment_label}: FFmpeg trim failed.)

```

[illegible][illegible]

Confirmed ? `compiler.py` has a `SyntaxError` on this branch. Let me check whether this is also on `master` and when it was introduced.

[illegible]

```
Exit code 1
    from backend.pipeline.stitcher.compiler import VideoCompiler
File "C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\stitcher\compiler.py", line 100
    logger.warning({segment_label}: start_time is negative ({start:.2f}s). Clamping to
0.)
                                                                    ^
SyntaxError: invalid decimal literal
```

16. user (2026-06-10T05:34:48.585Z)

```
.claude
.pytest_cache
.sixth
backend
docs
output
scratch
scripts
tests
.pytest_cache\v
.pytest_cache\v\cache
.sixth\skills
backend\annotations
backend\api
backend\audio
backend\config
backend\detectors
backend\eval
backend\indexer
backend\infrastructure
backend\pipeline
backend\providers
backend\reporting
backend\sports
backend\stitcher
backend\tools
backend\utils
backend\validators
backend\api\static
backend\eval\audio
backend\pipeline\audio
backend\pipeline\detectors
backend\pipeline\segmenters
backend\pipeline\stitcher
backend\pipeline\validators
output\Compressed Videos
output\rally_clips
```

17. user (2026-06-10T05:34:57.012Z)

FullName

Length

C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors__init__.py

1034

C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py

4593

C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\rally_gate.py

4497

C:\Users\avidu\Projects\badminton-highlight-

indexer\backend\pipeline\detectors\tracknet_runner.py 12532

C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\wasb_infer.py

18134

C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\wasb_runner.py

7242

C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting__init__.py

2831

C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\harvest.py

9753

C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml

7853

C:\Users\avidu\Projects\badminton-highlight-indexer\backend\providers__init__.py

```

975
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\providers\base.py
1085
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\providers\gemini.py
5527
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\__init__.py
662
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\base.py
2413
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\gemini.py
19333
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\motion.py
9653
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\tracknet_hybrid.py 11739
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\wasb_hybrid.py
10431
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\yolo_hybrid.py
26448

```

18. user (2026-06-10T05:35:03.629Z)

```

1      import os
2      import time
3      import json
4      import logging
5      from typing import Tuple
6      from google import genai
7      from google.genai import types
8      from backend.providers.base import AIVideoProvider
9
10     logger = logging.getLogger(__name__)
11
12     # The Gemini File API caps uploads at 2GB; stay well under it. Callers must chunk so
13     # each uploaded clip is below this. Oversize uploads are refused (not silently
attempted).
14     MAX_UPLOAD_MB = 500
15
16     class GeminiProvider(AIVideoProvider):
17         def __init__(self, api_key: str, model_name: str):
18             super().__init__(api_key, model_name)
19             # Initialize client lazily to avoid connection/auth issues during class
instantiation
20             self._client = None
21
22         @property
23         def client(self):
24             if self._client is None:
25                 self._client = genai.Client(api_key=self.api_key)
26             return self._client
27
28         def analyze_video(self, video_path: str, prompt: str,
29                           chunk_duration: float = 0.0,
30                           label: str = "") -> Tuple[list, dict]:
31             log_prefix = f"Gemini {label}" if label else "Gemini"
32
33             metrics = {
34                 "upload_time": 0.0,
35                 "process_time": 0.0,
36                 "gen_time": 0.0
37             }
38
39             # 0. Refuse oversize uploads (Gemini File API limit is 2GB; we cap lower for
safety/speed)
40             try:

```

```

41         size_mb = os.path.getsize(video_path) / (1024 * 1024)
42     except OSError:
43         size_mb = 0.0
44     if size_mb > MAX_UPLOAD_MB:
45         logger.error(f"[{log_prefix}] {video_path} is {size_mb:.0f}MB >
{MAX_UPLOAD_MB}MB cap ? "
46                     f"chunk smaller. Skipping upload.")
47         return [], metrics
48
49     # 1. Upload (retry transient network failures, e.g. WinError 10053 / reset
connections)
50     logger.info(f"[{log_prefix}] Uploading {video_path}...")
51     t_upload_start = time.time()
52     video_file = None
53     for attempt in range(3):
54         try:
55             video_file = self.client.files.upload(file=video_path)
56             metrics["upload_time"] = time.time() - t_upload_start
57             logger.info(f"[{log_prefix}] Upload complete in
{metrics['upload_time']:.1f}s. URI: {video_file.uri}")
58             break
59         except Exception as e:
60             logger.warning(f"[{log_prefix}] Upload attempt {attempt + 1}/3 failed:
{e}")
61             if attempt < 2:
62                 time.sleep(2 * (attempt + 1))
63     if video_file is None:
64         logger.error(f"[{log_prefix}] Upload failed after 3 attempts.")
65         return [], metrics
66
67     # 2. Wait for processing on Google servers
68     t_process_start = time.time()
69     try:
70         while video_file.state.name == "PROCESSING":
71             logger.info(f"[{log_prefix}] Processing on Google servers. Elapsed:
{time.time()-t_process_start:.1f}s. Retrying in 10s...")
72             time.sleep(10)
73             video_file = self.client.files.get(name=video_file.name)
74             if video_file.state.name == "FAILED":
75                 logger.error(f"[{log_prefix}] Video processing failed:
{video_file.error.message}")
76                 try:
77                     self.client.files.delete(name=video_file.name)
78                 except Exception:
79                     pass
80                 return [], metrics
81     except Exception as e:
82         logger.error(f"[{log_prefix}] Status check failed: {e}")
83         try:
84             self.client.files.delete(name=video_file.name)
85         except Exception:
86             pass
87         return [], metrics
88
89     metrics["process_time"] = time.time() - t_process_start
90     logger.info(f"[{log_prefix}] Processing finished in
{metrics['process_time']:.1f}s.")
91
92     # 3. Call Gemini (retry transient 503/UNAVAILABLE model-overload + network
blips)
93     logger.info(f"[{log_prefix}] Submitting multimodal request to
{self.model_name}...")
94     t_gen_start = time.time()
95     raw_text = None
96     for attempt in range(3):

```

```

97         try:
98             response = self.client.models.generate_content(
99                 model=self.model_name,
100                 contents=[video_file, prompt],
101                 config=types.GenerateContentConfig(
102                     response_mime_type="application/json",
103                     temperature=0.0
104                 )
105             )
106             raw_text = response.text
107             metrics["gen_time"] = time.time() - t_gen_start
108             logger.info(f"[{log_prefix}] Response received in
{metrics['gen_time']:.1f}s:\n{raw_text}")
109             break
110         except Exception as e:
111             logger.warning(f"[{log_prefix}] generate attempt {attempt + 1}/3 failed:
{e}")
112             if attempt < 2:
113                 time.sleep(3 * (attempt + 1))
114         try:
115             self.client.files.delete(name=video_file.name)
116         except Exception:
117             pass
118
119         if not raw_text:
120             return [], metrics
121
122         # 4. Parse JSON
123         try:
124             segments = json.loads(raw_text)
125             if not isinstance(segments, list):
126                 logger.error(f"[{log_prefix}] Response is not a JSON array.")
127                 return [], metrics
128             return segments, metrics
129         except Exception as e:
130             logger.error(f"[{log_prefix}] JSON parse failed: {e}")
131             return [], metrics
132

```

19. user (2026-06-10T05:35:03.700Z)

```

1     """Abstract base for detector runners (TrackNet, WASB, future native, etc.).
2
3     This completes the Detector Runner Abstraction (low-regret item 6 continuation / item 1
4     of the post-observability re-prioritization) to de-risk the WSL/conda dependency for
5     the high-accuracy shuttle trajectory substrate used by tracknet_hybrid / wasb_hybrid.
6
7     The ABC lets us:
8     - Keep the current WSL adapters working unchanged for Windows users.
9     - Swap in (or A/B test) a Linux-native implementation later with zero changes to
10        the hybrid segmenters or the model-agnostic windowing brain.
11     - Provide a single place to evolve the healthcheck / error contract and thread
12        detector readiness into per-video reports.
13
14     Contract (deliberately matches what the concrete runners + hybrids + tests already did):
15     - run_predict(video_win_path: str, output_win_dir: str) -> Optional[str]
16       Run detector, return the Windows-side path to the produced trajectory CSV
17       (e.g. ".../clip_predict.csv" or ".../clip_wasb.csv"), or None on any failure.
18       The caller (hybrid) is responsible for output dir; impls may stage the video
19       into WSL fs for perf.
20
21     - healthcheck() -> tuple[bool, str]
22       Return (ok, message). Never raises. "ok" decides whether to proceed to run_predict.
23       Message is logged (and can be surfaced to reports or UX in future).
24       Intended to be cheap (env existence, import, GPU visibility, weights present).

```

Existing behavior is 100% preserved. The only additions are the inheritance and the explicit common type for future implementers.

How to add a new runner (for docs / future contributors):

1. Subclass `DetectorRunner` in a new or existing module under `pipeline/detectors/`.
2. Implement `run_predict` and `healthcheck` (return (bool, str) tuple).
3. If it produces the same CSV schema (Frame,Visibility,X,Y), reuse or re-export `TrackNetRunner.parse_trajectory_csv` and `.trajectory_to_action_windows` (they are deliberately static and model-agnostic). If the new detector needs different windowing, it can implement its own or we generalize later.
4. Lazy-import heavy native deps inside the methods (so importing the segmenter registry never pulls torch/tf on a machine that doesn't have them).
5. Update the two hybrid segmenters (or introduce a detector= choice) to accept `DetectorRunner` instances (they already do the right thing via duck-typing + the type hints we add here).
6. Add unit tests that exercise the contract (isinstance, healthcheck shape, run returns str|None) using mocks for the external bits ? see `test_tracknet_runner.py`.
7. No change to eval/calibration code paths (they consume the CSV + the static windowing helpers directly).

Optional Linux-native path (the main de-risk goal):

A `LinuxNativeRunner` (or `ONNXRunner`) would:

- Skip all `wsl / _stage / conda activate`.
 - Load weights locally (torch or onnx) or call a native binary.
 - Write the exact same CSV format so `trajectory_to_action_windows` works verbatim.
 - `healthcheck` can just check "weights file exists and torch importable + CUDA?".
- This removes the WSL requirement for Linux users / future SaaS workers while keeping the exact same candidate quality characteristics (or better, once native weights are available).

Error surfacing into reports:

The healthcheck message is already logged by the hybrids on failure. The reporting harvest sees the outcome as "0 candidates" + detector name. When we wire richer segmenter Results (or pass `detector_health` into `emit_indexer_report` context), the (ok, msg) can be recorded explicitly under `segmentation.details` or a new detector stage. The ABC makes that future change local to one place.

See also:

- `backend/pipeline/segmenters/{tracknet_hybrid,wasb_hybrid}.py` (the two callers)
- `tests/test_tracknet_runner.py` (contract + pure logic tests; hybrids mock the runner)
- `docs/TRACKNET_WSL_SETUP.md` and `CODE_MAP.md` (2.2 Detectors section)

```
from abc import ABC, abstractmethod
from typing import Optional, Tuple
```

```
class DetectorRunner(ABC):
```

```
    """Common interface for shuttle/ball trajectory detectors (WSL or otherwise)."""
```

```
    @abstractmethod
```

```
    def run_predict(self, video_win_path: str, output_win_dir: str) -> Optional[str]:
```

```
        """Run inference and return path to trajectory CSV (or None on failure)."""
```

```
        pass
```

```
    @abstractmethod
```

```
    def healthcheck(self) -> Tuple[bool, str]:
```

```
        """Quick check if the runner (env, weights, etc.) is usable.
```

```
        Returns (ok, message). Never raises. Cheap to call before a long run.
```

```
        """
```

```
        pass
```

20. user (2026-06-10T05:35:18.027Z)

```
1      """TrackNet shuttle-trajectory runner (WSL2 subprocess adapter).
2
3      This bridges the Windows-side indexer to a TensorFlow TrackNetV4 install living
4      inside WSL2 (ADR-001 in docs/DECISIONS.md). The flow is:
5
6          1. Translate the Windows video path to its WSL-visible form (and optionally
7             stage it into the WSL filesystem to avoid slow /mnt/c per-frame reads).
8          2. Invoke ``src/predict.py`` inside the TrackNetV4 conda env via ``wsl``.
9          3. Read back the per-frame ``Frame,Visibility,X,Y`` CSV it writes.
10         4. Convert that trajectory into high-motion "action windows" ? the same
11            candidate-rally shape HybridYoloSegmenter produces ? so the rest of the
12            pipeline (AI handoff, DB population) is unchanged.
13
14      Nothing here imports TensorFlow; all model code stays in WSL.
15      """
16
17      import os
18      import csv
19      import shlex
20      import logging
21      import subprocess
22      from dataclasses import dataclass
23      from typing import List, Dict, Any, Optional, Tuple
24
25      from backend.pipeline.detectors.base import DetectorRunner
26
27      logger = logging.getLogger(__name__)
28
29
30      @dataclass
31      class TrackNetConfig:
32          """Resolved settings for a TrackNet WSL run (built from config.json ->
indexing.tracknet)."""
33          distro: str = "Ubuntu"
34          repo_dir: str = "~/models/TrackNetV4"          # WSL path to the cloned repo
35          conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
36          conda_env: str = "TrackNetV4"
37          weights_path: str = ""                          # WSL path to the .h5/.keras weights
38          queue_length: int = 5
39          stage_in_wsl: bool = True                        # copy video into WSL fs first
40          wsl_stage_dir: str = "~/clips"                   # where staged videos/outputs live
41          timeout_sec: int = 0                             # 0 = no timeout
42
43      @classmethod
44      def from_indexing_cfg(cls, idx_cfg: Dict[str, Any]) -> "TrackNetConfig":
45          tn = (idx_cfg or {}).get("tracknet", {}) or {}
46          return cls(
47              distro=tn.get("wsl_distro", "Ubuntu"),
48              repo_dir=tn.get("repo_dir", "~/models/TrackNetV4"),
49              conda_sh=tn.get("conda_sh", "~/miniconda3/etc/profile.d/conda.sh"),
50              conda_env=tn.get("conda_env", "TrackNetV4"),
51              weights_path=tn.get("weights_path", ""),
52              queue_length=int(tn.get("queue_length", 5)),
53              stage_in_wsl=bool(tn.get("stage_in_wsl", True)),
54              wsl_stage_dir=tn.get("wsl_stage_dir", "~/clips"),
55              timeout_sec=int(tn.get("timeout_sec", 0)),
56          )
57
58
59      def to_wsl_mnt_path(win_path: str) -> str:
```

```

60     """Translate a Windows path (C:\\Users\\x) to its WSL /mnt form (/mnt/c/Users/x).
61
62     Already-POSIX paths (starting with / or ~) are returned unchanged so the
63     same helper is safe to call on either kind of path.
64     """
65     p = str(win_path)
66     if p.startswith("/") or p.startswith("~"):
67         return p
68     p = p.replace("\\", "/")
69     if len(p) >= 2 and p[1] == ":":
70         drive = p[0].lower()
71         return f"/mnt/{drive}{p[2:]}"
72     return p
73
74
75 @dataclass
76 class TrajectoryPoint:
77     frame: int
78     visible: bool
79     x: float
80     y: float
81
82
83 class TrackNetRunner(DetectorRunner): # DetectorRunner ABC (finish item 6 / new item 1)
84     """Runs TrackNet predict.py in WSL and turns the output CSV into action windows.
85
86     Implements the common DetectorRunner contract so a future Linux-native or
87     alternative trajectory runner can be substituted without touching the hybrids.
88     """
89
90     def __init__(self, cfg: TrackNetConfig):
91         self.cfg = cfg
92
93     # ----- WSL exec
94
95     def _wsl_bash(self, inner_cmd: str) -> subprocess.CompletedProcess:
96         """Run a bash command string inside the configured WSL distro."""
97         full = f"source {self.cfg.conda_sh} && {inner_cmd}"
98         argv = ["wsl", "-d", self.cfg.distro, "bash", "-c", full]
99         logger.debug("WSL exec: %s", full)
100         return subprocess.run(
101             argv, capture_output=True, text=True,
102             timeout=(self.cfg.timeout_sec or None),
103         )
104
105     def healthcheck(self) -> Tuple[bool, str]:
106         """Verify the WSL env is usable: conda env exists, TF imports, GPU visible.
107
108         Returns (ok, message). Cheap to call before a long run.
109         """
110         cmd = (
111             f"conda activate {self.cfg.conda_env} && "
112             f"python -c \"import tensorflow as tf; "
113             f"print('TF', tf.__version__, 'GPUs', "
114             f"len(tf.config.list_physical_devices('GPU'))))\"")
115         try:
116             res = self._wsl_bash(cmd)
117         except FileNotFoundError:
118             return False, "wsl` not found ? is this running on Windows with WSL2
119             installed?"
120         except subprocess.TimeoutExpired:
121             return False, "WSL healthcheck timed out."
122         out = (res.stdout or "").strip()
123         if res.returncode != 0:

```

```

123         return False, f"WSL env check failed: {(res.stderr or out).strip()}"
124     return True, out
125
126 # ----- inference
127
128 def run_predict(self, video_win_path: str, output_win_dir: str) -> Optional[str]:
129     """Run predict.py on a video. Returns the Windows path to the produced CSV, or
None.
130
131     ``output_win_dir`` is a Windows directory (under the repo's output/) that
132     WSL writes into via its /mnt view, so the Windows process can read the CSV
133     back with no copy.
134     """
135     if not self.cfg.weights_path:
136         logger.error(
137             "TrackNet weights_path is not set. Set indexing.tracknet.weights_path "
138             "in config.json to the WSL path of your trained TrackNetV4 weights."
139         )
140         return None
141
142     os.makedirs(output_win_dir, exist_ok=True)
143     # Normalize separators so basename/splitext work when tests (or collector)
144     # pass Windows-style paths on a Linux host.
145     video_win_path = str(video_win_path).replace("\\", "/")
146     base = os.path.basename(video_win_path)
147     stem, ext = os.path.splitext(base)
148
149     # predict.py only accepts .avi/.mp4 and names the CSV "<stem>_predict.csv".
150     if ext.lower() not in (".mp4", ".avi"):
151         logger.error("TrackNet predict.py only supports .mp4/.avi (got %s)", ext)
152         return None
153
154     wsl_out = to_wsl_mnt_path(output_win_dir)
155     expected_csv_win = os.path.join(output_win_dir, f"{stem}_predict.csv")
156
157     # Resolve the video path WSL will read.
158     if self.cfg.stage_in_wsl:
159         staged = self._stage_video(video_win_path, base)
160         if staged is None:
161             return None
162         wsl_video = staged
163     else:
164         wsl_video = to_wsl_mnt_path(video_win_path)
165
166     cmd = (
167         f"conda activate {self.cfg.conda_env} && "
168         f"cd {self.cfg.repo_dir}/src && "
169         f"python predict.py "
170         f"--video_path {shlex.quote(wsl_video)} "
171         f"--model_weights {shlex.quote(self.cfg.weights_path)} "
172         f"--output_dir {shlex.quote(wsl_out)} "
173         f"--queue_length {self.cfg.queue_length}"
174     )
175     logger.info("Running TrackNet inference on %s ...", base)
176     try:
177         res = self._wsl_bash(cmd)
178     except subprocess.TimeoutExpired:
179         logger.error("TrackNet inference timed out after %ss.",
self.cfg.timeout_sec)
180         return None
181
182     if res.returncode != 0:
183         logger.error("TrackNet predict.py failed:\n%s", (res.stderr or
res.stdout)[-2000:])
184         return None

```

```

185
186         if not os.path.exists(expected_csv_win):
187             logger.error("TrackNet finished but expected CSV not found at %s",
expected_csv_win)
188             return None
189             logger.info("TrackNet trajectory CSV: %s", expected_csv_win)
190             return expected_csv_win
191
192     def _stage_video(self, video_win_path: str, base: str) -> Optional[str]:
193         """Copy the input video into the WSL filesystem (avoids slow /mnt/c reads).
194
195         Returns the WSL path to the staged copy, or None on failure.
196         """
197         src_mnt = to_wsl_mnt_path(video_win_path)
198         dest = f"{self.cfg.wsl_stage_dir}/{base}"
199         cmd = f"mkdir -p {self.cfg.wsl_stage_dir} && cp {shlex.quote(src_mnt)}
{shlex.quote(dest)} && echo OK"
200         try:
201             res = self._wsl_bash(cmd)
202         except subprocess.TimeoutExpired:
203             logger.error("Staging video into WSL timed out.")
204             return None
205         if res.returncode != 0 or "OK" not in (res.stdout or ""):
206             logger.error("Failed to stage video into WSL: %s", (res.stderr or
res.stdout).strip())
207             return None
208             return dest
209
210     # ----- CSV -> windows
211
212     @staticmethod
213     def parse_trajectory_csv(csv_win_path: str) -> List[TrajectoryPoint]:
214         """Parse a TrackNet predict CSV (columns: Frame,Visibility,X,Y)."""
215         points: List[TrajectoryPoint] = []
216         with open(csv_win_path, newline="") as f:
217             reader = csv.DictReader(f)
218             for row in reader:
219                 try:
220                     points.append(TrajectoryPoint(
221                         frame=int(row["Frame"]),
222                         visible=int(row["Visibility"]) == 1,
223                         x=float(row["X"]),
224                         y=float(row["Y"]),
225                     ))
226                 except (KeyError, ValueError):
227                     continue
228             points.sort(key=lambda p: p.frame)
229             return points
230
231     @staticmethod
232     def trajectory_to_action_windows(
233         points: List[TrajectoryPoint],
234         fps: float,
235         frame_width: float,
236         chunk_start: float = 0.0,
237         velocity_thresh: float = 0.02,
238         merge_gap: float = 2.0,
239         min_window_duration: float = 2.0,
240         chunk_index: Optional[int] = None,
241     ) -> List[Dict[str, Any]]:
242         """Convert a shuttle trajectory into candidate rally windows.
243
244         A frame is "active" when the shuttle is visible AND its inter-frame
245         displacement (normalised by frame width, per second) exceeds
246         ``velocity_thresh`` ? i.e. the shuttle is in flight. Active frames within

```

```

247     ``merge_gap`` seconds of each other are grouped into one window; short
248     windows are dropped. Mirrors HybridYoloSegmenter's windowing so the dict
249     shape feeding the AI-handoff phase is identical.
250     """
251     if fps <= 0:
252         fps = 30.0
253     if frame_width <= 0:
254         frame_width = 1280.0
255
256     # Compute per-frame normalised velocity from the last *visible* point.
257     active = [] # (frame_idx, velocity)
258     prev: Optional[TrajectoryPoint] = None
259     for p in points:
260         if not p.visible:
261             prev = None # break the trajectory; absent shuttle = not in flight
262             continue
263         if prev is not None:
264             dframes = p.frame - prev.frame
265             if dframes > 0:
266                 dist = ((p.x - prev.x) ** 2 + (p.y - prev.y) ** 2) ** 0.5
267                 velocity = (dist / frame_width) * (fps / dframes)
268                 if velocity > velocity_thresh:
269                     active.append((p.frame, velocity))
270             prev = p
271
272     if not active:
273         return []
274
275     max_gap_frames = int(merge_gap * fps)
276
277     def _finalize(group: List[Tuple[int, float]]) -> Dict[str, Any]:
278         vels = [v for _, v in group]
279         return {
280             "start": group[0][0] / fps + chunk_start,
281             "end": group[-1][0] / fps + chunk_start,
282             "active_frame_count": len(group),
283             "peak_velocity": max(vels),
284             "mean_velocity": sum(vels) / len(vels),
285             "track_id_count": 1, # single shuttle; kept for window-shape parity
286             "chunk_index": chunk_index,
287         }
288
289     windows = []
290     group = [active[0]]
291     for af in active[1:]:
292         if af[0] - group[-1][0] <= max_gap_frames:
293             group.append(af)
294         else:
295             windows.append(_finalize(group))
296             group = [af]
297     windows.append(_finalize(group))
298
299     return [w for w in windows if (w["end"] - w["start"]) >= min_window_duration]
300

```

21. user (2026-06-10T05:35:18.286Z)

```

1     """WASB shuttle-trajectory runner (WSL2 subprocess adapter).
2
3     Windows-side counterpart to backend/detectors/wasb_infer.py. It:
4     1. stages the video into the WSL filesystem (fast local reads, not /mnt),
5     2. copies wasb_infer.py into the WASB repo's src/ dir,
6     3. runs it in the WSL `wasb` conda env -> a Frame,Visibility,X,Y trajectory CSV,
7     4. reuses the (model-agnostic) trajectory->action-window logic from tracknet_runner.
8

```

```

9      So the only WASB-specific bits live here; the windowing + CSV parsing are shared
10     with the TrackNet path. Verified inference core: wasb_infer.py (see its docstring).
11     """
12     import os
13     import shlex
14     import logging
15     import subprocess
16     from dataclasses import dataclass
17     from typing import Dict, Any, Optional, Tuple
18
19     # Reuse the model-agnostic helpers ? trajectory shape + windowing are identical.
20     # parse_trajectory_csv / trajectory_to_action_windows are @staticmethods on
TrackNetRunner.
21     from backend.pipeline.detectors.tracknet_runner import to_wsl_mnt_path, TrackNetRunner
22     from backend.pipeline.detectors.base import DetectorRunner
23
24     logger = logging.getLogger(__name__)
25
26     # Windows path to the inference wrapper we stage into WSL.
27     _INFER_WIN = os.path.join(os.path.dirname(os.path.abspath(__file__)), "wasb_infer.py")
28
29
30     @dataclass
31     class WasbConfig:
32         """Resolved settings for a WASB WSL run (built from config.json ->
indexing.wasb)."""
33         distro: str = "Ubuntu"
34         repo_dir: str = "~/models/WASB-SBDT"
35         conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
36         conda_env: str = "wasb"
37         weights_path: str = "~/models/WASB-
SBDT/pretrained_weights/wasb_badminton_best.pth.tar"
38         sport: str = "badminton"
39         wsl_stage_dir: str = "~/clips"
40         timeout_sec: int = 0 # 0 = no timeout
41
42         @classmethod
43         def from_indexing_cfg(cls, idx_cfg: Dict[str, Any]) -> "WasbConfig":
44             w = (idx_cfg or {}).get("wasb", {}) or {}
45             return cls(
46                 distro=w.get("wsl_distro", "Ubuntu"),
47                 repo_dir=w.get("repo_dir", "~/models/WASB-SBDT"),
48                 conda_sh=w.get("conda_sh", "~/miniconda3/etc/profile.d/conda.sh"),
49                 conda_env=w.get("conda_env", "wasb"),
50                 weights_path=w.get("weights_path",
~/models/WASB-
SBDT/pretrained_weights/wasb_badminton_best.pth.tar"),
51                 sport=w.get("sport", "badminton"),
52                 wsl_stage_dir=w.get("wsl_stage_dir", "~/clips"),
53                 timeout_sec=int(w.get("timeout_sec", 0)),
54             )
55
56
57
58     class WasbRunner(DetectorRunner): # DetectorRunner ABC (finish item 6 / new item 1)
59         def __init__(self, cfg: WasbConfig):
60             self.cfg = cfg
61
62         def _wsl_bash(self, inner_cmd: str) -> subprocess.CompletedProcess:
63             full = f"source {self.cfg.conda_sh} && {inner_cmd}"
64             argv = ["wsl", "-d", self.cfg.distro, "bash", "-c", full]
65             logger.debug("WSL exec: %s", full)
66             return subprocess.run(argv, capture_output=True, text=True,
67                                   timeout=(self.cfg.timeout_sec or None))
68
69         def healthcheck(self) -> Tuple[bool, str]:

```

```

70         """Verify the WSL `wasb` env imports torch and sees a GPU. Returns (ok, msg)."""
71         cmd = (f"conda activate {self.cfg.conda_env} && "
72               f"python -c \"import torch; print('torch', torch.__version__, "
73               f"'cuda', torch.cuda.is_available())\"")
74         try:
75             res = self._wsl_bash(cmd)
76         except FileNotFoundError:
77             return False, "`wsl` not found ? Windows + WSL2 required."
78         except subprocess.TimeoutExpired:
79             return False, "WSL healthcheck timed out."
80         out = (res.stdout or "").strip()
81         if res.returncode != 0:
82             return False, f"WSL `wasb` env check failed: {(res.stderr or out).strip()}"
83         return True, out
84
85     def _stage_video(self, video_win_path: str, base: str) -> Optional[str]:
86         src_mnt = to_wsl_mnt_path(video_win_path)
87         dest = f"{self.cfg.wsl_stage_dir}/{base}"
88         cmd = f"mkdir -p {self.cfg.wsl_stage_dir} && cp {shlex.quote(src_mnt)} "
89         {shlex.quote(dest)} && echo OK"
90         try:
91             res = self._wsl_bash(cmd)
92         except subprocess.TimeoutExpired:
93             logger.error("Staging video into WSL timed out.")
94             return None
95         if res.returncode != 0 or "OK" not in (res.stdout or ""):
96             logger.error("Failed to stage video into WSL: %s", (res.stderr or
97 res.stdout).strip())
98             return None
99         return dest
100
101     def _stage_infer_script(self) -> bool:
102         """Copy wasb_infer.py into the WASB repo src/ so it can import WASB modules."""
103         src_mnt = to_wsl_mnt_path(_INFER_WIN)
104         cmd = f"cp {shlex.quote(src_mnt)} {self.cfg.repo_dir}/src/wasb_infer.py && echo
105 OK"
106         res = self._wsl_bash(cmd)
107         return res.returncode == 0 and "OK" in (res.stdout or "")
108
109     def run_predict(self, video_win_path: str, output_win_dir: str) -> Optional[str]:
110         """Run WASB on a video. Returns the Windows path to the trajectory CSV, or
111 None."""
112         os.makedirs(output_win_dir, exist_ok=True)
113         # Normalize for cross-platform basename (tests use Windows paths on Linux).
114         video_win_path = str(video_win_path).replace("\\", "/")
115         base = os.path.basename(video_win_path)
116         stem = os.path.splitext(base)[0]
117         wsl_out_dir = to_wsl_mnt_path(output_win_dir)
118         expected_csv_win = os.path.join(output_win_dir, f"{stem}_wasb.csv")
119
120         if not self._stage_infer_script():
121             logger.error("Could not stage wasb_infer.py into WSL.")
122             return None
123         staged_video = self._stage_video(video_win_path, base)
124         if staged_video is None:
125             return None
126         frames_dir = f"{self.cfg.wsl_stage_dir}/{stem}_frames"
127
128         cmd = (
129             f"conda activate {self.cfg.conda_env} && cd {self.cfg.repo_dir}/src && "
130             f"python wasb_infer.py "
131             f"--video {shlex.quote(staged_video)} "
132             f"--frames_out_dir {shlex.quote(frames_dir)} "
133             f"--weights {shlex.quote(self.cfg.weights_path)} "
134             f"--sport {shlex.quote(self.cfg.sport)} "

```

```

131         f"--out {shlex.quote(wsl_out_dir + '/' + stem + '_wasb.csv'}}"
132     )
133     logger.info("Running WASB inference on %s ...", base)
134     try:
135         res = self._wsl_bash(cmd)
136     except subprocess.TimeoutExpired:
137         logger.error("WASB inference timed out after %ss.", self.cfg.timeout_sec)
138         return None
139     if res.returncode != 0:
140         logger.error("WASB inference failed:\n%s", (res.stderr or
res.stdout)[-2000:])
141         return None
142     if not os.path.exists(expected_csv_win):
143         logger.error("WASB finished but expected CSV not found at %s",
expected_csv_win)
144         return None
145     logger.info("WASB trajectory CSV: %s", expected_csv_win)
146     return expected_csv_win
147
148     # Trajectory parsing + windowing are inherited from tracknet_runner (model-
agnostic).
149     parse_trajectory_csv = staticmethod(TrackNetRunner.parse_trajectory_csv)
150     trajectory_to_action_windows =
staticmethod(TrackNetRunner.trajectory_to_action_windows)
151

```

22. user (2026-06-10T05:35:44.067Z)

```

1     import os
2     import time
3     import json
4     import logging
5     import subprocess
6     import concurrent.futures
7     from typing import List, Dict, Any, Tuple
8     from backend.infrastructure.database import Database
9     from backend.pipeline.segmenters.base import BasePlaySegmenter, segmenter_registry
10    from backend.utils.video import get_video_duration, extract_video_chunk
11    from backend.sports import sport_registry
12    from backend.providers import provider_registry
13
14    logger = logging.getLogger(__name__)
15
16    class GeminiPlaySegmenter(BasePlaySegmenter):
17        @property
18        def name(self) -> str:
19            return "gemini"
20
21        @property
22        def description(self) -> str:
23            return "High-precision AI-powered semantic play segment detection using Google
Gemini Multimodal API."
24
25        def process_video(self, video_id: str, video_path: str, player_pool: List[str] =
None, max_frames: int = None) -> List[Dict[str, Any]]:
26            """
27            Processes a video to detect play segments using Google Gemini Multimodal Video
API,
28            populates SQLite, and indexes detailed events.
29            """
30            # Get sport profile config
31            sport_name = self.config.get("sport", "badminton")
32            try:
33                sport_profile = sport_registry.get(sport_name)
34            except KeyError as e:

```

```

35         logger.error(str(e))
36         return []
37
38         # Get AI provider and model config
39         idx_cfg = self.config.get("indexing", {})
40         provider_name = self.config.get("ai_provider", idx_cfg.get("ai_provider",
"gemini"))
41         model_name = self.config.get("ai_model", idx_cfg.get("ai_model",
"gemini-2.5-flash"))
42
43         # Get appropriate API key based on the provider
44         api_key_env_var = f"{provider_name.upper()}_API_KEY"
45         api_key = os.environ.get(api_key_env_var)
46         if not api_key:
47             logger.error(
48                 f"{api_key_env_var} environment variable is not set! "
49                 f"To run the {provider_name} segmenter, set your API key. "
50                 f"In PowerShell: $env:{api_key_env_var}=\"your_api_key_here\""
51             )
52             return []
53
54         try:
55             provider = provider_registry.get(provider_name, api_key=api_key,
model_name=model_name)
56         except KeyError as e:
57             logger.error(str(e))
58             return []
59
60         # A: Check for debug_duration to trim the video first
61         debug_duration = self.config.get("indexing", {}).get("debug_duration")
62         video_to_upload = video_path
63         is_trimmed = False
64         temp_trimmed_path = None
65         is_compressed = False
66         temp_compressed_path = None
67
68         if debug_duration:
69             logger.info(f"DEBUG FEATURE: Trimming video to first {debug_duration}
seconds for fast processing...")
70             temp_trimmed_path = os.path.join("output", f"temp_trimmed_{video_id}.mp4")
71             os.makedirs("output", exist_ok=True)
72             if os.path.exists(temp_trimmed_path):
73                 try:
74                     os.remove(temp_trimmed_path)
75                 except Exception:
76                     pass
77
78             cmd = [
79                 "ffmpeg", "-y",
80                 "-ss", "0",
81                 "-t", str(debug_duration),
82                 "-i", video_path,
83                 "-c", "copy",
84                 temp_trimmed_path
85             ]
86             logger.debug(f"Running command: {' '.join(cmd)}")
87             try:
88                 subprocess.run(cmd, check=True, stdout=subprocess.DEVNULL,
stderr=subprocess.DEVNULL)
89                 logger.info(f"Trimmed video successfully created at:
{temp_trimmed_path}")
90                 video_to_upload = temp_trimmed_path
91                 is_trimmed = True
92             except Exception as e:
93                 logger.warning(f"Failed to trim video using ffmpeg: {e}. Falling back to

```

```

full video.")
94
95
96
97     # Helper to clean up local temp files
98     def cleanup_temp_files():
99         if is_trimmed and temp_trimmed_path and os.path.exists(temp_trimmed_path):
100             try:
101                 os.remove(temp_trimmed_path)
102             except Exception:
103                 pass
104
105     # 1. Determine actual video duration
106     video_duration = get_video_duration(video_to_upload)
107     if video_duration > 0:
108         logger.info(f"Detected video duration: {video_duration:.1f}s")
109     else:
110         logger.warning("Could not detect video duration. Prompt will not include
duration context.")
111
112     # 2. Decide whether to use chunked processing
113     idx_cfg = self.config.get("indexing", {})
114     chunk_duration = idx_cfg.get("chunk_duration", 300)    # 5 minutes default
115     chunk_overlap = idx_cfg.get("chunk_overlap", 30)       # 30 seconds overlap
116     use_chunking = video_duration > 0 and video_duration > chunk_duration
117
118     ai_segments = []
119     chunks_dir = os.path.join("output", "chunks")
120
121     if use_chunking:
122         # --- CHUNKED PROCESSING ---
123         step = chunk_duration - chunk_overlap
124         chunk_starts = []
125         t = 0.0
126         while t < video_duration:
127             chunk_starts.append(t)
128             t += step
129         num_chunks = len(chunk_starts)
130         logger.info(
131             f"Video is {video_duration:.1f}s ? splitting into {num_chunks}
overlapping chunks "
132             f"({chunk_duration}s each, {chunk_overlap}s overlap).")
133         )
134
135         os.makedirs(chunks_dir, exist_ok=True)
136
137         def process_single_chunk(ci: int, chunk_start: float) -> Tuple[str, list,
dict]:
138             chunk_end = min(chunk_start + chunk_duration, video_duration)
139             actual_chunk_dur = chunk_end - chunk_start
140             chunk_label = f"Chunk {ci+1}/{num_chunks}"
141             chunk_path = os.path.join(chunks_dir, f"chunk_{video_id}_{ci}.mp4")
142
143             # Extract chunk with ffmpeg
144             logger.info(f"Extracting {chunk_label}: {chunk_start:.1f}s -
{chunk_end:.1f}s ({actual_chunk_dur:.1f}s)")
145             success = extract_video_chunk(video_to_upload, chunk_start,
actual_chunk_dur, chunk_path)
146             if not success:
147                 logger.error(f"Failed to extract {chunk_label}. Skipping.")
148                 return chunk_label, [], {}
149
150             # Measure chunk size
151             try:
152                 chunk_size_mb = os.path.getsize(chunk_path) / (1024 * 1024)

```

```

153         except Exception:
154             chunk_size_mb = 0.0
155
156         # Get prompt from sport profile
157         prompt = sport_profile.get_prompt(chunk_duration=actual_chunk_dur)
158
159         # Process this chunk through the provider
160         chunk_segments, metrics = provider.analyze_video(
161             chunk_path, prompt, chunk_duration=actual_chunk_dur,
label=chunk_label
162         )
163         metrics["file_size_mb"] = chunk_size_mb
164         metrics["segments_found"] = len(chunk_segments)
165
166         # Offset timestamps back to full-video coordinates
167         for segment in chunk_segments:
168             if "start_time" in segment:
169                 try:
170                     segment["start_time"] = float(segment["start_time"]) +
chunk_start
171                 except (ValueError, TypeError):
172                     pass
173             if "end_time" in segment:
174                 try:
175                     segment["end_time"] = float(segment["end_time"]) +
chunk_start
176                 except (ValueError, TypeError):
177                     pass
178
179         logger.info(f"{chunk_label} returned {len(chunk_segments)} play
segments.")
180
181         # Clean up chunk file
182         try:
183             os.remove(chunk_path)
184         except Exception:
185             pass
186
187         return chunk_label, chunk_segments, metrics
188
189     # Run chunks in parallel
190     all_metrics = {}
191     logger.info("Starting parallel processing with up to 5 concurrent
workers...")
192     t_chunking_start = time.time()
193     with concurrent.futures.ThreadPoolExecutor(max_workers=5) as executor:
194         futures = [executor.submit(process_single_chunk, ci, start) for ci,
start in enumerate(chunk_starts)]
195         for future in concurrent.futures.as_completed(futures):
196             label, segments, metrics = future.result()
197             ai_segments.extend(segments)
198             if metrics:
199                 all_metrics[label] = metrics
200
201     # Print Summary Log
202     summary_lines = [
203         "",
204         "=" * 60,
205         "                GEMINI CHUNKING SUMMARY",
206         "=" * 60,
207         f"Total time: {time.time() - t_chunking_start:.1f}s",
208         f"Total chunks: {num_chunks}",
209         f"Total play segments found: {len(ai_segments)}",
210         "-" * 60,
211         f"{'Chunk':<12} | {'Size':<8} | {'Upload':<8} | {'Process':<8} |

```

```

{'Gen.':<8} | {'Segments'}",
212         "-" * 60,
213     ]
214     for i in range(num_chunks):
215         lbl = f"Chunk {i+1}/{num_chunks}"
216         m = all_metrics.get(lbl, {})
217         sz = f"{m.get('file_size_mb', 0):.1f}MB" if m else "N/A"
218         upl = f"{m.get('upload_time', 0):.1f}s" if m else "N/A"
219         prc = f"{m.get('process_time', 0):.1f}s" if m else "N/A"
220         gen = f"{m.get('gen_time', 0):.1f}s" if m else "N/A"
221         seg = m.get('segments_found', 0) if m else 0
222         summary_lines.append(f" {lbl:<12} | {sz:<8} | {upl:<8} | {prc:<8} |
{'gen:<8} | {seg}")
223         summary_lines.append("-" * 60)
224         logger.info("\n".join(summary_lines))
225
226         # Clean up local temp files
227         cleanup_temp_files()
228         try:
229             os.rmdir(chunks_dir)
230         except Exception:
231             pass
232
233         logger.info(f"All chunks processed. {len(ai_segments)} total play segments
before deduplication.")
234
235     else:
236         # --- SINGLE-FILE PROCESSING (short videos) ---
237         try:
238             file_sz = os.path.getsize(video_to_upload) / (1024 * 1024)
239         except Exception:
240             file_sz = 0.0
241
242         # Get prompt from sport profile
243         prompt = sport_profile.get_prompt(chunk_duration=video_duration)
244         ai_segments, metrics = provider.analyze_video(
245             video_to_upload, prompt, chunk_duration=video_duration,
label="SingleFile"
246         )
247         summary_lines = [
248             "",
249             "-" * 50,
250             "          GEMINI SINGLE FILE SUMMARY",
251             "-" * 50,
252             f" File Size: {file_sz:.1f} MB",
253             f" Upload time: {metrics.get('upload_time', 0):.1f}s",
254             f" API Processing time: {metrics.get('process_time', 0):.1f}s",
255             f" Prompt Generation time: {metrics.get('gen_time', 0):.1f}s",
256             f" Play segments found: {len(ai_segments)}",
257             "-" * 50,
258         ]
259         logger.info("\n".join(summary_lines))
260         cleanup_temp_files()
261
262     if not ai_segments:
263         return []
264
265     # Populate SQLite DB with parsed play segments
266     segments_data = []
267     logger.info(f"Processing {len(ai_segments)} play segments through
validation...")
268
269     def parse_time(val) -> float:
270         """Converts a timestamp value to total float seconds.
271         Handles float/int pass-through, plain numeric strings, and

```

```

272         colon-separated formats (MM:SS, HH:MM:SS). Logs warnings
273         when unexpected formats are encountered."""
274         if isinstance(val, (int, float)):
275             return float(val)
276         if isinstance(val, str):
277             val = val.strip()
278             if ":" in val:
279                 # Gemini returned MM:SS or HH:MM:SS despite being asked for decimal
seconds.
280                 # Convert it, but warn so the issue is visible in logs.
281                 parts = val.split(":")
282                 parts.reverse()
283                 total = 0.0
284                 for i, p in enumerate(parts):
285                     try:
286                         total += float(p) * (60 ** i)
287                     except ValueError:
288                         pass
289                 logger.warning(f"Timestamp '{val}' was in colon-separated format
(MM:SS/HH:MM:SS) instead of decimal seconds. Converted to {total:.2f}s.")
290                 return total
291             try:
292                 return float(val)
293             except ValueError:
294                 logger.warning(f"Could not parse timestamp string '{val}' as a
number. Defaulting to 0.0.")
295                 return 0.0
296                 logger.warning(f"Unexpected timestamp type {type(val).__name__} for value
'{val}'. Defaulting to 0.0.")
297                 return 0.0
298
299         # 7. Parse timestamps and build candidate segment list
300         idx_cfg = self.config.get("indexing", {})
301         min_segment_duration = idx_cfg.get("min_segment_duration", 3.0)
302         max_segment_duration = 90.0 # Singles play segments almost never exceed this
303
304         candidates = []
305         for idx, segment in enumerate(ai_segments):
306             start = parse_time(segment.get("start_time", 0.0))
307             end = parse_time(segment.get("end_time", 0.0))
308             if start >= end:
309                 logger.info(f"Skipping segment #{idx+1}: start_time ({start:.2f}s) >=
end_time ({end:.2f}s)")
310                 continue
311                 candidates.append({
312                     "idx": idx,
313                     "start": start,
314                     "end": end,
315                     "raw": segment
316                 })
317
318         # 8. Post-hoc validation pass
319         logger.info(f"Running validation on {len(candidates)} candidate play
segments...")
320         validated = []
321         rejected = []
322
323         for c in candidates:
324             label = f"Segment #{c['idx']+1} [{c['start']:.2f}s - {c['end']:.2f}s]"
325
326             # A: Reject segments that start beyond the video
327             if video_duration > 0 and c["start"] >= video_duration:
328                 rejected.append((label, f"start_time ({c['start']:.2f}s) is beyond video
duration ({video_duration:.1f}s)"))
329                 continue

```

```

330
331         # B: Clamp end_time to video duration
332         if video_duration > 0 and c["end"] > video_duration:
333             logger.info(f"{label}: end_time ({c['end']:.2f}s) exceeds video duration
({video_duration:.1f}s). Clamping.")
334             c["end"] = video_duration
335
336         dur = c["end"] - c["start"]
337
338         # C: Reject segments shorter than min_segment_duration
339         if dur < min_segment_duration:
340             rejected.append((label, f"duration ({dur:.2f}s) is below minimum
({min_segment_duration}s)"))
341             continue
342
343         # D: Flag suspiciously long segments
344         if dur > max_segment_duration:
345             logger.warning(
346                 f"{label}: duration ({dur:.1f}s) is unusually long
(>{max_segment_duration}s). "
347                 f"This may be a detection error. Keeping but flagging."
348             )
349
350         validated.append(c)
351
352     # E: Resolve overlapping segments (keep the longer one)
353     if len(validated) > 1:
354         # Sort by start time
355         validated.sort(key=lambda r: r["start"])
356         non_overlapping = [validated[0]]
357         for curr in validated[1:]:
358             prev = non_overlapping[-1]
359             # Check if current segment overlaps with the previous one
360             if curr["start"] < prev["end"]:
361                 prev_dur = prev["end"] - prev["start"]
362                 curr_dur = curr["end"] - curr["start"]
363                 kept, dropped = (prev, curr) if prev_dur >= curr_dur else (curr,
prev)
364                 kept_label = f"Segment #{kept['idx']+1} [{kept['start']:.2f}s -
{kept['end']:.2f}s]"
365                 dropped_label = f"Segment #{dropped['idx']+1}
[{dropped['start']:.2f}s - {dropped['end']:.2f}s]"
366                 rejected.append((dropped_label, f"overlaps with {kept_label} ? kept
the longer segment"))
367                 if kept == curr:
368                     non_overlapping[-1] = curr # Replace previous with current
369                 else:
370                     non_overlapping.append(curr)
371             validated = non_overlapping
372
373     # Log validation summary
374     if rejected:
375         reject_lines = [f" ? {label}: {reason}" for label, reason in rejected]
376         logger.info(f"{len(rejected)} play segments were REJECTED:\n" +
"\n".join(reject_lines))
377         logger.info(f"{len(validated)} play segments passed validation (out of
{len(ai_segments)} from Gemini).")
378
379     stitch_cfg = self.config.get("stitching", {})
380     begin_pad = stitch_cfg.get("begin_padding", 0.5)
381     end_pad = stitch_cfg.get("end_padding", 0.5)
382
383     total_highlight_dur = 0.0
384     for c in validated:
385         total_highlight_dur += (c["end"] - c["start"]) + begin_pad + end_pad

```

```

386
387         mins = int(total_highlight_dur // 60)
388         secs = int(total_highlight_dur % 60)
389         logger.info(
390             f"Estimated final highlight video duration: {mins:02d}:{secs:02d} "
391             f"({total_highlight_dur:.1f}s total; includes {begin_pad}s begin +
{end_pad}s end padding per segment)"
392         )
393
394     # 9. Populate SQLite DB with validated play segments
395     for c in validated:
396         start = c["start"]
397         end = c["end"]
398         segment = c["raw"]
399         dur = end - start
400
401         # Extract shot_count from Gemini or compute estimate if not present
402         shot_count = segment.get("shuttle_exchange_count")
403         if shot_count is None:
404             shot_count = max(3, int(dur / 0.8)) # 1 shot every 0.8 seconds as
fallback estimate
405         else:
406             try:
407                 shot_count = int(shot_count)
408             except Exception:
409                 shot_count = max(3, int(dur / 0.8))
410
411         # Only store metrics we actually know ? no fabricated data
412         metadata = {
413             "shot_count": shot_count,
414             "shot_count_confidence": segment.get("shot_count_confidence",
"Unknown"),
415             "detector": model_name
416         }
417
418         segment_id = self.db.add_play_segment(video_id, start, end, None, None,
metadata)
419
420         if segment_id:
421             segments_data.append({
422                 "id": segment_id,
423                 "start_time": start,
424                 "end_time": end,
425                 "duration": dur,
426                 "metadata": metadata
427             })
428
429     return segments_data
430
431     # Register the segmenter
432     segmenter_registry.register(GeminiPlaySegmenter)
433

```

23. user (2026-06-10T05:36:05.029Z)

```

1     """WASB-based hybrid play segmenter.
2
3     Identical three-phase shape to TrackNetHybridSegmenter, but Phase 2 uses **WASB**
4     (nttcom/WASB-SBDT, MIT) shuttle-trajectory detection via WSL2
5     (backend/detectors/wasb_runner.py) instead of TrackNetV4. WASB ships pretrained
6     badminton weights and runs today, so this is the live shuttle-motion substrate.
7     Phase 3 (AI semantic boundaries) and Phase 4 (DB) are shared with the other hybrids.
8
9     For the per-video *calibration* goal we mainly use Phase 2 (trajectory -> candidate
10    windows); the windowing thresholds (velocity_thresh / merge_gap / min_window_duration)

```

```

11     are what get tuned against golden rally labels.
12     """
13     import os
14     import time
15     import logging
16     import concurrent.futures
17     from typing import List, Dict, Any
18
19     import cv2
20
21     from backend.pipeline.segmenters.base import BasePlaySegmenter, segmenter_registry
22     from backend.utils.video import get_video_duration, extract_video_chunk
23     from backend.sports import sport_registry
24     from backend.providers import provider_registry
25     from backend.pipeline.detectors.wasb_runner import WasbRunner, WasbConfig
26     from backend.pipeline.detectors.base import DetectorRunner # ABC for the de-risked
detector seam
27
28     logger = logging.getLogger(__name__)
29
30
31     def _fmt_ts(seconds: float) -> str:
32         seconds = max(0, int(seconds))
33         h, rem = divmod(seconds, 3600)
34         m, s = divmod(rem, 60)
35         return f"{h:02d}:{m:02d}:{s:02d}"
36
37
38     class WasbHybridSegmenter(BasePlaySegmenter):
39         @property
40         def name(self) -> str:
41             return "wasb_hybrid"
42
43         @property
44         def description(self) -> str:
45             return ("Two-stage pipeline: WASB shuttle-trajectory detection (WSL2, pretrained
badminton "
46                     "weights) for precise candidate rallies + AI provider for semantic
boundaries.")
47
48         @staticmethod
49         def _probe_fps_width(video_path: str) -> tuple:
50             cap = cv2.VideoCapture(video_path)
51             fps = cap.get(cv2.CAP_PROP_FPS) if cap.isOpened() else 0.0
52             width = cap.get(cv2.CAP_PROP_FRAME_WIDTH) if cap.isOpened() else 0.0
53             cap.release()
54             return (fps if fps > 0 else 30.0, width if width > 0 else 1280.0)
55
56         def process_video(self, video_id: str, video_path: str, player_pool: List[str] =
None,
57                         max_frames: int = None) -> List[Dict[str, Any]]:
58             sport_name = self.config.get("sport", "badminton")
59             try:
60                 sport_profile = sport_registry.get(sport_name)
61             except KeyError as e:
62                 logger.error(str(e))
63                 return []
64
65             idx_cfg = self.config.get("indexing", {})
66             provider_name = self.config.get("ai_provider", idx_cfg.get("ai_provider",
"gemini"))
67             model_name = self.config.get("ai_model", idx_cfg.get("ai_model",
"gemini-2.5-flash"))
68
69             api_key_env_var = f"{provider_name.upper()}_API_KEY"

```

```

70         api_key = os.environ.get(api_key_env_var)
71         if not api_key:
72             logger.error(f"{api_key_env_var} not set; cannot run the {provider_name}
handoff.")
73             return []
74         try:
75             provider = provider_registry.get(provider_name, api_key=api_key,
model_name=model_name)
76         except KeyError as e:
77             logger.error(str(e))
78             return []
79
80         video_duration = get_video_duration(video_path)
81         if video_duration <= 0:
82             logger.error("Invalid video duration.")
83             return []
84         fps, frame_width = self._probe_fps_width(video_path)
85
86         wasb_cfg = WasbConfig.from_indexing_cfg(idx_cfg)
87         runner: DetectorRunner = WasbRunner(wasb_cfg)
88
89         w = idx_cfg.get("wasb", {})
90         velocity_thresh = w.get("velocity_threshold", 0.02)
91         merge_gap = idx_cfg.get("dead_time_merge_gap", 2.0)
92         min_window_duration = idx_cfg.get("min_action_window_duration", 2.0)
93         handoff_padding = idx_cfg.get("ai_handoff_padding", 2.0)
94         ai_max_workers = idx_cfg.get("ai_max_workers", 5)
95         log_candidates = idx_cfg.get("log_yolo_candidates", True)
96         store_candidates = idx_cfg.get("store_yolo_candidates", True)
97
98         detection_params = {
99             "detector": "wasb_hybrid",
100             "velocity_thresh": velocity_thresh,
101             "merge_gap": merge_gap,
102             "min_action_window_duration": min_window_duration,
103             "weights_path": wasb_cfg.weights_path,
104         }
105
106         ok, msg = runner.healthcheck()
107         if not ok:
108             logger.error("WASB WSL environment not ready: %s", msg)
109             return []
110         logger.info("WASB WSL env OK: %s", msg)
111
112         # --- Phase 2: shuttle trajectory -> candidate rally windows ---
113         t_start = time.time()
114         out_dir = os.path.join("output", "wasb")
115         csv_path = runner.run_predict(video_path, out_dir)
116         if not csv_path:
117             logger.error("WASB produced no trajectory; aborting.")
118             return []
119
120         points = runner.parse_trajectory_csv(csv_path)
121         logger.info("Parsed %d trajectory points (%d visible).",
122                     len(points), sum(1 for p in points if p.visible))
123
124         all_candidate_windows = runner.trajectory_to_action_windows(
125             points, fps=fps, frame_width=frame_width, chunk_start=0.0,
126             velocity_thresh=velocity_thresh, merge_gap=merge_gap,
127             min_window_duration=min_window_duration, chunk_index=0,
128         )
129         logger.info("Phase 2 (WASB) completed in %.1fs. Candidate windows: %d",
130                     time.time() - t_start, len(all_candidate_windows))
131
132         if log_candidates:

```

```

133         for cw in all_candidate_windows:
134             logger.info(f"WASB rally] {_fmt_ts(cw['start'])}-{_fmt_ts(cw['end'])} "
135                         f"({cw['end'] - cw['start']:.1f}s) |
frames={cw['active_frame_count']} "
136                         f"peak_v={cw['peak_velocity']:.3f}
mean_v={cw['mean_velocity']:.3f}")
137
138         candidate_ids = [None] * len(all_candidate_windows)
139         if store_candidates:
140             for i, cw in enumerate(all_candidate_windows):
141                 stats = {
142                     "peak_velocity": cw["peak_velocity"], "mean_velocity":
cw["mean_velocity"],
143                     "active_frame_count": cw["active_frame_count"], "track_id_count":
cw["track_id_count"],
144                 }
145                 candidate_ids[i] = self.db.add_candidate_segment(
146                     video_id, detector="wasb_hybrid", start_time=cw["start"],
end_time=cw["end"],
147                     chunk_index=cw["chunk_index"], confidence=min(1.0,
cw["peak_velocity"]),
148                     stats=stats, detection_params=detection_params,
149                 )
150
151         if not all_candidate_windows:
152             return []
153
154         # --- Phase 3: AI provider handoff (shared pattern) ---
155         handoff_dir = os.path.join("output", f"chunks_{provider_name}_handoff")
156         os.makedirs(handoff_dir, exist_ok=True)
157         logger.info("Phase 3: %s validation on %d candidate windows...",
158                   provider_name, len(all_candidate_windows))
159
160         def process_candidate_window(wi: int, window: Dict[str, Any]) -> List[dict]:
161             w_start = max(0, window["start"] - handoff_padding)
162             w_end = min(video_duration, window["end"] + handoff_padding)
163             w_dur = w_end - w_start
164             w_path = os.path.join(handoff_dir, f"handoff_{video_id}_{wi}.mp4")
165             if not extract_video_chunk(video_path, w_start, w_dur, w_path,
reencode=True):
166                 return []
167             prompt = sport_profile.get_prompt(chunk_duration=w_dur)
168             segments, _ = provider.analyze_video(w_path, prompt, chunk_duration=w_dur,
169                                                label=f"Handoff {wi+1}")
170             valid = []
171             for seg in segments:
172                 try:
173                     seg["start_time"] = float(seg["start_time"]) + w_start
174                     seg["end_time"] = float(seg["end_time"]) + w_start
175                     seg["_candidate_id"] = candidate_ids[wi]
176                     valid.append(seg)
177                 except (ValueError, TypeError, KeyError) as e:
178                     logger.warning("Dropping segment with unparseable timestamps: %s ?
%s", seg, e)
179             try:
180                 os.remove(w_path)
181             except OSError:
182                 pass
183             return valid
184
185         ai_segments: List[dict] = []
186         with concurrent.futures.ThreadPoolExecutor(max_workers=ai_max_workers) as
executor:
187             futures = [executor.submit(process_candidate_window, wi, cw)
188                        for wi, cw in enumerate(all_candidate_windows)]

```

```

189         for future in futures:
190             ai_segments.extend(future.result())
191     try:
192         os.rmdir(handoff_dir)
193     except OSError:
194         pass
195     logger.info("Phase 3 completed. AI confirmed %d semantic play segments.",
len(ai_segments))
196
197     # --- Phase 4: DB population (shared pattern) ---
198     segments_data = []
199     for segment in ai_segments:
200         try:
201             start = float(segment["start_time"])
202             end = float(segment["end_time"])
203             dur = end - start
204             shot_count = segment.get("shuttle_exchange_count")
205             try:
206                 shot_count = int(shot_count)
207             except (ValueError, TypeError):
208                 shot_count = max(3, int(dur / 0.8))
209             metadata = {"shot_count": shot_count,
210                        "shot_count_confidence":
segment.get("shot_count_confidence", "Unknown"),
211                        "detector": f"wasb-hybrid-{model_name}"}
212             segment_id = self.db.add_play_segment(video_id, start, end, None, None,
metadata)
213             if segment_id:
214                 candidate_id = segment.get("_candidate_id")
215                 if candidate_id is not None:
216                     self.db.link_candidate_to_segment(candidate_id, segment_id)
217                 segments_data.append({"id": segment_id, "start_time": start,
"end_time": end,
218                                     "duration": dur, "metadata": metadata})
219         except Exception as e:
220             logger.warning("Skipping invalid segment struct: %s", e)
221
222     return segments_data
223
224
225     segmenter_registry.register(WasbHybridSegmenter)
226

```

24. user (2026-06-10T05:36:05.568Z)

```

1     """TrackNet-based hybrid play segmenter.
2
3     Same three-phase shape as HybridYoloSegmenter, but Phase 2 (candidate
4     generation) uses a TrackNetV4 *shuttle* trajectory instead of YOLO *person*
5     motion. TrackNet runs in WSL2 (see backend/detectors/tracknet_runner.py and
6     docs/DECISIONS.md ADR-001); the shuttle trajectory is converted into candidate
7     rally windows, which are then handed to the AI provider for semantic boundary
8     confirmation (Phase 3) and written to the DB (Phase 4).
9
10    Rationale: a tiny, fast, motion-blurred shuttle is exactly what general
11    detectors miss; a purpose-built trajectory model localises it far more
12    precisely, giving cleaner candidate rallies than person-velocity heuristics.
13    """
14
15    import os
16    import time
17    import logging
18    import concurrent.futures
19    from typing import List, Dict, Any
20

```

```

21     import cv2
22
23     from backend.pipeline.segmenters.base import BasePlaySegmenter, segmenter_registry
24     from backend.utils.video import get_video_duration, extract_video_chunk
25     from backend.sports import sport_registry
26     from backend.providers import provider_registry
27     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner, TrackNetConfig
28     from backend.pipeline.detectors.base import DetectorRunner # ABC for the de-risked
detector seam
29
30     logger = logging.getLogger(__name__)
31
32
33     def _fmt_ts(seconds: float) -> str:
34         seconds = max(0, int(seconds))
35         h, rem = divmod(seconds, 3600)
36         m, s = divmod(rem, 60)
37         return f"{h:02d}:{m:02d}:{s:02d}"
38
39
40     class TrackNetHybridSegmenter(BasePlaySegmenter):
41         @property
42         def name(self) -> str:
43             return "tracknet_hybrid"
44
45         @property
46         def description(self) -> str:
47             return ("Two-stage pipeline: TrackNetV4 shuttle-trajectory detection (WSL2) for
precise "
48                     "candidate rallies + AI provider for high-precision semantic
boundaries.")
49
50         @staticmethod
51         def _probe_fps_width(video_path: str) -> tuple:
52             """Return (fps, frame_width) for a video, with sensible fallbacks."""
53             cap = cv2.VideoCapture(video_path)
54             fps = cap.get(cv2.CAP_PROP_FPS) if cap.isOpened() else 0.0
55             width = cap.get(cv2.CAP_PROP_FRAME_WIDTH) if cap.isOpened() else 0.0
56             cap.release()
57             return (fps if fps > 0 else 30.0, width if width > 0 else 1280.0)
58
59         def process_video(self, video_id: str, video_path: str, player_pool: List[str] =
None,
60                         max_frames: int = None) -> List[Dict[str, Any]]:
61             # --- Sport profile + AI provider wiring (identical to the other segmenters) ---
62             sport_name = self.config.get("sport", "badminton")
63             try:
64                 sport_profile = sport_registry.get(sport_name)
65             except KeyError as e:
66                 logger.error(str(e))
67                 return []
68
69             idx_cfg = self.config.get("indexing", {})
70             provider_name = self.config.get("ai_provider", idx_cfg.get("ai_provider",
"gemini"))
71             model_name = self.config.get("ai_model", idx_cfg.get("ai_model",
"gemini-2.5-flash"))
72
73             api_key_env_var = f"{provider_name.upper()}_API_KEY"
74             api_key = os.environ.get(api_key_env_var)
75             if not api_key:
76                 logger.error(
77                     f"{api_key_env_var} environment variable is not set! "
78                     f"To run the {provider_name} handoff, set your API key. "
79                     f"In PowerShell: $env:{api_key_env_var}=\"your_api_key_here\"")

```

```

80         )
81     return []
82     try:
83         provider = provider_registry.get(provider_name, api_key=api_key,
model_name=model_name)
84     except KeyError as e:
85         logger.error(str(e))
86         return []
87
88     video_duration = get_video_duration(video_path)
89     if video_duration <= 0:
90         logger.error("Invalid video duration.")
91         return []
92     fps, frame_width = self._probe_fps_width(video_path)
93
94     # --- TrackNet runner config + thresholds ---
95     tn_cfg = TrackNetConfig.from_indexing_cfg(idx_cfg)
96     runner: DetectorRunner = TrackNetRunner(tn_cfg)
97
98     tn = idx_cfg.get("tracknet", {})
99     velocity_thresh = tn.get("velocity_threshold", 0.02)    # normalised shuttle
displacement/sec
100     merge_gap = idx_cfg.get("dead_time_merge_gap", 2.0)
101     min_window_duration = idx_cfg.get("min_action_window_duration", 2.0)
102     handoff_padding = idx_cfg.get("ai_handoff_padding", 2.0)
103     ai_max_workers = idx_cfg.get("ai_max_workers", 5)
104     log_candidates = idx_cfg.get("log_yolo_candidates", True)
105     store_candidates = idx_cfg.get("store_yolo_candidates", True)
106
107     detection_params = {
108         "detector": "tracknet_hybrid",
109         "velocity_thresh": velocity_thresh,
110         "merge_gap": merge_gap,
111         "min_action_window_duration": min_window_duration,
112         "weights_path": tn_cfg.weights_path,
113         "queue_length": tn_cfg.queue_length,
114     }
115
116     # --- Phase 1: env healthcheck (fail fast with a clear message) ---
117     ok, msg = runner.healthcheck()
118     if not ok:
119         logger.error("TrackNet WSL environment not ready: %s", msg)
120         return []
121     logger.info("TrackNet WSL env OK: %s", msg)
122
123     # --- Phase 2: shuttle trajectory -> candidate rally windows ---
124     # TrackNet processes the whole video in one pass (it carries its own
125     # frame buffering), so unlike the YOLO segmenter we don't pre-chunk here.
126     t_start = time.time()
127     tn_out_dir = os.path.join("output", "tracknet")
128     csv_path = runner.run_predict(video_path, tn_out_dir)
129     if not csv_path:
130         logger.error("TrackNet produced no trajectory; aborting.")
131         return []
132
133     points = runner.parse_trajectory_csv(csv_path)
134     logger.info("Parsed %d trajectory points (%d visible).",
135                 len(points), sum(1 for p in points if p.visible))
136
137     all_candidate_windows = runner.trajectory_to_action_windows(
138         points, fps=fps, frame_width=frame_width, chunk_start=0.0,
139         velocity_thresh=velocity_thresh, merge_gap=merge_gap,
140         min_window_duration=min_window_duration, chunk_index=0,
141     )
142     logger.info("Phase 2 (TrackNet) completed in %.1fs. Candidate windows: %d",

```

```

143         time.time() - t_start, len(all_candidate_windows))
144
145     if log_candidates:
146         for w in all_candidate_windows:
147             logger.info(
148                 f"[TrackNet rally] {_fmt_ts(w['start'])}-{_fmt_ts(w['end'])} "
149                 f"({w['end'] - w['start']:.1f}s) | frames={w['active_frame_count']}
150                 "
151                 f"peak_v={w['peak_velocity']:.3f} mean_v={w['mean_velocity']:.3f}"
152             )
153
154     # Persist raw candidates for the fine-tuning dataset (same table as YOLO).
155     candidate_ids = [None] * len(all_candidate_windows)
156     if store_candidates:
157         for i, w in enumerate(all_candidate_windows):
158             stats = {
159                 "peak_velocity": w["peak_velocity"],
160                 "mean_velocity": w["mean_velocity"],
161                 "active_frame_count": w["active_frame_count"],
162                 "track_id_count": w["track_id_count"],
163             }
164             confidence = min(1.0, w["peak_velocity"])
165             candidate_ids[i] = self.db.add_candidate_segment(
166                 video_id, detector="tracknet_hybrid",
167                 start_time=w["start"], end_time=w["end"],
168                 chunk_index=w["chunk_index"], confidence=confidence,
169                 stats=stats, detection_params=detection_params,
170             )
171
172     if not all_candidate_windows:
173         return []
174
175     # --- Phase 3: AI provider handoff (identical pattern to HybridYoloSegmenter)
176     ---
177     handoff_dir = os.path.join("output", f"chunks_{provider_name}_handoff")
178     os.makedirs(handoff_dir, exist_ok=True)
179     logger.info("Phase 3: %s validation on %d candidate windows...",
180                 provider_name, len(all_candidate_windows))
181
182     def process_candidate_window(wi: int, window: Dict[str, Any]) -> List[dict]:
183         w_start = max(0, window["start"] - handoff_padding)
184         w_end = min(video_duration, window["end"] + handoff_padding)
185         w_dur = w_end - w_start
186         w_path = os.path.join(handoff_dir, f"handoff_{video_id}_{wi}.mp4")
187         if not extract_video_chunk(video_path, w_start, w_dur, w_path,
188                                   reencode=True):
189             return []
190         prompt = sport_profile.get_prompt(chunk_duration=w_dur)
191         segments, _ = provider.analyze_video(w_path, prompt, chunk_duration=w_dur,
192                                              label=f"Handoff {wi+1}")
193
194         valid = []
195         for seg in segments:
196             try:
197                 seg["start_time"] = float(seg["start_time"]) + w_start
198                 seg["end_time"] = float(seg["end_time"]) + w_start
199                 seg["candidate_id"] = candidate_ids[wi]
200                 valid.append(seg)
201             except (ValueError, TypeError, KeyError) as e:
202                 logger.warning("Dropping segment with unparseable timestamps: %s ?
203                                %s", seg, e)
204
205         try:
206             os.remove(w_path)
207         except OSError:
208             pass
209         return valid

```

```

204
205     ai_segments: List[dict] = []
206     with concurrent.futures.ThreadPoolExecutor(max_workers=ai_max_workers) as
executor:
207         futures = [executor.submit(process_candidate_window, wi, w)
208                     for wi, w in enumerate(all_candidate_windows)]
209         for future in futures:
210             ai_segments.extend(future.result())
211
212     try:
213         os.rmdir(handoff_dir)
214     except OSError:
215         pass
216     logger.info("Phase 3 completed. AI confirmed %d semantic play segments.",
len(ai_segments))
217
218     # --- Phase 4: DB population (identical pattern to HybridYoloSegmenter) ---
219     segments_data = []
220     for segment in ai_segments:
221         try:
222             start = float(segment["start_time"])
223             end = float(segment["end_time"])
224             dur = end - start
225             shot_count = segment.get("shuttle_exchange_count")
226             if shot_count is None:
227                 shot_count = max(3, int(dur / 0.8))
228             else:
229                 try:
230                     shot_count = int(shot_count)
231                 except (ValueError, TypeError):
232                     shot_count = max(3, int(dur / 0.8))
233             metadata = {
234                 "shot_count": shot_count,
235                 "shot_count_confidence": segment.get("shot_count_confidence",
"Unknown"),
236                 "detector": f"tracknet-hybrid-{model_name}",
237             }
238             segment_id = self.db.add_play_segment(video_id, start, end, None, None,
metadata)
239             if segment_id:
240                 candidate_id = segment.get("_candidate_id")
241                 if candidate_id is not None:
242                     self.db.link_candidate_to_segment(candidate_id, segment_id)
243                 segments_data.append({
244                     "id": segment_id, "start_time": start, "end_time": end,
245                     "duration": dur, "metadata": metadata,
246                 })
247         except Exception as e:
248             logger.warning("Skipping invalid segment struct: %s", e)
249
250     return segments_data
251
252
253     # Register the segmenter
254     segmenter_registry.register(TrackNetHybridSegmenter)
255

```

25. user (2026-06-10T05:36:19.926Z)

```

1     """WASB shuttle-trajectory inference on an arbitrary frame sequence (runs in WSL).
2
3     WASB-SBDT only ships *dataset* evaluation; this is a thin wrapper that reuses its
4     own building blocks (ImageDataset transform + detector + online tracker) to run on
5     any folder of extracted frames and emit a trajectory CSV in the exact shape the
6     indexer's `tracknet_runner.parse_trajectory_csv` already consumes:

```

```

7
8     Frame, Visibility, X, Y
9
10 It must run INSIDE the WSL `wasb` conda env, from the WASB-SBDT `src/` dir (it
11 imports WASB's `detectors`/`trackers`/`dataloaders`). The Windows-side adapter
12 (`wasb_runner.py`) stages this file + the frames into WSL and invokes it.
13
14 Resumable across reboots
15 -----
16 The GPU detector pass (one forward per sliding window ? ~21k for a 6-min clip) is
17 the slow, expensive part; the CPU tracker pass that turns detections into a
18 trajectory is cheap and *stateful* (it must see every frame in order, so it can't
19 resume mid-stream). We exploit that split:
20
21 * The detector pass writes its per-window raw outputs incrementally to
22   ``<cache>/det_raw.jsonl`` (flushed+fsync'd every ``--flush-every`` windows) and
23   records progress in ``<cache>/manifest.json`` (atomic write). A reboot loses at
24   most ``--flush-every`` windows of GPU work instead of the whole pass.
25 * On restart we replay the cached windows, skip the DataLoader ahead to the first
26   un-cached window, and continue. Once every window is cached, the detector pass
27   is skipped entirely and we just re-run the (cheap, deterministic) tracker over
28   the full ordered cache -> trajectory CSV. So a lost trajectory CSV costs seconds,
29   not the GPU hour.
30
31 The cache is keyed by the output path (``<out>_wasbcache/`` by default), lives next
32 to the output (NOT in %TEMP%), and is invalidated automatically if the frames dir,
33 weights, sport, window size, or frame count change.
34
35 Usage (in WSL, from ~/models/WASB-SBDT/src):
36     python wasb_infer.py --frames_dir /path/frames --weights
37     ../pretrained_weights/wasb_badminton_best.pth.tar \
38     --out /path/traj.csv [--sport badminton] [--limit N] [--cache-dir DIR] [--flush-
39     every 1000] [--fresh]
40     """
41     import os
42     import os.path as osp
43     import csv
44     import glob
45     import json
46     import argparse
47     import logging
48     from collections import defaultdict
49
50     logger = logging.getLogger(__name__)
51
52     CACHE_VERSION = 1
53     _DET_FILE = "det_raw.jsonl"
54     _MANIFEST_FILE = "manifest.json"
55
56     def _build_cfg(weights: str, sport: str):
57         """Compose the WASB eval config via Hydra, overriding model/weights/device."""
58         from hydra import compose, initialize
59         with initialize(config_path="configs", version_base=None):
60             cfg = compose(config_name="eval", overrides=[
61                 f"dataset={sport}",
62                 "model=wasb",
63                 f"detector.model_path={weights}",
64                 "runner.device=cuda",
65                 "runner.gpus=[0]", # this box has a single GPU; default config assumes
66
67         ])
68         return cfg

```

```

69     def _frame_id(path: str) -> int:
70         """Best-effort integer frame id from a frame filename (e.g. frame_000180.png ->
180)."""
71         stem = osp.splitext(osp.basename(path))[0]
72         digits = "".join(ch for ch in stem if ch.isdigit())
73         return int(digits) if digits else -1
74
75
76     # ----- #
77     # Resumable detector cache (pure-Python, no torch/numpy ? unit-testable)
78     # ----- #
79     def _cache_paths(cache_dir: str):
80         return osp.join(cache_dir, _DET_FILE), osp.join(cache_dir, _MANIFEST_FILE)
81
82
83     def default_cache_dir(out_csv: str) -> str:
84         """Stable cache dir next to the trajectory output (survives reboots; not %TEMP%)."""
85         d = osp.dirname(osp.abspath(out_csv))
86         stem = osp.splitext(osp.basename(out_csv))[0]
87         return osp.join(d, stem + "_wasbcache")
88
89
90     def _atomic_write_text(path: str, text: str) -> None:
91         """Write then rename so a crash mid-write can't leave a half-written file."""
92         tmp = path + ".tmp"
93         with open(tmp, "w") as f:
94             f.write(text)
95             f.flush()
96             os.fsync(f.fileno())
97         os.replace(tmp, path)
98
99
100    def _det_plain(det) -> list:
101        """A detector candidate dict {'xy': np.array([x,y]), 'score', 'scale'} -> JSON-able
list."""
102        xy = det["xy"]
103        return [float(xy[0]), float(xy[1]), float(det["score"]), int(det["scale"])]
104
105
106    def _dets_to_tracker(plain_list):
107        """Inverse of _det_plain: rebuild the dicts the tracker expects (xy MUST be a numpy
array,
108        the tracker does vector math / np.linalg.norm on it)."""
109        import numpy as np
110        return [{"xy": np.array([p[0], p[1]], dtype=float), "score": p[2], "scale": p[3]}
111                for p in plain_list]
112
113
114    def write_manifest(cache_dir: str, manifest: dict) -> None:
115        _, mpath = _cache_paths(cache_dir)
116        _atomic_write_text(mpath, json.dumps(manifest, indent=2))
117
118
119    def read_manifest(cache_dir: str):
120        _, mpath = _cache_paths(cache_dir)
121        if not osp.exists(mpath):
122            return None
123        try:
124            with open(mpath) as f:
125                return json.load(f)
126        except (json.JSONDecodeError, OSError):
127            return None
128
129
130    def manifest_compatible(manifest: dict, *, frames_dir: str, weights: str, sport: str,

```

```

131             frames_in: int, frames_total: int) -> bool:
132         """A cache is reusable only if the run that produced it matches this run's
inputs."""
133         if not manifest or manifest.get("version") != CACHE_VERSION:
134             return False
135         return (
136             manifest.get("frames_dir") == osp.abspath(frames_dir)
137             and manifest.get("weights") == weights
138             and manifest.get("sport") == sport
139             and manifest.get("frames_in") == frames_in
140             and manifest.get("frames_total") == frames_total
141         )
142
143
144     def reset_cache(cache_dir: str) -> None:
145         """Drop any existing cache so the next run starts clean."""
146         det_jsonl, mpath = _cache_paths(cache_dir)
147         for p in (det_jsonl, mpath):
148             if osp.exists(p):
149                 os.remove(p)
150
151
152     def load_and_clean_cache(cache_dir: str):
153         """Read the longest *contiguous* (w == line index) valid prefix of det_raw.jsonl,
154         rebuild the per-frame detection accumulation, and rewrite the file to exactly that
155         prefix so a trailing half-written line from a crash can't corrupt future appends.
156
157         Returns (windows_done, det_by_fid) where det_by_fid maps frame_id -> list of
158         [x, y, score, scale] candidates (accumulated across every window the frame is in,
159         in window order ? identical to a non-resumed run).
160         """
161         det_jsonl, _ = _cache_paths(cache_dir)
162         det_by_fid = defaultdict(list)
163         valid = []
164         if osp.exists(det_jsonl):
165             with open(det_jsonl, "r") as f:
166                 for line in f:
167                     s = line.strip()
168                     if not s:
169                         continue
170                     try:
171                         obj = json.loads(s)
172                     except json.JSONDecodeError:
173                         break # truncated trailing line (crash mid-
flush)
174                     if obj.get("w") != len(valid): # non-contiguous -> stop at the gap
175                         break
176                     valid.append(s)
177                     for fid, plain in obj["f"]:
178                         det_by_fid[int(fid)].extend([list(p) for p in plain])
179                     # Guarantee a clean append boundary by rewriting only the good prefix.
180                     _atomic_write_text(det_jsonl, ("\n".join(valid) + "\n") if valid else "")
181         return len(valid), det_by_fid
182
183
184     def _append_windows(fh, objs) -> None:
185         """Append window records as JSONL and durably flush (bounded loss on crash)."""
186         for o in objs:
187             fh.write(json.dumps(o, separators=(",", ":")) + "\n")
188         fh.flush()
189         os.fsync(fh.fileno())
190
191
192     def _atomic_write_trajectory(out_csv: str, rows) -> None:
193         os.makedirs(osp.dirname(osp.abspath(out_csv)), exist_ok=True)

```

```

194     tmp = out_csv + ".tmp"
195     with open(tmp, "w", newline="") as f:
196         w = csv.writer(f)
197         w.writerow(["Frame", "Visibility", "X", "Y"])
198         w.writerows(rows)
199         f.flush()
200         os.fsync(f.fileno())
201     os.replace(tmp, out_csv)
202
203
204     # ----- #
205     # Inference
206     # ----- #
207     def run(frames_dir: str, weights: str, out_csv: str, sport: str = "badminton",
208             limit: int = 0, batch_size: int = 8, num_workers: int = 4,
209             log_every_batches: int = 50, cache_dir: str = None,
210             flush_every: int = 1000, fresh: bool = False) -> str:
211         import time
212         import torch
213         from torch.utils.data import DataLoader
214         from detectors import build_detector
215         from trackers import build_tracker
216         from dataloaders import build_img_transforms, build_seq_transforms
217         from dataloaders.dataset_loader import ImageDataset
218         from utils import Center
219
220         cfg = _build_cfg(weights, sport)
221         frames_in = int(cfg["model"]["frames_in"])
222         input_wh = (int(cfg["model"]["inp_width"]), int(cfg["model"]["inp_height"]))
223         output_wh = (int(cfg["model"]["out_width"]), int(cfg["model"]["out_height"]))
224
225         frames = sorted(glob.glob(osp.join(frames_dir, "*.png")) +
226                        glob.glob(osp.join(frames_dir, "*.jpg")))
227         if limit:
228             frames = frames[:limit]
229         if len(frames) < frames_in:
230             raise SystemExit(f"need >= {frames_in} frames, found {len(frames)} in
231 {frames_dir}")
232
233         # Sliding windows of `frames_in` consecutive frames (the unit of GPU work +
234         # checkpoint).
235         samples = []
236         for i in range(len(frames) - frames_in + 1):
237             window = frames[i:i + frames_in]
238             annos = [{"frame_path": p, "center": Center(is_visible=False, x=-1.0, y=-1.0)}
239                     for p in window]
240             samples.append({"frames": window, "annos": annos})
241         windows_total = len(samples)
242
243         # ---- cache setup / resume decision ----- #
244         if cache_dir is None:
245             cache_dir = default_cache_dir(out_csv)
246             os.makedirs(cache_dir, exist_ok=True)
247             det_jsonl, _ = _cache_paths(cache_dir)
248             manifest = None if fresh else read_manifest(cache_dir)
249             resume = manifest_compatible(
250                 manifest or {}, frames_dir=frames_dir, weights=weights, sport=sport,
251                 frames_in=frames_in, frames_total=len(frames),
252             )
253         if resume:
254             windows_done, det_by_fid = load_and_clean_cache(cache_dir)
255             logger.info(f"resuming from cache: {windows_done}/{windows_total} windows
256 already detected")
257         else:
258             if manifest is not None:

```

```

254         logger.info("cache incompatible with this run -> starting fresh")
255         reset_cache(cache_dir)
256         windows_done, det_by_fid = 0, defaultdict(list)
257
258     base_manifest = {
259         "version": CACHE_VERSION,
260         "frames_dir": osp.abspath(frames_dir),
261         "weights": weights,
262         "sport": sport,
263         "frames_in": frames_in,
264         "frames_total": len(frames),
265         "windows_total": windows_total,
266         "windows_done": windows_done,
267     }
268     write_manifest(cache_dir, base_manifest)
269
270     # ---- detector pass over the un-cached windows ----- #
271     remaining = samples[windows_done:]
272     if remaining:
273         _, transform_test = build_img_transforms(cfg)
274         try:
275             _, seq_transform_test = build_seq_transforms(cfg)
276         except Exception:
277             seq_transform_test = None
278         ds = ImageDataset(cfg, remaining, input_wh, output_wh,
279                          transform=transform_test, seq_transform=seq_transform_test,
280                          is_train=False)
281         loader = DataLoader(ds, batch_size=batch_size, shuffle=False,
282                             num_workers=num_workers)
283         detector = build_detector(cfg)
284
285         t0 = time.time()
286         done = windows_done
287         win_idx = windows_done
288         log_every = max(1, log_every_batches)
289         flush_n = max(1, flush_every)
290         pending = []
291         logger.info(f"detecting on {len(remaining)} remaining windows "
292                    f"({total {windows_total}, batch={batch_size},
293                    workers={num_workers}})...")
294         det_f = open(det_jsonl, "a")
295         try:
296             with torch.no_grad():
297                 for bi, batch in enumerate(loader):
298                     imgs, _hms, trans = batch[0], batch[1], batch[2]
299                     img_paths = [list(t) for t in batch[-1]] # [frame_in][batch] ->
300                     path
301
302                     batch_results, _ = detector.run_tensor(imgs, trans)
303                     nb = imgs.shape[0]
304                     for ib in range(nb):
305                         frames_payload = []
306                         for ie in sorted(batch_results[ib].keys()):
307                             p = img_paths[ie][ib]
308                             fid = _frame_id(p)
309                             plain = [_det_plain(d) for d in batch_results[ib][ie]]
310                             frames_payload.append([fid, plain])
311                             det_by_fid[fid].extend(plain)
312                             pending.append({"w": win_idx, "f": frames_payload})
313                             win_idx += 1
314                         done += nb
315                     if len(pending) >= flush_n or done >= windows_total:
316                         _append_windows(det_f, pending)
317                         pending = []
318                         base_manifest["windows_done"] = done
319                         write_manifest(cache_dir, base_manifest)

```

```

315         if (bi + 1) % log_every == 0 or done >= windows_total:
316             el = time.time() - t0
317             new_done = done - windows_done
318             rate = new_done / el if el > 0 else 0.0
319             eta = (windows_total - done) / rate if rate > 0 else 0.0
320             logger.info(f"{done}/{windows_total} "
321                 f"({100 * done // max(1, windows_total)}%) |
{rate:.1f} win/s | "
322                 f"elapsed {el:.0f}s | ETA {eta:.0f}s")
323         if pending:
324             _append_windows(det_f, pending)
325             base_manifest["windows_done"] = done
326             write_manifest(cache_dir, base_manifest)
327         finally:
328             det_f.close()
329             logger.info(f"detection done in {time.time() - t0:.0f}s; running tracker...")
330     else:
331         logger.info("detector cache already complete -> tracker-only re-run")
332
333     # ---- tracker re-run over the full ordered cache (cheap, deterministic) ---- #
334     tracker = build_tracker(cfg)
335     tracker.refresh()
336     rows = []
337     for fid in sorted(det_by_fid.keys()):
338         r = tracker.update(_dets_to_tracker(det_by_fid[fid]))
339         rows.append((fid, 1 if r.get("visi") else 0, r.get("x", -1), r.get("y", -1)))
340
341     _atomic_write_trajectory(out_csv, rows)
342     visible = sum(1 for _, v, *_ in rows if v)
343     logger.info(f"wrote {len(rows)} rows -> {out_csv}")
344     logger.info(f"visible-shuttle frames: {visible}/{len(rows)}")
345     return out_csv
346
347
348 def extract_frames(video_path: str, frames_dir: str) -> str:
349     """Decode a video to PNG frames named by source frame index (00000000.png ...).
350
351     Idempotent: if the dir already holds the expected number of frames (matching the
352     container's reported frame count) we skip re-decoding ? re-extraction after a
353     reboot is wasteful. NOTE: cv2 PNG extraction is slow; for big clips prefer
354     extracting with ffmpeg on the host and passing --frames_dir.
355     """
356     import cv2
357     os.makedirs(frames_dir, exist_ok=True)
358     cap = cv2.VideoCapture(video_path)
359     if not cap.isOpened():
360         raise SystemExit(f"cannot open video: {video_path}")
361     expected = int(cap.get(cv2.CAP_PROP_FRAME_COUNT) or 0)
362     existing = len(glob.glob(osp.join(frames_dir, "*.png")))
363     if expected > 0 and existing >= expected:
364         cap.release()
365         logger.info(f"reusing {existing} already-extracted frames -> {frames_dir}")
366         return frames_dir
367     i = 0
368     while True:
369         ok, frame = cap.read()
370         if not ok:
371             break
372         cv2.imwrite(osp.join(frames_dir, f"{i:08d}.png"), frame)
373         i += 1
374     cap.release()
375     logger.info(f"extracted {i} frames -> {frames_dir}")
376     return frames_dir
377
378

```

```

379     def main():
380         ap = argparse.ArgumentParser()
381         ap.add_argument("--frames_dir", help="folder of pre-extracted frames")
382         ap.add_argument("--video", help="video file; frames are extracted internally (in
WSL)")
383         ap.add_argument("--frames_out_dir", help="where to extract frames when --video is
used")
384         ap.add_argument("--weights", required=True)
385         ap.add_argument("--out", required=True)
386         ap.add_argument("--sport", default="badminton")
387         ap.add_argument("--limit", type=int, default=0, help="cap #frames (for quick
tests)")
388         ap.add_argument("--batch-size", type=int, default=8)
389         ap.add_argument("--num-workers", type=int, default=4)
390         ap.add_argument("--log-every-batches", type=int, default=50, help="progress log
cadence")
391         ap.add_argument("--cache-dir", default=None,
392                         help="resumable detector cache dir (default: <out>_wasbcache next to
--out)")
393         ap.add_argument("--flush-every", type=int, default=1000,
394                         help="fsync the detector cache every N windows (bounds reboot
loss)")
395         ap.add_argument("--fresh", action="store_true", help="ignore any existing cache and
start over")
396         args = ap.parse_args()
397
398         frames_dir = args.frames_dir
399         if args.video:
400             frames_dir = extract_frames(args.video, args.frames_out_dir or (args.video +
"_frames"))
401         if not frames_dir:
402             raise SystemExit("provide --frames_dir or --video")
403         run(frames_dir, args.weights, args.out, args.sport, args.limit,
404             batch_size=args.batch_size, num_workers=args.num_workers,
405             log_every_batches=args.log_every_batches, cache_dir=args.cache_dir,
406             flush_every=args.flush_every, fresh=args.fresh)
407
408
409     if __name__ == "__main__":
410         main()
411

```

26. user (2026-06-10T05:36:35.684Z)

```

1     """Per-video observability reports for the indexer.
2
3     `obsreport` is a SOFT dependency: if it is not installed, this module degrades to a
4     no-op and the pipeline runs exactly as before. The single entrypoint
5     `emit_indexer_report(...)` is designed to be called from a `finally` block and
6     NEVER raises ? emitting a report must not be able to break a processing run.
7     """
8     from __future__ import annotations
9
10    import os
11    from typing import Any, Dict, List, Optional
12
13    try: # soft dependency ? never hard-fail the pipeline on a reporting import
14        import obsreport # noqa: F401
15        from . import harvest
16
17        OBSREPORT_AVAILABLE = True
18    except Exception: # pragma: no cover - exercised only when obsreport is absent
19        OBSREPORT_AVAILABLE = False
20
21

```

```

22     def report_enabled(config: Optional[Dict[str, Any]]) -> bool:
23         return bool((config or {}).get("report", {}).get("enabled", True))
24
25
26     def emit_indexer_report(
27         *,
28         db,
29         video_id: str,
30         video_path: Optional[str],
31         config: Optional[Dict[str, Any]],
32         val_results: List[Any],
33         val_context: Optional[Dict[str, Any]] = None,
34         segmenter_requested: Optional[str] = None,
35         segmenter_actual: Optional[str] = None,
36         was_reingest: bool = False,
37         segmentation_ran: bool = True,
38     ) -> Optional[Dict[str, str]]:
39         """Harvest + write the report. Returns the written file paths, or None if
40         reporting is unavailable/disabled/failed. Never raises."""
41         config = config or {}
42         # Accept either a plain dict or a typed config object (e.g.
43         backend.config.AppConfig).
44         # Normalize to a plain nested dict so harvest's dict access works uniformly.
45         if hasattr(config, "model_dump"):
46             try:
47                 config = config.model_dump(mode="json")
48             except Exception:
49                 config = {}
50         if not OBSREPORT_AVAILABLE or not report_enabled(config):
51             return None
52         try:
53             play_segments = db.query_play_segments(video_id, {}) if db is not None else []
54             candidates = db.query_candidate_segments(video_id) if db is not None else []
55             rec = harvest.record_run(
56                 video_id=video_id,
57                 video_path=video_path,
58                 config=config,
59                 val_results=val_results,
60                 val_context=val_context,
61                 play_segments=play_segments,
62                 candidates=candidates,
63                 segmenter_requested=segmenter_requested,
64                 segmenter_actual=segmenter_actual,
65                 was_reingest=was_reingest,
66                 segmentation_ran=segmentation_ran,
67             )
68             output_dir = (config.get("report") or {}).get("output_dir")
69             res = rec.write(output_dir=output_dir)
70             return res.paths or None
71         except Exception: # pragma: no cover - defensive; reporting must never break a run
72             return None

```

27. user (2026-06-10T05:36:36.098Z)

```

1     """Build an observability report for one indexer run.
2
3     Pure-ish harvest: given the validation results/context, the persisted segments,
4     and run metadata, populate an `obsreport.RunRecorder` with the raw signals. The
5     *footgun rules* (which turn signals into flags) live declaratively in spec.yaml,
6     so this module only records facts faithfully ? it does not decide warnings.
7
8     Imported only after `backend.reporting` confirms obsreport is installed, so it may
9     import obsreport freely.
10    """

```

```

11     from __future__ import annotations
12
13     import os
14     from pathlib import Path
15     from typing import Any, Dict, List, Optional
16
17     try:
18         import obsreport # soft dep; only needed for RunRecorder / load_spec at runtime
19     except ImportError:
20         obsreport = None # type: ignore[assignment]
21
22     _AI_SEGMENTERS = {"yolo_hybrid", "tracknet_hybrid", "wasb_hybrid", "gemini"}
23     _SPEC_PATH = Path(__file__).resolve().parent / "spec.yaml"
24
25
26     def load_spec() -> "obsreport.ReportSpec":
27         return obsreport.load_spec(_SPEC_PATH)
28
29
30     # --- media metadata -----
31     def video_meta_from_context(val_context: Optional[Dict[str, Any]], video_path:
Optional[str]) -> Dict[str, Any]:
32         """Derive VideoMeta fields from the validators' shared ffprobe_meta. fps_source
33         distinguishes a real probe from the absence of one (the fallback footgun)."""
34         meta: Dict[str, Any] = {"fps_source": "unknown"}
35         if video_path and os.path.exists(video_path):
36             try:
37                 meta["size_mb"] = round(os.path.getsize(video_path) / (1024 * 1024), 1)
38             except OSError:
39                 pass
40
41         ffmata = (val_context or {}).get("ffprobe_meta") or {}
42         streams = ffmata.get("streams") or []
43         fmt = ffmata.get("format") or {}
44         if streams:
45             s = streams[0]
46             w, h = s.get("width"), s.get("height")
47             meta["width"], meta["height"] = w, h
48             if w and h:
49                 meta["resolution"] = f"{w}x{h}"
50             fps_str = s.get("r_frame_rate", "0/1")
51             try:
52                 num, den = (int(x) for x in fps_str.split("/"))
53                 if den:
54                     meta["fps"] = round(num / den, 2)
55                     meta["fps_source"] = "probed"
56             except (ValueError, ZeroDivisionError):
57                 pass
58             if s.get("codec_name"):
59                 meta["container"] = fmt.get("format_name") or s.get("codec_name")
60         if fmt.get("duration"):
61             try:
62                 meta["duration_s"] = round(float(fmt["duration"]), 1)
63             except (ValueError, TypeError):
64                 pass
65         if fmt.get("format_name"):
66             meta["container"] = fmt.get("format_name")
67
68         # Audio readiness signal (for Input Quality UX extension per review item 2).
69         # Guidelines make "record with audio on, mic unobstructed" a first-class requirement
70         # (ADR-007) because shuttle thwacks are a strong rally cue for recall.
71         # We only record presence here (cheap, from ffprobe already in context); the actual
72         # thwack onset probe (pipeline/audio/hit_detector) remains a diagnostic /
73         calibration
74         # tool for now.

```

```

74     audio_streams = [s for s in streams if (s or {}).get("codec_type") == "audio"]
75     meta["audio_present"] = len(audio_streams) > 0
76     return meta
77
78
79 # --- stage builders -----
80 def _validation_stage(rec: "obsreport.RunRecorder", val_results: List[Any]) -> None:
81     with rec.stage("validation") as st:
82         total = len(val_results)
83         passed = sum(1 for r in val_results if getattr(r, "passed", False))
84         st.add_metric("checks_passed", passed)
85         st.add_metric("checks_total", total)
86         failures = [r for r in val_results if not getattr(r, "passed", True)]
87         for r in failures:
88             st.messages.append(f"FAILED {getattr(r, 'validator_name', '?')}: {getattr(r,
89 'message', '')}")
89         st.status = "error" if failures else ("ok" if total else "not_run")
90
91
92 def _segmentation_stage(
93     rec: "obsreport.RunRecorder",
94     play_segments: List[Dict[str, Any]],
95     candidates: List[Dict[str, Any]],
96     segmenter_requested: Optional[str],
97     segmenter_actual: Optional[str],
98     config_default: Optional[str],
99     was_reingest: bool,
100     ran: bool,
101 ) -> None:
102     with rec.stage("segmentation") as st:
103         if not ran:
104             st.status = "not_run"
105             seg_count = len(play_segments)
106             total_play = round(sum(float(s.get("duration", 0.0)) for s in play_segments), 1)
107             detector_tag = None
108             if play_segments:
109                 detector_tag = (play_segments[0].get("metadata") or {}).get("detector")
110             st.add_metric("candidate_windows", len(candidates))
111             st.add_metric("segment_count", seg_count, audience="both")
112             st.add_metric("total_play_s", total_play, unit="s")
113             # "diverges" only when a segmenter actually RAN, the user did NOT explicitly
114             # pick it, and it differs from config's default (the genuine surprise case).
115             # An explicit --segmenter override or a halted run is not a divergence.
116             matches_default = True
117             if ran and segmenter_actual and config_default:
118                 matches_default = (segmenter_actual == config_default)
119             st.details.update({
120                 "segmenter_requested": segmenter_requested,
121                 "segmenter_actual": segmenter_actual,
122                 "config_default": config_default,
123                 "segmenter_explicitly_requested": segmenter_requested is not None,
124                 "matches_config_default": matches_default,
125                 "was_reingest": was_reingest,
126                 "detector": detector_tag,
127             })
128             # Surface metadata trustworthiness at the data level (not just in a warning):
129             # the motion baseline fabricates per-segment metadata; everything else measures
130             it.
131             if ran and segmenter_actual:
132                 st.details["metadata_source"] = (
133                     "synthetic/heuristic ? NOT measured" if segmenter_actual == "motion"
134                     else "measured (detector/AI)"
135                 )
136

```

```

137     def _ai_stage(rec: "obsreport.RunRecorder", config: Dict[str, Any], segmenter_actual:
Optional[str],
138                 play_segments: List[Dict[str, Any]], ran: bool) -> None:
139         """Only meaningful for AI-based segmenters. Records whether the provider key was
140         present ? the signal the silent_provider_noop rule keys on."""
141         ai_based = segmenter_actual in _AI_SEGMENTERS
142         provider = (config.get("ai_provider") or "gemini")
143         key_env = f"{provider.upper()}_API_KEY"
144         with rec.stage("ai_handoff") as st:
145             if not ran or not ai_based:
146                 st.status = "not_run" if not ran else "skipped"
147             st.details.update({
148                 "ai_based": ai_based,
149                 "provider": provider if ai_based else None,
150                 "model": config.get("ai_model") if ai_based else None,
151                 "api_key_present": bool(os.environ.get(key_env)) if ai_based else None,
152             })
153             if ai_based:
154                 st.add_metric("confirmed_segments", len(play_segments))
155
156
157     def _highlights(rec: "obsreport.RunRecorder", play_segments: List[Dict[str, Any]],
158                   video_meta: Dict[str, Any]) -> None:
159         total = round(sum(float(s.get("duration", 0.0)) for s in play_segments), 1)
160         coverage = None
161         dur = video_meta.get("duration_s")
162         if dur:
163             coverage = round(100.0 * total / dur, 1)
164         notes = []
165         if not play_segments:
166             notes.append("No highlights were produced for this video.")
167         rec.set_highlights(segment_count=len(play_segments), total_highlight_s=total,
168                           coverage_pct=coverage, notes=notes)
169
170
171     # --- top-level -----
172     def record_run(
173         *,
174         video_id: str,
175         video_path: Optional[str],
176         config: Dict[str, Any],
177         val_results: List[Any],
178         val_context: Optional[Dict[str, Any]],
179         play_segments: List[Dict[str, Any]],
180         candidates: List[Dict[str, Any]],
181         segmenter_requested: Optional[str] = None,
182         segmenter_actual: Optional[str] = None,
183         was_reingest: bool = False,
184         segmentation_ran: bool = True,
185         tool_version: Optional[str] = None,
186         spec: Optional["obsreport.ReportSpec"] = None,
187         **recorder_kwargs: Any,
188     ) -> "obsreport.RunRecorder":
189         """Assemble a finalized recorder for an indexer run (does NOT write)."""
190         rec = obsreport.RunRecorder(
191             "badminton-highlight-indexer",
192             tool_version=tool_version,
193             video_id=video_id,
194             video_id_kind="file_md5",
195             video_path=video_path,
196             spec=spec if spec is not None else load_spec(),
197             **recorder_kwargs,
198         )
199         vm = video_meta_from_context(val_context, video_path)
200         # If we never probed an fps but segmentation still ran, the segmenter must have

```

```

201      # used its internal fallback (30fps / 1280px) ? surface that as the footgun it is,
202      # AND show the assumed numeric value so the report reads "30 (fallback)" not "?"
(fallback)".
203      if vm.get("fps_source") == "unknown" and segmentation_ran:
204          vm["fps_source"] = "fallback"
205          if vm.get("fps") is None:
206              vm["fps"] = 30.0 # matches the runners' fallback (_probe_fps_width -> 30.0
/ 1280)
207      rec.set_video_meta(**vm)
208
209      _validation_stage(rec, val_results)
210      config_default = (config.get("indexing") or {}).get("default_segmenter")
211      _segmentation_stage(rec, play_segments, candidates, segmenter_requested,
212                          segmenter_actual, config_default, was_reingest,
segmentation_ran)
213      _ai_stage(rec, config, segmenter_actual, play_segments, segmentation_ran)
214      _highlights(rec, play_segments, vm)
215
216      rec.finalize()
217      return rec
218

```

28. user (2026-06-10T05:36:49.941Z)

```

1      # Observability report spec for the badminton-highlight-indexer.
2      # The report ENGINE (config language, renderers) lives in the shared `obsreport`
3      # library; this file is the indexer-specific layout + footgun rules. Every
4      # actionable rule cites a README/FAQ section so a reader can self-debug.
5      name: indexer
6      title: Badminton Highlight Indexer ? Video Report
7
8      # Customer here = whoever receives the highlight reel.
9      customer_verdict:
10         pass: "? Your highlights are ready."
11         pass_with_warnings: "? Your highlights are ready. (A couple of minor notes above.)"
12         fail: "?? We couldn't finish generating highlights for this video. Please contact
support."
13
14         sections:
15             - { id: run,          title: "Run",          kind: identity,  audience: both }
16             - { id: video,       title: "Video",       kind: video_meta, audience: both }
17             - { id: highlights,  title: "Highlights",    kind: highlights, audience: customer
}
18             - { id: cust_notes, title: "Things to know", kind: flags,      audience: customer
}
19             - { id: stages,     title: "Processing stages", kind: stages,    audience:
technical }
20             - { id: findings,   title: "Findings & warnings", kind: flags,      audience:
technical }
21             - { id: verdict,    title: "Verdict",        kind: verdict,   audience: both }
22
23         rules:
24             # The classic silent failure: 0 segments because the AI provider no-op'd on a
25             # missing API key ? NOT because the video has no rallies.
26             - code: silent_provider_noop
27               severity: error
28               when:
29                 - { path: "stages.ai_handoff.details.api_key_present", op: eq, value: false }
30                 - { path: "stages.segmentation.metrics.segment_count", op: eq, value: 0 }
31             message: >-
32                 0 segments AND no AI provider API key was set. The segmenter (see Processing
33                 stages > segmentation > segmenter_actual) hands its candidate windows to the AI
34                 provider, which silently did nothing without a key ? so this is a MISSING
35                 CREDENTIAL, not 'the video has no rallies'. Fix: put GEMINI_API_KEY=<key> in a
36                 .env file in the project root and re-run (see README#Q7).

```

```

37     faq_ref: "README#Q7"
38     customer_message: >-
39         We couldn't generate highlights for this video due to a configuration issue
40         on our side. Please contact support.
41
42 # API key WAS present but an AI segmenter still produced nothing ? likely a
43 # provider error / quota / WSL healthcheck issue rather than a truly empty video.
44 - code: ai_empty_vs_no_rallies
45     severity: warn
46     when:
47         - { path: "stages.ai_handoff.details.ai_based", op: eq, value: true }
48         - { path: "stages.ai_handoff.details.api_key_present", op: eq, value: true }
49         - { path: "stages.segmentation.metrics.segment_count", op: eq, value: 0 }
50     message: >-
51         An AI-based segmenter returned 0 segments even though the API key is set.
52         Empty can mean a provider error, exhausted quota, or (for WASB/TrackNet) a
53         failed WSL healthcheck ? these all return [] silently. Re-run with -u for
54         unbuffered logs and check the provider/WSL output before concluding 'no rallies'.
55     faq_ref: "README#Q7"
56
57 # fps was assumed rather than probed -> velocity thresholds + timing may be wrong.
58 - code: fps_fallback_used
59     severity: warn
60     when:
61         - { path: "video_meta.fps_source", op: eq, value: fallback }
62     message: >-
63         fps was NOT probed from the file ? a fallback default (shown as the fps value)
64         was used. Because rally timing and motion thresholds scale with fps, boundaries
65         and motion sensitivity may be miscalibrated. This usually means validation didn't
66         run (it probes the file) or ffprobe couldn't read r_frame_rate. Confirm the real
67         fps with ffprobe and re-run.
68     faq_ref: "README#Q14"
69
70 # Audio stream missing or disabled. Per VIDEO_RECORDING_GUIDELINES (and ADR-007)
71 # the shuttle "thwack" is a first-class rally cue for recall; recording with audio
72 # on (mic unobstructed, no aggressive noise suppression) is now a hard requirement
73 # for good results with the shuttle-tracking substrates.
74 - code: audio_disabled_or_missing
75     severity: warn
76     when:
77         - { path: "video_meta.audio_present", op: eq, value: false }
78     message: >-
79         No audio stream was detected in the input (or it could not be read). The shuttle
80         "thwack" impact sound is a strong cue that improves rally recall (especially for
81         the WASB/TrackNet detector paths). Record with audio enabled, keep the camera
82         mic unobstructed, and avoid heavy wind/noise-reduction filters that flatten the
83         transient. See VIDEO_RECORDING_GUIDELINES.md (the "Audio (REQUIRED)" row) and
84         AUDIO_RALLY_DETECTION_PLAN.md.
85     faq_ref: "VIDEO_RECORDING_GUIDELINES"
86     customer_message: >-
87         For best highlight results, record with audio on and the mic unobstructed.
88         The shuttle impact sound helps the detector find rallies. Re-record or
89         re-process after enabling audio (see VIDEO_RECORDING_GUIDELINES.md).
90
91 # The motion baseline fabricates its per-segment metadata/events ? they are not
92 # measured. Warn so nobody trusts them as ground truth.
93 - code: synthetic_motion_metadata
94     severity: warn
95     when:
96         - { path: "stages.segmentation.details.segmenter_actual", op: eq, value: motion }
97     message: >-
98         The 'motion' segmenter was used: its per-segment metadata (intensity, speed,
99         events, server/receiver) are PLACEHOLDER values (unmeasured) ? fabricated by the
100        heuristic, NOT derived from the video ? so a PASS with segments here does not mean
101        the metadata is trustworthy. Use yolo_hybrid/gemini for real shot metadata.

```

```

102         faq_ref: "README#Q10"
103
104     # The segmenter that ran differs from config's default -> you may be looking at a
105     # different detector than expected (config/setup/CLI defaults are known to diverge).
106     - code: segmenter_divergence
107       severity: info
108       when:
109         - { path: "stages.segmentation.details.matches_config_default", op: eq, value:
false }
110         - { path: "stages.segmentation.details.segmenter_explicitly_requested", op: eq,
value: false }
111       message: >-
112         The segmenter that ran differs from config.json's default_segmenter, and no
113         explicit --segmenter was given ? so a fallback default kicked in. The indexer's
114         defaults are known to diverge across config.json / setup.py / the CLI; confirm
115         you ran the detector you meant.
116       faq_ref: "README#Q7"
117
118     # Re-processing an already-indexed video replaced its prior segments (INSERT OR
119     # REPLACE cascades). Note it so a changed result isn't mistaken for instability.
120     - code: reingest_replaced_prior
121       severity: info
122       when:
123         - { path: "stages.segmentation.details.was_reingest", op: eq, value: true }
124       message: >-
125         This video was already in the database; re-processing REPLACED its previous
126         segments (videos are keyed by file MD5; re-ingest cascades a delete). Expected
127         when you intend to re-run; just know the earlier results are gone.
128       faq_ref: "README#Q11"
129
130     # A validation check failed -> segmentation was halted. Point the reader at the
131     # report's Validation section + the per-check FAQ.
132     - code: validation_failed
133       severity: error
134       when:
135         - { path: "stages.validation.status", op: eq, value: error }
136       message: >-
137         One or more validation checks failed, so indexing was halted. See the
138         Processing stages > validation block above: each failed check states the
139         reason (relevance -> README#Q2, fps/keyframes -> README#Q5).
140       faq_ref: "README#Q13"
141       customer_message: >-
142         This video couldn't be processed because it didn't pass our quality checks
143         (for example: too blurry, too low resolution, or not a badminton court).
144

```

29. user (2026-06-10T05:36:50.630Z)

No files found

30. user (2026-06-10T05:36:59.335Z)

No files found

31. user (2026-06-10T05:37:01.754Z)

FullName

```

C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\__init__.py
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\loader.py
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\utils\__init__.py
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\utils\geometry.py
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\utils\hashing.py

```

```

C:\Users\avidu\Projects\badminton-highlight-indexer\backend\utils\video.py
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\validators\__init__.py
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\validators\base.py
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\validators\blur.py
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\validators\format.py
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\validators\pts_continuity.py
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\validators\relevance.py
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\validators\resolution_fps.py
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\validators\scene_cut.py
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\providers\__init__.py
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\providers\base.py
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\providers\gemini.py
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\sports\__init__.py
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\sports\badminton.py
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\sports\base.py

```

32. user (2026-06-10T05:37:06.562Z)

```

1      """Pydantic models for the application configuration.
2
3      This module defines the canonical, validated, typed configuration for the
4      Badminton Highlight Indexer. Built-in defaults live here (single source of truth).
5
6      The models support graceful migration from old config.json files via extra="allow".
7      """
8
9      from __future__ import annotations
10
11     from typing import Any, Literal
12
13     from pydantic import BaseModel, Field, field_validator
14
15     Sport = Literal["badminton", "tennis", "pickleball", "table_tennis", "cricket"]
16
17
18     class ValidationRelevanceConfig(BaseModel):
19         court_green_lower: list[int] = Field(default=[35, 40, 40])
20         court_green_upper: list[int] = Field(default=[85, 255, 255])
21         court_pixels_min_ratio: float = Field(default=0.05, ge=0.0, le=1.0)
22
23
24     class ValidationConfig(BaseModel):
25         min_resolution_width: int = 1280
26         min_resolution_height: int = 720
27         min_fps: float = 29.9
28         min_blur_threshold: float = 20.0
29         max_scene_cuts_per_minute: float = 0.5
30         relevance: ValidationRelevanceConfig =
31         Field(default_factory=ValidationRelevanceConfig)
32
33     class TrackNetConfig(BaseModel):
34         wsl_distro: str = "Ubuntu"
35         repo_dir: str = "~/models/TrackNetV4"
36         conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
37         conda_env: str = "TrackNetV4"
38         weights_path: str = ""
39         queue_length: int = 5
40         stage_in_wsl: bool = True
41         wsl_stage_dir: str = "~/clips"
42         timeout_sec: int = 0
43         velocity_threshold: float = 0.02
44

```

```

45
46 class IndexingConfig(BaseModel):
47     default_segmenter: str = "yolo_hybrid"
48     use_gpu: bool = True
49     motion_threshold: float = 0.08
50     min_segment_duration: float = 3.0
51     dead_time_merge_gap: float = 2.0
52     chunk_duration: float = 90.0
53     chunk_overlap: float = 10.0
54     detector_model: str = "fasterrcnn_resnet50_fpn"
55     detector_score_threshold: float = 0.5
56     yolo_velocity_threshold: float = 0.15
57     yolo_frame_skip: int = 3
58     yolo_tracker: str = "bytetrack.yaml"
59     yolo_parallel_chunks: bool = False
60     yolo_chunk_workers: int = 2
61     min_action_window_duration: float = 2.0
62     log_yolo_candidates: bool = True
63     store_yolo_candidates: bool = True
64     ai_handoff_padding: float = 2.0
65     ai_max_workers: int = 5
66
67     tracknet: TrackNetConfig = Field(default_factory=TrackNetConfig)
68
69     @field_validator("default_segmenter")
70     @classmethod
71     def segmenter_must_be_registered(cls, v: str) -> str:
72         # Registry check happens at runtime in main/segmenter selection.
73         # Here we just allow any string for forward compatibility.
74         return v
75
76
77 class OutputConfig(BaseModel):
78     default_highlights_name: str = "highlights.mp4"
79     db_name: str = "sports_indexer.db"
80     # Directory where the DB, highlight reels, and processed clips are written.
81     # The CLI's --output-dir overrides this at startup (see backend/main.py::main).
82     output_dir: str = "output"
83
84
85 class StitchingConfig(BaseModel):
86     begin_padding: float = 0.5
87     end_padding: float = 0.5
88     use_cache: bool = False
89     min_shot_count: int = 1
90
91
92 class AppConfig(BaseModel):
93     """Root configuration object.
94
95     All runtime configuration should be accessed via an instance of this class
96     (or the loader) rather than raw dict lookups.
97     """
98
99     sport: Sport = "badminton"
100     ai_provider: str = "gemini"
101     ai_model: str = "gemini-2.5-flash"
102
103     validation: ValidationConfig = Field(default_factory=ValidationConfig)
104     indexing: IndexingConfig = Field(default_factory=IndexingConfig)
105     output: OutputConfig = Field(default_factory=OutputConfig)
106     stitching: StitchingConfig = Field(default_factory=StitchingConfig)
107
108     # Allow extra keys during transition so existing config.json files do not break.
109     model_config = {

```

```

110         "extra": "allow",
111         "validate_assignment": True,
112     }
113
114     # Temporary migration shim for legacy dict-style access (config.get("section",
115     {}).get(...)).
116     # Will be removed once all call sites (validators, segmenters, etc.) move to typed
117     # attribute access / the typed config. A nested *section* is returned as a plain
dict
118     # via model_dump, so legacy chained access keeps working ? returning the Pydantic
119     # section model instead breaks it (a model has no .get), which bit every validator
on
120     # real runs. Scalar leaves are returned as-is.
121     def get(self, key: str, default: Any = None) -> Any:
122         _MISSING = object()
123         val: Any = _MISSING
124         if key in type(self).model_fields or hasattr(self, key):
125             val = getattr(self, key, _MISSING)
126         if val is _MISSING:
127             # Legacy convenience: find a leaf key inside any nested section. Sections
are
128             # discovered dynamically from the model fields, so a future top-level
129             # (e.g. `audio`, `jobs`) is covered automatically without editing a
section
130             # hardcoded list.
131             for fname in type(self).model_fields:
132                 section = getattr(self, fname, None)
133                 if isinstance(section, BaseModel) and hasattr(section, key):
134                     val = getattr(section, key)
135                     break
136             if val is _MISSING:
137                 return default
138             if isinstance(val, BaseModel):
139                 return val.model_dump(mode="json")
140             return val
141
142     def __getitem__(self, key: str) -> Any:
143         val = self.get(key)
144         if val is None and key not in type(self).model_fields:
145             # Re-raise like a real dict for unknown top-level
146             raise KeyError(key)
147         return val
148
149     # Convenience type alias for callers that want to annotate
Config = AppConfig

```

33. user (2026-06-10T05:37:07.045Z)

```

1     """Configuration loader and saver.
2
3     Provides a single place to load validated AppConfig with built-in defaults
4     merged with optional file overrides (config.json).
5
6     This replaces scattered json.load + .get() patterns throughout the codebase.
7     """
8
9     from __future__ import annotations
10
11     import json
12     import logging
13     from pathlib import Path
14     from typing import Any
15
16     from .models import AppConfig

```

```

17
18     logger = logging.getLogger(__name__)
19
20     DEFAULT_CONFIG_PATH = Path("config.json")
21
22
23     def _get_builtin_defaults() -> dict[str, Any]:
24         """Return a plain dict of the current built-in defaults.
25
26         The AppConfig model is the single source of truth for defaults.
27         """
28         return AppConfig().model_dump(mode="json")
29
30
31     def load_config(path: str | Path | None = None) -> AppConfig:
32         """Load configuration with the following precedence:
33
34         1. Built-in defaults defined in the Pydantic models.
35         2. Values from the JSON file (if present and readable).
36         3. (Future) environment / CLI overrides.
37
38         Missing file or unreadable file ? returns a fully defaulted AppConfig.
39         Unknown keys in the file are tolerated (extra="allow" on the model).
40         """
41         cfg_path = Path(path) if path is not None else DEFAULT_CONFIG_PATH
42
43         file_data: dict[str, Any] = {}
44         if cfg_path.exists():
45             try:
46                 with open(cfg_path, "r", encoding="utf-8") as f:
47                     file_data = json.load(f)
48             except Exception as exc:
49                 # Non-fatal: continue with defaults + whatever we could parse.
50                 # In production we might log here.
51                 logger.warning(f"Failed to read {cfg_path}: {exc}. Using defaults + partial
data.")
52
53         # Merge: start from defaults, overlay file data (Pydantic will validate/coerce).
54         merged = {**_get_builtin_defaults(), **file_data}
55
56         return AppConfig.model_validate(merged)
57
58
59     def save_default_config(path: str | Path | None = None) -> Path:
60         """Write a fresh default config.json.
61
62         Used by setup.py diagnostics and the --setup CLI path.
63         Overwrites if the file exists.
64         """
65         cfg_path = Path(path) if path is not None else DEFAULT_CONFIG_PATH
66         cfg = AppConfig()
67         cfg_path.parent.mkdir(parents=True, exist_ok=True)
68         cfg_path.write_text(cfg.model_dump_json(indent=4), encoding="utf-8")
69         return cfg_path
70
71
72     # Backwards-compatible alias used by some older call sites during transition.
73     def get_default_config() -> dict[str, Any]:
74         """Return defaults as a plain dict (for transitional use only)."""
75         return _get_builtin_defaults()
76

```

34. user (2026-06-10T05:37:07.556Z)

```

1     import subprocess

```

```

2     import os
3     import shutil
4     import logging
5
6     logger = logging.getLogger(__name__)
7
8     def get_video_duration(video_path: str) -> float:
9         """Returns the exact duration of a video file in seconds using ffprobe.
10        Returns 0.0 on failure."""
11        try:
12            duration_output = subprocess.check_output([
13                "ffprobe", "-v", "error", "-show_entries", "format=duration",
14                "-of", "default=noprint_wrappers=1:nokey=1", video_path
15            ]).decode().strip()
16            return float(duration_output)
17        except Exception as e:
18            logger.error(f"Could not detect video duration for {video_path}: {e}")
19            return 0.0
20
21    def extract_video_chunk(video_path: str, start_time: float, duration: float,
22                           output_path: str, reencode: bool = False,
23                           scale_width: int = None) -> bool:
24        """Extracts a chunk from a video file.
25
26        Args:
27            reencode: If True, re-encodes with libx264 (slower but frame-accurate).
28                      If False, uses stream-copy (fast but cuts at nearest keyframe).
29                      Use True for short clips where keyframe drift matters.
30            scale_width: If set (requires reencode), downscale to this width (height auto,
31                        even). Useful for high-res sources headed to an API that analyses at
32                        low res anyway ? shrinks uploads with no analysis loss.
33
34        """
35        cmd = [
36            "ffmpeg", "-y",
37            "-ss", str(start_time),
38            "-t", str(duration),
39            "-i", video_path,
40        ]
41        if reencode:
42            if scale_width:
43                cmd += ["-vf", f"scale={int(scale_width)}:-2"]
44            cmd += ["-c:v", "libx264", "-preset", "ultrafast", "-crf", "23", "-c:a", "aac"]
45        else:
46            cmd += ["-c", "copy"]
47        cmd.append(output_path)
48        try:
49            subprocess.run(cmd, check=True, stdout=subprocess.DEVNULL,
50                           stderr=subprocess.DEVNULL)
51            return True
52        except Exception as e:
53            logger.error(f"Failed to extract chunk {start_time}s - {start_time+duration}s: {e}")
54            return False

```

35. user (2026-06-10T05:37:27.457Z)

```

1     from typing import Dict, Type
2     from backend.providers.base import AIVideoProvider
3     from backend.providers.gemini import GeminiProvider
4
5     class ProviderRegistry:
6         def __init__(self):
7             self._registry: Dict[str, Type[AIVideoProvider]] = {}
8             # Register Gemini provider

```

```

9         self.register("gemini", GeminiProvider)
10
11     def register(self, name: str, provider_cls: Type[AIVideoProvider]):
12         self._registry[name] = provider_cls
13
14     def get(self, name: str, api_key: str, model_name: str) -> AIVideoProvider:
15         if name not in self._registry:
16             raise KeyError(f"AI Provider '{name}' is not registered in the
ProviderRegistry. Available: {list(self._registry.keys())}")
17         return self._registry[name](api_key=api_key, model_name=model_name)
18
19     def list_available(self) -> Dict[str, str]:
20         return {name: cls.__doc__ or f"{name.capitalize()} provider" for name, cls in
self._registry.items()}
21
22     provider_registry = ProviderRegistry()
23

```

36. user (2026-06-10T05:37:27.978Z)

```

1     from abc import ABC, abstractmethod
2     from typing import Tuple
3
4     class AIVideoProvider(ABC):
5         def __init__(self, api_key: str, model_name: str):
6             self.api_key = api_key
7             self.model_name = model_name
8
9         @abstractmethod
10        def analyze_video(self, video_path: str, prompt: str,
11                        chunk_duration: float = 0.0,
12                        label: str = "") -> Tuple[list, dict]:
13            """Sends a video file to the AI model along with a prompt, and returns a tuple
of
14            (segments_list, metrics_dict).
15
16            Args:
17                video_path: Path to the video file to analyze.
18                prompt: Prompt instructions.
19                chunk_duration: Duration of this chunk in seconds, or 0.0 if not chunked.
20                label: Diagnostic label for logging.
21
22            Returns:
23                Tuple containing:
24                - A list of detected segment dicts (e.g. [{"start_time": ..., "end_time":
...}])
25                - A dictionary of performance metrics (e.g. {"upload_time": ...,
"process_time": ..., "gen_time": ...})
26            """
27            pass
28

```

37. user (2026-06-10T05:37:28.395Z)

```

1     from abc import ABC, abstractmethod
2     from pydantic import BaseModel
3     from typing import List, Dict, Any, Optional
4
5     class ValidationResult(BaseModel):
6         validator_name: str
7         passed: bool
8         message: str
9         details: Dict[str, Any] = {}
10
11     class VideoValidator(ABC):

```

```

12     @property
13     @abstractmethod
14     def name(self) -> str:
15         """Unique identifier for this validator."""
16         pass
17
18     @property
19     @abstractmethod
20     def description(self) -> str:
21         """Human-readable explanation of what this validator checks."""
22         pass
23
24     @abstractmethod
25     def validate(self, video_path: str, config: Dict[str, Any], context: Dict[str, Any])
-> ValidationResult:
26         """
27         Executes validation.
28         - video_path: Local path to the MP4 file.
29         - config: Dict parsed from config.json.
30         - context: Dynamic context dict shared across validators (stores shared values
like ffprobe metadata).
31         """
32         pass
33
34     class ValidationRegistry:
35     def __init__(self):
36         self._validators: List[VideoValidator] = []
37
38     def register(self, validator: VideoValidator):
39         """Registers a new validator."""
40         self._validators.append(validator)
41
42     def run_all_with_context(self, video_path: str, config: Dict[str, Any]):
43         """Runs all validators and returns ``(results, context)``.
44
45         The per-run ``context`` accumulates shared values (notably
46         ``context['ffprobe_meta']`` produced by FormatValidator). ``run_all``
47         discards it; callers that want media metadata for an observability report
48         use this variant instead of re-probing.
49         """
50         results = []
51         context = {} # Shared context dictionary (e.g. for caching ffprobe outputs)
52         for val in self._validators:
53             try:
54                 res = val.validate(video_path, config, context)
55                 results.append(res)
56             except Exception as e:
57                 results.append(ValidationResult(
58                     validator_name=val.name,
59                     passed=False,
60                     message=f"Validator threw an unexpected error: {str(e)}"
61                 ))
62         return results, context
63
64     def run_all(self, video_path: str, config: Dict[str, Any]) ->
List[ValidationResult]:
65         """Runs all registered validators in sequence, accumulating results."""
66         results, _ = self.run_all_with_context(video_path, config)
67         return results
68
69     # Shared global registry
70     registry = ValidationRegistry()
71

```

```

1     import os
2     import sys
3     import json
4     import logging
5     from dotenv import load_dotenv
6
7     load_dotenv()
8
9     # Configure root logger ? all backend modules use logging.getLogger(__name__)
10    logging.basicConfig(
11        level=logging.INFO,
12        format="%(asctime)s [%(name)s] %(levelname)s: %(message)s",
13        datefmt="%H:%M:%S",
14    )
15
16    from backend.infrastructure.database import Database
17    from backend.config import load_config
18    from backend.utils.hashing import compute_video_id
19    from backend.pipeline.validators.base import registry
20    from backend.pipeline.segmenters import segmenter_registry
21    from backend.pipeline.stitcher.compiler import VideoCompiler
22
23    def main():
24        video_path = r"C:\Users\avidu\Videos\Testing\KushagraYashAviFirstVideo.mp4"
25        output_dir = r"C:\Users\avidu\Videos\Testing"
26        output_filename = "Final_2_OutputOfKushagraYashAviFirstVideo.mp4"
27
28        print(f"Loading config...")
29        cfg = load_config() # typed AppConfig ? single source of defaults
30
31        print(f"Verifying files exist...")
32        if not os.path.exists(video_path):
33            print(f"Error: Video file not found at {video_path}")
34            return
35
36        os.makedirs(cfg.output.output_dir, exist_ok=True)
37        db_path = os.path.join(cfg.output.output_dir, cfg.output.db_name)
38        db = Database(db_path)
39
40        # Calculate MD5 of the video to uniquely identify it
41        import time
42        print("Calculating video MD5 (this might take a few seconds on large files)...",
43flush=True)
44        t0 = time.time()
45        video_id = compute_video_id(video_path)
46        print(f"Calculated MD5: {video_id} (took {time.time()-t0:.1f}s)", flush=True)
47
48        segments = []
49        cached_video = db.get_video(video_id)
50        cache_hit = False
51
52        if cached_video and cached_video.get("status") == "processed":
53            segments = db.query_play_segments(video_id, {})
54            if len(segments) > 0:
55                print(f"\n[CACHE] Found cached indexed video in database (MD5: {video_id})
56with {len(segments)} parsed play segments.", flush=True)
57                choice = input("Would you like to (1) Run stitching on the cached segments,
58or (2) Reprocess the video with Gemini from scratch? [1/2]: ").strip()
59                if choice == '1':
60                    print("[CACHE] Using cached play segments. Skipping segmenter API
61call.", flush=True)
62                    cache_hit = True
63                else:
64                    print("[CACHE] Ignoring cache...", flush=True)
65            else:

```

```

62         print(f"[CACHE] Video exists in DB but contains 0 play segments. Treating as
cache miss.", flush=True)
63
64     if not cache_hit:
65         print("[CACHE] Initializing clean indexing...", flush=True)
66         db.clear_video_data(video_id)
67
68         print("Bypassing OpenCV validations for performance on massive files...",
flush=True)
69         db.add_video(video_id, video_path, "processed", [])
70
71         print("Running segmenter & indexing...", flush=True)
72         seg_name = cfg.indexing.default_segmenter
73         segmenter_cls = segmenter_registry.get(seg_name)
74         if not segmenter_cls:
75             print(f"[FAIL] Segmenter '{seg_name}' not found.")
76             return
77         print(f"Using segmenter: {seg_name}", flush=True)
78         segmenter = segmenter_cls(db, cfg)
79         segments = segmenter.process_video(video_id, video_path)
80         print(f"Successfully indexed {len(segments)} play segments.", flush=True)
81
82     if not segments:
83         print("No play segments detected. Cannot compile highlights.")
84         return
85
86     end_padding = cfg.stitching.end_padding
87     begin_padding = cfg.stitching.begin_padding
88     min_shot_count = cfg.stitching.min_shot_count
89
90
91     if cache_hit:
92         print("\n--- Stitching Customizations ---")
93         try:
94             bpad_input = input(f"Enter begin_padding in seconds (default
{begin_padding}): ").strip()
95             if bpad_input:
96                 begin_padding = float(bpad_input)
97
98             pad_input = input(f"Enter end_padding in seconds (default {end_padding}):
").strip()
99             if pad_input:
100                 end_padding = float(pad_input)
101
102             shot_input = input(f"Enter min_shot_count to filter by (default
{min_shot_count}): ").strip()
103             if shot_input:
104                 min_shot_count = int(shot_input)
105         except ValueError:
106             print("[WARNING] Invalid input detected, falling back to defaults from
config.json.")
107
108     print(f"\nCompiling highlights (with begin_padding={begin_padding}s,
end_padding={end_padding}s, min_shot_count={min_shot_count})...", flush=True)
109
110     # Filter segments where shot_count >= min_shot_count
111     time_pairs = []
112     skipped_count = 0
113     for s in segments:
114         meta = s.get("metadata", {})
115         if isinstance(meta, str):
116             try:
117                 meta = json.loads(meta)
118             except Exception:
119                 meta = {}

```

```

120         shot_count = meta.get("shot_count", 0)
121         if shot_count >= min_shot_count:
122             time_pairs.append((s["start_time"], s["end_time"]))
123         else:
124             skipped_count += 1
125
126         if skipped_count > 0:
127             print(f"[FILTER] Excluded {skipped_count} play segments with fewer than
{min_shot_count} shots.", flush=True)
128         print(f"[FILTER] Retaining {len(time_pairs)} play segments for stitching.",
flush=True)
129
130         if not time_pairs:
131             print("No play segments passed the filtering criteria. Cannot compile
highlights.")
132             return
133
134         compiler = VideoCompiler(output_dir, end_padding=end_padding,
begin_padding=begin_padding)
135         try:
136             out_path = compiler.compile_highlights(video_path, time_pairs, output_filename)
137             print(f"Stitched highlight video saved to: {out_path}", flush=True)
138         except Exception as e:
139             print(f"Error during stitching: {e}", flush=True)
140
141     if __name__ == "__main__":
142         main()
143

```

39. user (2026-06-10T05:37:36.643Z)

```

1     """Canonical file-identity hashing for the indexer.
2
3     `video_id` is the MD5 of the *whole* video file. Historically each entriypoint
4     (`backend/main.py`, `run_eval.py`, `run_phase3_eval.py`, `run_process_and_stitch.py`)
5     defined its own `get_file_md5` with a different read-chunk size (64KB / 8KB / 1MB).
6     The hash values agreed (same bytes hashed), but the duplicated code was a footgun:
7     any future change to one copy would silently diverge the DB key.
8
9     This module is the single source of truth. Import `compute_video_id` everywhere a
10    video needs to be keyed.
11    """
12
13    from __future__ import annotations
14
15    import hashlib
16
17    # 64 KB matches the value used by the live pipeline (backend/main.py) before this
18    # was centralized. The chunk size does not affect the resulting digest ? it only
19    # bounds memory use while streaming large files.
20    _CHUNK_SIZE = 65536
21
22
23    def compute_video_id(filepath: str) -> str:
24        """Return the MD5 hex digest of the whole file ? the canonical video_id.
25
26        Streams the file in fixed-size chunks so multi-gigabyte recordings never load
27        into memory at once.
28        """
29        hasher = hashlib.md5()
30        with open(filepath, "rb") as f:
31            for chunk in iter(lambda: f.read(_CHUNK_SIZE), b""):
32                hasher.update(chunk)
33        return hasher.hexdigest()
34

```

40. assistant (2026-06-10T05:37:46.697Z)

Now let me run the offline test suite to verify whether it passes despite the broken compiler (that would confirm a coverage hole), and check run_all_tests.py.

41. user (2026-06-10T05:37:46.932Z)

```
1      #!/usr/bin/env python
2      """Run the FULL test suite for the badminton-highlight-indexer.
3
4      A single, discoverable entry point so you (or a reviewer / CI) can verify the
5      whole suite before opening a PR:
6
7          python run_all_tests.py          # full suite, quiet
8          python run_all_tests.py -v       # verbose (passthrough to pytest)
9          python run_all_tests.py -k config # filter by keyword (passthrough)
10         python run_all_tests.py --cov=backend --cov-report=term-missing # coverage
11
12     Anything after the script name is passed straight through to pytest.
13
14     Notes
15     -----
16     - The suite is fully mocked (no GPU, no network, no WSL) and runs offline.
17     - The observability **reporting** tests skip cleanly if the optional ``obsreport``
18       package isn't installed (they use ``pytest.importorskip``). To exercise them too:
19         pip install -e ../sports-obsreport --no-build-isolation
20     - Exit code is pytest's own, so this is CI / pre-PR friendly (0 = all green).
21     """
22     from __future__ import annotations
23
24     import importlib.util
25     import subprocess
26     import sys
27
28
29     def _installed(module: str) -> bool:
30         try:
31             return importlib.util.find_spec(module) is not None
32         except (ImportError, ValueError):
33             return False
34
35
36     def main() -> int:
37         args = list(sys.argv[1:])
38         # Default to quiet unless the caller chose a verbosity.
39         if not any(a in ("-q", "-v", "-vv", "-vvv", "--quiet", "--verbose") for a in args):
40             args = ["-q", *args]
41         cmd = [sys.executable, "-m", "pytest", *args]
42
43         obs = "available" if _installed("obsreport") else "NOT installed ? reporting tests
will SKIP"
44         bar = "=" * 72
45         print(bar)
46         print(" Full test suite: " + " ".join(cmd))
47         print(f" obsreport (optional reporting dep): {obs}")
48         print(bar, flush=True)
49
50         rc = subprocess.run(cmd).returncode
51
52         print(bar)
53         print(" RESULT: " + ("PASS - all tests green" if rc == 0 else f"FAIL - pytest exit
{rc}"))
54         print(bar)
55         return rc
56
```

```
57
58     if __name__ == "__main__":
59         raise SystemExit(main())
60
```

42. user (2026-06-10T05:38:06.832Z)

```
=====
Full test suite: C:\Users\avidu\AppData\Local\Programs\Python\Python313\python.exe -m pytest
-q
obsreport (optional reporting dep): available
=====
..... [ 28%]
..... [ 56%]
..... [ 85%]
..... [100%]
254 passed in 11.87s
=====
RESULT: PASS - all tests green
=====
```

43. user (2026-06-10T05:38:16.354Z)

No files found

44. user (2026-06-10T05:38:18.334Z)

```
__init__.py
test_annotation_providers.py
test_automated_provider.py
test_base.py
test_calibrate_local.py
test_calibrate_wasb.py
test_calibration.py
test_config.py
test_database.py
test_distill_local.py
test_eval_annotations.py
test_eval_batch.py
test_eval_classifier.py
test_eval_harness.py
test_eval_metrics.py
test_eval_partial_labels.py
test_eval_regression.py
test_eval_training.py
test_export_wasb_segments.py
test_gemini_refine.py
test_gemini.py
test_geometry.py
test_hashing.py
test_hit_detector.py
test_provider_gemini.py
test_providers.py
test_rally_gate.py
test_rally_seq_proto.py
test_rally_slicer.py
test_reporting_harvest.py
test_reporting_spec_faq.py
test_reporting.py
test_sport_profiles.py
test_tracknet_hybrid.py
test_tracknet_runner.py
test_validators_base.py
test_video.py
```

```
test_wasb_cache.py
test_yolo_hybrid.py
```

45. user (2026-06-10T05:38:40.892Z)

```
1:import pytest
2:from unittest.mock import MagicMock, patch
3:from backend.pipeline.segmenters.gemini import GeminiPlaySegmenter
```

46. user (2026-06-10T05:38:44.590Z)

done scanning

47. assistant (2026-06-10T05:38:49.753Z)

Confirmed: zero test references the compiler or `backend.main`. The SyntaxError is invisible to the entire 254-test suite. Let me check the `.env`/API-key handling and whether reports/logs could leak the key, plus the YOLO hybrid for the same `bash -c` injection surface, and confirm the WSL `bash -c` injection via filename.

48. user (2026-06-10T05:38:51.192Z)

```
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\providers\base.py:5:    def
__init__(self, api_key: str, model_name: str):
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\providers\base.py:6:
self.api_key = api_key
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\providers\gemini.py:17:    def
__init__(self, api_key: str, model_name: str):
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\providers\gemini.py:18:
super().__init__(api_key, model_name)
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\providers\gemini.py:25:
self._client = genai.Client(api_key=self.api_key)
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\providers\__init__.py:14:    def
get(self, name: str, api_key: str, model_name: str) -> AIVideoProvider:
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\providers\__init__.py:17:
return self._registry[name](api_key=api_key, model_name=model_name)
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\gemini.py:44:
api_key_env_var = f"{provider_name.upper()}_API_KEY"
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\gemini.py:45:
api_key = os.environ.get(api_key_env_var)
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\gemini.py:46:
if not api_key:
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\gemini.py:48:
f"{api_key_env_var} environment variable is not set! "
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\gemini.py:50:
f"In PowerShell: $env:{api_key_env_var}=\"your_api_key_here\""
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\gemini.py:55:
provider = provider_registry.get(provider_name, api_key=api_key, model_name=model_name)
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\tracknet_hybrid.py:73:
api_key_env_var =
f"{provider_name.upper()}_API_KEY"
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\tracknet_hybrid.py:74:
api_key =
os.environ.get(api_key_env_var)
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\tracknet_hybrid.py:75:
if not api_key:
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\tracknet_hybrid.py:77:
f"{api_key_env_var}
environment variable is not set! "
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\tracknet_hybrid.py:79:
f"In PowerShell:
$env:{api_key_env_var}=\"your_api_key_here\""
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\tracknet_hybrid.py:83:
provider =
```

```

provider_registry.get(provider_name, api_key=api_key, model_name=model_name)
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml:29: - {
path: "stages.ai_handoff.details.api_key_present", op: eq, value: false }
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml:35:
CREDENTIAL, not 'the video has no rallies'. Fix: put GEMINI_API_KEY=<key> in a
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml:48: - {
path: "stages.ai_handoff.details.api_key_present", op: eq, value: true }
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\wasb_hybrid.py:69:         api_key_env_var =
f"{provider_name.upper()}_API_KEY"
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\wasb_hybrid.py:70:         api_key =
os.environ.get(api_key_env_var)
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\wasb_hybrid.py:71:         if not api_key:
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\wasb_hybrid.py:72:
logger.error(f"{api_key_env_var} not set; cannot run the {provider_name} handoff.")
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\wasb_hybrid.py:75:         provider =
provider_registry.get(provider_name, api_key=api_key, model_name=model_name)
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\yolo_hybrid.py:280:         api_key_env_var =
f"{provider_name.upper()}_API_KEY"
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\yolo_hybrid.py:281:         api_key =
os.environ.get(api_key_env_var)
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\yolo_hybrid.py:282:         if not api_key:
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\yolo_hybrid.py:284:         f"{api_key_env_var}
environment variable is not set! "
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\yolo_hybrid.py:286:         f"In PowerShell:
$env:{api_key_env_var}=\"your_api_key_here\""
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\yolo_hybrid.py:291:         provider =
provider_registry.get(provider_name, api_key=api_key, model_name=model_name)
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\harvest.py:143:     key_env
= f"{provider.upper()}_API_KEY"
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\harvest.py:151:
"api_key_present": bool(os.environ.get(key_env)) if ai_based else None,
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\gemini_refine.py:17:Run
(GEMINI_API_KEY in env):
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\gemini_refine.py:71:def
refine_window(api_key: str, model: str, sport_profile, video: str, window: Interval,
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\gemini_refine.py:76:
provider = provider_registry.get("gemini", api_key=api_key, model_name=model)
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\gemini_refine.py:105:
api_key = os.environ.get("GEMINI_API_KEY")
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\gemini_refine.py:106:     if not
api_key:
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\gemini_refine.py:107:
raise SystemExit("GEMINI_API_KEY not set in env")
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\gemini_refine.py:122:
futs = [ex.submit(refine_window, api_key, model, sport_profile, video, w, pad, duration,
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\gemini_refine.py:156:
api_key = os.environ.get("GEMINI_API_KEY")
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\gemini_refine.py:157:     if not
api_key:
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\gemini_refine.py:158:
raise SystemExit("GEMINI_API_KEY not set in env")
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\gemini_refine.py:174:
provider = provider_registry.get("gemini", api_key=api_key, model_name=model)

```

49. assistant (2026-06-10T05:38:54.707Z)

The API key handling looks clean (key never logged; reports only record presence as a bool). Now let me examine the YOLO hybrid's WSL/subprocess surface and the validators that run on the request path (format validator ? does it shell out with the filename?), to map the command-injection surface from `video\_path`.

50. user (2026-06-10T05:38:55.676Z)

```

1 import subprocess
2 import json
3 import os
4 from backend.pipeline.validators.base import VideoValidator, ValidationResult
5 from typing import Dict, Any
6
7 class FormatValidator(VideoValidator):
8     @property
9     def name(self) -> str:
10         return "format_check"
11
12     @property
13     def description(self) -> str:
14         return "Verifies the video is a valid, uncorrupted MP4 container and extracts
metadata."
15
16     def validate(self, video_path: str, config: Dict[str, Any], context: Dict[str, Any])
-> ValidationResult:
17         if not os.path.exists(video_path):
18             return ValidationResult(
19                 validator_name=self.name,
20                 passed=False,
21                 message=f"File not found: {video_path}"
22             )
23
24         cmd = [
25             "ffprobe",
26             "-v", "error",
27             "-select_streams", "v:0",
28             "-show_format",
29             "-show_streams",
30             "-of", "json",
31             video_path
32         ]
33
34         try:
35             result = subprocess.run(cmd, capture_output=True, text=True, check=True)
36             info = json.loads(result.stdout)
37
38             if not info.get("streams") or not info.get("format"):
39                 return ValidationResult(
40                     validator_name=self.name,
41                     passed=False,
42                     message="Failed to retrieve valid video streams from the file."
43                 )
44
45             # Store the metadata in context for subsequent validators to utilize
46             context["ffprobe_meta"] = info
47
48             stream = info["streams"][0]
49             fmt = info["format"]
50
51             details = {
52                 "format_name": fmt.get("format_name"),
53                 "duration": float(fmt.get("duration", 0)),
54                 "size_bytes": int(fmt.get("size", 0)),

```

```

55         "codec_name": stream.get("codec_name"),
56     }
57
58     # Check if container name includes mp4
59     if "mp4" not in fmt.get("format_name", "").lower():
60         return ValidationResult(
61             validator_name=self.name,
62             passed=False,
63             message=f"Invalid format: Found '{fmt.get('format_name')}', but MP4
is required.",
64             details=details
65         )
66
67     return ValidationResult(
68         validator_name=self.name,
69         passed=True,
70         message="Video format is valid MP4 container.",
71         details=details
72     )
73
74     except subprocess.CalledProcessError as e:
75         return ValidationResult(
76             validator_name=self.name,
77             passed=False,
78             message="Video stream is corrupted or unreadable by ffprobe.",
79             details={"error": e.stderr}
80         )
81     except Exception as e:
82         return ValidationResult(
83             validator_name=self.name,
84             passed=False,
85             message=f"Unexpected error running ffprobe: {str(e)}"
86         )
87

```

51. user (2026-06-10T05:38:55.995Z)

```

1     from abc import ABC, abstractmethod
2     from typing import List, Dict, Any, Type, Optional
3     from backend.infrastructure.database import Database
4     from backend.results import Result # new standardized contract (Result[Value] ==
Success[Value] | Failure)
5
6     class BasePlaySegmenter(ABC):
7         def __init__(self, db: Database, config: Dict[str, Any]):
8             self.db = db
9             self.config = config
10
11         @property
12         @abstractmethod
13         def name(self) -> str:
14             """Unique identifier for this segmenter."""
15             pass
16
17         @property
18         @abstractmethod
19         def description(self) -> str:
20             """Human-readable explanation of this segmenter."""
21             pass
22
23         @abstractmethod
24         def process_video(self, video_id: str, video_path: str, player_pool: List[str] =
None, max_frames: int = None) -> "Result[List[Dict[str, Any]]]":
25             """
26             Processes a video to detect play segments, populates SQLite database,

```

```

27         and returns a Result.
28
29         Use isinstance(result, Failure) (or result.is_success()) to distinguish
30         "no segments found" from "something went wrong" (missing key, healthcheck
31         failure, etc.). This is the start of the standardized error/result contract.
32         """
33         pass
34
35     class SegmenterRegistry:
36         def __init__(self):
37             self._segmenters: Dict[str, Type[BasePlaySegmenter]] = {}
38
39         def register(self, segmenter_cls: Type[BasePlaySegmenter]):
40             """Registers a new segmenter class under its name property."""
41             try:
42                 temp_inst = segmenter_cls(None, {})
43                 name = temp_inst.name
44             except Exception as e:
45                 raise ValueError(f"Failed to resolve name property from segmenter class
{segmenter_cls.__name__}: {e}")
46             self._segmenters[name] = segmenter_cls
47
48         def get(self, name: str) -> Optional[Type[BasePlaySegmenter]]:
49             return self._segmenters.get(name)
50
51         def list_available(self) -> Dict[str, str]:
52             """Returns a dict of registered segmenter names and their descriptions."""
53             res = {}
54             for name, cls in self._segmenters.items():
55                 try:
56                     temp_inst = cls(None, {})
57                     desc = temp_inst.description
58                 except Exception:
59                     desc = "No description available."
60                 res[name] = desc
61             return res
62
63     segmenter_registry = SegmenterRegistry()
64

```

52. assistant (2026-06-10T05:39:03.506Z)

Now I have the key concurrency/contract picture. The segmenters return bare `List` (and `[ ]` on failure), but `main.py` checks `isinstance(result, Failure)` ? they never return `Failure`. Let me confirm the segmenters don't return `Result`/`Failure` anywhere and check the `results` module, then verify the WSL injection more concretely and the `add\_video` re-ingest cascade behavior.

53. user (2026-06-10T05:39:04.476Z)

```

1     """Lightweight Result type for standardizing success vs. failure across the pipeline.
2
3     This is the start of the Error/Result Contract standardization (low-regret item #3).
4
5     The goal is to make it impossible to confuse "no data / no rallies" with "something
actually failed"
6     (e.g. missing API key, WSL healthcheck failure, provider error, etc.).
7
8     We use simple dataclasses so callers can use isinstance checks or structural pattern
matching.
9     This is intentionally small and incremental ? we start by applying it to the segmenter
and provider layers (the biggest sources of silent [ ] returns today).
10
11     The types are generic where it makes sense. Callers can still access .value on Success
or .message on Failure.
12
13

```

```

14
15     This module has no dependencies on the rest of the app.
16     """
17
18     from __future__ import annotations
19
20     from dataclasses import dataclass, field
21     from typing import Generic, TypeVar, List, Optional, Any, Union
22
23     T = TypeVar("T")
24
25
26     @dataclass
27     class Success(Generic[T]):
28         """Represents a successful outcome with an optional list of non-fatal warnings."""
29
30         value: T
31         warnings: List[str] = field(default_factory=list)
32
33         def is_success(self) -> bool:
34             return True
35
36         def unwrap(self) -> T:
37             """Return the value. Only safe to call if you already know it's a Success."""
38             return self.value
39
40
41     @dataclass
42     class Failure:
43         """Represents a failure with a human-readable message and optional details."""
44
45         message: str
46         details: Optional[dict[str, Any]] = None
47         is_recoverable: bool = False # e.g. missing API key (recoverable) vs internal crash
48
49         def is_success(self) -> bool:
50             return False
51
52         def unwrap(self) -> Any:
53             """Raise to make misuse obvious in code that assumes success."""
54             raise RuntimeError(f"Attempted to unwrap a Failure: {self.message}")
55
56
57     # Convenient alias for use in type hints: Result[list[dict], Failure]
58     # Result[ValueType] means Success[ValueType] | Failure
59     # (the Failure "error" type is fixed for this contract)
60     Result = Union[Success[T], Failure]
61

```

54. user (2026-06-10T05:39:05.714Z)

```

C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\motion.py:33:
return Failure(message=f"Could not open video: {video_path}", is_recoverable=False)
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\motion.py:226:
return Success(value=segments_data)

```

55. assistant (2026-06-10T05:39:11.228Z)

This is a critical finding: only `motion.py` returns `Result`/`Failure`. The four AI segmenters (`gemini`, `tracknet\_hybrid`, `wasb\_hybrid`, `yolo\_hybrid`) all return a bare `list` and `[]` on every failure. So in `main.py`, `isinstance(result, Failure)` is **never true** for the AI segmenters ? every failure mode (missing API key, WSL healthcheck fail, provider error, JSON parse fail) becomes `status: "success"` with `play\_segments: []`. Let me confirm the yolo_hybrid return type and signature, and the motion segmenter's signature, to be precise.

56. user (2026-06-10T05:39:12.693Z)

```
250         if af[0] - group[-1][0] <= max_gap_frames:
251             group.append(af)
252         else:
253             windows.append(_finalize(group))
254             group = [af]
255     windows.append(_finalize(group))
256
257     # Filter out extremely short action windows
258     min_window_duration = self.config.get("indexing",
259     {}).get("min_action_window_duration", 2.0)
260     windows = [w for w in windows if (w["end"] - w["start"]) >= min_window_duration]
261
262     return windows
263
264     def process_video(self, video_id: str, video_path: str, player_pool: List[str] =
None, max_frames: int = None) -> List[Dict[str, Any]]:
265         self._lazy_load_model()
266
267         # Get sport profile config
268         sport_name = self.config.get("sport", "badminton")
269         try:
270             sport_profile = sport_registry.get(sport_name)
271         except KeyError as e:
272             logger.error(str(e))
273             return []
274
275         # Get AI provider and model config
276         idx_cfg = self.config.get("indexing", {})
277         provider_name = self.config.get("ai_provider", idx_cfg.get("ai_provider",
"gemini"))
278         model_name = self.config.get("ai_model", idx_cfg.get("ai_model",
"gemini-2.5-flash"))
279
280         # Get appropriate API key based on the provider
281         api_key_env_var = f"{provider_name.upper()}_API_KEY"
282         api_key = os.environ.get(api_key_env_var)
283         if not api_key:
284             logger.error(
285                 f"{api_key_env_var} environment variable is not set! "
286                 f"To run the {provider_name} segmenter, set your API key. "
287                 f"In PowerShell: $env:{api_key_env_var}=\"your_api_key_here\"")
288             return []
289
290         try:
291             provider = provider_registry.get(provider_name, api_key=api_key,
model_name=model_name)
292         except KeyError as e:
293             logger.error(str(e))
294             return []
295
296         logger.info(f"Initializing Hybrid Segmenter (torchvision + ByteTrack) for video:
{video_path}")
297
298         video_duration = get_video_duration(video_path)
299         if video_duration <= 0:
300             logger.error("Invalid video duration.")
301             return []
302
303         chunk_duration = idx_cfg.get("chunk_duration", 300)
304         chunk_overlap = idx_cfg.get("chunk_overlap", 30)
305         velocity_thresh = idx_cfg.get("yolo_velocity_threshold", 0.15) # Fraction of
frame width per second
```

```

306         merge_gap = idx_cfg.get("dead_time_merge_gap", 2.0)
307         frame_skip = idx_cfg.get("yolo_frame_skip", 3)
308         tracker_config = idx_cfg.get("yolo_tracker", "bytetrack.yaml")
309         handoff_padding = idx_cfg.get("ai_handoff_padding",
idx_cfg.get("gemini_handoff_padding", 2.0))

```

57. user (2026-06-10T05:39:13.182Z)

```

20         return "Traditional CV pixel-motion heuristics tracking frame differences."
21
22     def process_video(self, video_id: str, video_path: str, player_pool: List[str] =
None, max_frames: int = None) -> List[Dict[str, Any]]:
23         """
24         Processes a video to detect play segments using motion heuristics,
25         populates SQLite with detected segments, and indexes detailed events.
26         """
27         if not player_pool:
28             player_pool = ["Avi", "Suresh", "Ramesh", "Deepak"]
29
30         cap = cv2.VideoCapture(video_path)
31         if not cap.isOpened():
32             logger.error(f"Segmenter could not open video: {video_path}")
33             return Failure(message=f"Could not open video: {video_path}",
is_recoverable=False)
34
35         fps = cap.get(cv2.CAP_PROP_FPS)
36         total_frames = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))
37         duration = total_frames / fps if fps > 0 else 0
38
39         # Sub-sample settings (5 FPS for analysis)
40         analysis_fps = 5.0
41         frame_interval = max(1, int(fps / analysis_fps))
42
43         idx_cfg = self.config.get("indexing", {})
44         motion_threshold = idx_cfg.get("motion_threshold", 0.08)

```

58. assistant (2026-06-10T05:39:22.786Z)

Confirmed. The default segmenter is `yolo\_hybrid` (config) / `motion` (main.py fallback), and the four AI segmenters all `return []` on failure while only `motion` returns `Failure`. Let me verify the WSL injection concern concretely by checking how `wsl\_bash` interpolates the staged path? the staged dest is `~/clips/{base}` where `base` is the basename of the video. The video_path comes from the API request unvalidated. Let me also check the `.env`/dotenv and the `clear\_video\_data` / re-ingest interaction with `add\_video` (INSERT OR REPLACE).

The injection vector: in `run\_predict`, the WSL command embeds `self.cfg.repo\_dir`, `self.cfg.weights\_path`, `wsl\_out` (= `to\_wsl\_mnt\_path(output\_win\_dir)`), and the staged video path. `wsl\_out` is derived from `output\_win\_dir` which is hardcoded (`output/wasb`), so safe. The staged path uses `shlex.quote`. But `stage\_video` builds `dest = f"{self.cfg.wsl\_stage\_dir}/{base}"` and then `cp ... {shlex.quote(dest)}`? `base` is inside the quoted dest, so safe from shell injection. However `self.cfg.repo\_dir`, `conda\_sh`, `conda\_env`, `wsl\_stage\_dir` are interpolated **unquoted** and come from config. Let me verify there's no filename-injection path and assess the config-driven injection.

59. user (2026-06-10T05:39:23.792Z)

```

1     """Rally-start gate: suppress between-point false fires using shuttle net-crossings.
2
3     WASB (shuttle-only) over-detects: when a player flicks/tosses the shuttle back to
4     the opponent for service after losing a point, that single trajectory looks like a
5     rally. The discriminator is structural, not duration-based: a real rally has the
6     shuttle crossing the net MULTIPLE times (each shot lands on the other side); a
7     single service feed crosses ~once.
8

```

```

9      This module is camera-agnostic and needs NO new model ? it works on the trajectory
10     we already produce (Frame,Visibility,X,Y). It estimates the play axis (the image
11     axis with the larger shuttle spread) and the net position (median shuttle coord on
12     that axis ? the shuttle clusters at the two ends and crosses the net region often,
13     so the median sits near the net), then counts sign changes of (coord - net) across
14     the visible points inside a candidate window, with a deadband to ignore jitter.
15
16     Gate A in the over-fire fix spectrum (B = player-stance state machine, future).
17     Pure functions only ? no I/O, unit-tested; intended as a post-filter on the
18     windows from trajectory_to_action_windows, or to fold into it later.
19     """
20     from __future__ import annotations
21
22     from dataclasses import dataclass
23     from typing import List, Sequence, Tuple
24
25     Interval = Tuple[float, float]
26
27
28     @dataclass
29     class GatedWindow:
30         start: float
31         end: float
32         crossings: int
33         visible_points: int
34         kept: bool
35
36
37     def _spread(vals: Sequence[float]) -> float:
38         """Robust spread = p95 - p5 (ignores a few outlier detections)."""
39         if len(vals) < 2:
40             return 0.0
41         s = sorted(vals)
42         lo = s[max(0, int(0.05 * (len(s) - 1)))]
43         hi = s[min(len(s) - 1, int(0.95 * (len(s) - 1)))]
44         return hi - lo
45
46
47     def _median(vals: Sequence[float]) -> float:
48         if not vals:
49             return 0.0
50         s = sorted(vals)
51         n = len(s)
52         return s[n // 2] if n % 2 else 0.5 * (s[n // 2 - 1] + s[n // 2])
53
54
55     def estimate_play_axis(points) -> Tuple[str, float, float]:
56         """Return (axis 'x'/'y', net_value, span) from visible points.
57
58         Play axis = the image axis along which the shuttle travels most (larger spread).
59         For a baseline camera that's Y (players top/bottom); for a side camera, X.
60         """
61         xs = [p.x for p in points if p.visible]
62         ys = [p.y for p in points if p.visible]
63         sx, sy = _spread(xs), _spread(ys)
64         if sx >= sy:
65             return "x", _median(xs), sx
66         return "y", _median(ys), sy
67
68
69     def count_net_crossings(window_points, axis: str, net: float, deadband: float) -> int:
70         """Count sign changes of (coord - net) across visible points, ignoring the
71         deadband zone near the net (so jitter at the net doesn't inflate the count)."""
72         crossings = 0
73         last_side = None

```

```

74     for p in window_points:
75         if not p.visible:
76             continue
77         v = (p.x if axis == "x" else p.y) - net
78         if abs(v) < deadband:
79             continue
80         side = v > 0
81         if last_side is not None and side != last_side:
82             crossings += 1
83         last_side = side
84     return crossings
85
86
87 def _points_in(points, fps: float, start: float, end: float):
88     return [p for p in points if start * fps <= p.frame <= end * fps]
89
90
91 def gate_windows(points, windows: List[Interval], fps: float,
92                 min_crossings: int = 2, deadband_frac: float = 0.06) ->
List[GatedWindow]:
93     """Annotate each (start,end) window with its shuttle net-crossing count and a
94     keep decision (crossings >= min_crossings). Deadband is a fraction of the play
95     axis span. Net/axis are estimated once over the whole trajectory (stable)."""
96     axis, net, span = estimate_play_axis(points)
97     deadband = deadband_frac * span
98     out: List[GatedWindow] = []
99     for (s, e) in windows:
100         wp = _points_in(points, fps, s, e)
101         c = count_net_crossings(wp, axis, net, deadband)
102         vis = sum(1 for p in wp if p.visible)
103         out.append(GatedWindow(start=s, end=e, crossings=c, visible_points=vis,
104                               kept=c >= min_crossings))
105     return out
106
107
108 def apply_gate(points, windows: List[Interval], fps: float,
109               min_crossings: int = 2, deadband_frac: float = 0.06) -> List[Interval]:
110     """Convenience: return only the windows that pass the crossing gate."""
111     return [(g.start, g.end) for g in
112             gate_windows(points, windows, fps, min_crossings, deadband_frac) if g.kept]
113

```

60. user (2026-06-10T05:39:24.363Z)

```

1     """Low-level moving-object detectors that run outside the main Python process.
2
3     Houses:
4     - DetectorRunner ABC (the seam for WSL vs future Linux-native trajectory detectors).
5     - TrackNetRunner / WasbRunner (current WSL adapters; both inherit DetectorRunner).
6     - rally_gate (pure post-filter, no WSL).
7     - wasb_infer.py (the actual code that runs *inside* the WSL wasb env).
8
9     The ABC de-risks the WSL dependency (low-regret item 6 continuation): new
10    runners can be added without changing hybrid segmenters or windowing logic.
11    See base.py docstring for the exact contract + "how to add a runner" guide.
12
13    Keeping the heavy TF/torch deps quarantined in WSL (or a native subprocess)
14    is deliberate: the main project env stays free of them.
15    """
16
17    from .base import DetectorRunner # re-export the ABC for easy "from
backend.pipeline.detectors import DetectorRunner"
18
19    # The concrete runners are imported directly where needed (hybrids, eval/*).
20    # This keeps heavy optional deps from being pulled at import time of the package.

```

21
22

61. assistant (2026-06-10T05:39:28.392Z)

I have enough to finalize. Let me do two quick confirmations: (1) the `/api/compile` path ? does the `output\_filename` from the request flow into a filesystem path without sanitization (path traversal)? and (2) confirm there's no WAL mode anywhere and check whether two `add\_video` on the same id with INSERT OR REPLACE cascade-deletes child rows.

62. user (2026-06-10T05:39:30.430Z)

No matches found

63. user (2026-06-10T05:39:31.718Z)

```
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py:84:   output_filename: str
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py:203:       out_path =
compiler.compile_highlights(video["filepath"], time_pairs, req.output_filename)
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\api\static\app.js:356:
output_filename: outputName
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\stitcher\compiler.py:64:
def compile_highlights(self, raw_video_path: str, segments: List[Tuple[float, float]],
output_filename: str) -> str:
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\stitcher\compiler.py:75:
output_path = os.path.join(self.output_dir, output_filename)
```

64. assistant (2026-06-10T05:39:37.895Z)

Confirmed both: no WAL/busy_timeout/check_same_thread anywhere (default SQLite isolation, no concurrency hardening), and `output\_filename` flows from the request straight into `os.path.join(output\_dir, output\_filename)` with no sanitization ? path traversal (`../..`) or absolute path writes the highlight reel anywhere. The INSERT OR REPLACE cascade with `foreign\_keys=ON` will delete child `play\_segments`/`events`/`candidate\_segments` on every re-ingest.

Let me verify the INSERT OR REPLACE cascade behavior is real (SQLite REPLACE = DELETE + INSERT, triggers FK cascade) and check the `add\_video` ordering in `process\_video` ? it writes the "processed" video row *before* segmentation runs, which on a re-ingest wipes prior good segments even if the new run then fails/crashes.

65. user (2026-06-10T05:39:43.883Z)

```
Exit code 1
ParserError:
Line |
  10 |   c.execute(\"INSERT OR REPLACE INTO videos(id,fp)
VALUES('A','f2')\")
      |
~
      | Missing argument
in parameter list.
```

66. assistant (2026-06-10T05:39:46.520Z)

Quoting issue in PowerShell. Let me write it via the Bash tool with a heredoc-free one-liner using a temp file.

67. user (2026-06-10T05:39:50.845Z)

segs before re-ingest: 1
segs after INSERT OR REPLACE same id: 0

68. assistant (2026-06-10T05:39:55.985Z)

Proven: `INSERT OR REPLACE` on the `videos` row with `foreign\_keys=ON` cascade-deletes all child `play\_segments`, `events`, and `candidate\_segments`. In `main.py::process\_video`, `add\_video(... "processed" ...)` is called at line 123 ****before**** segmentation runs ? so on any re-ingest, the prior good segments are destroyed up-front, and if the new segmentation then crashes or returns `[]` (e.g. transient Gemini outage), the customer is left with an empty DB and no recovery.

I have a complete picture. Let me compile the final structured findings.

69. user (2026-06-10T05:41:46.884Z)

Output does not match required schema: root: must have required property 'strengths', root: must have required property 'findings'

70. assistant (2026-06-10T05:41:51.180Z)

The arrays need to use `item` keys correctly ? the schema expects plain arrays. Let me resend with the correct array shape (the issue was my use of `` XML-style elements; this tool takes JSON-style arrays).

71. user (2026-06-10T05:43:00.762Z)

Output does not match required schema: root: must have required property 'strengths', root: must have required property 'findings'

72. assistant (2026-06-10T05:43:04.985Z)

The schema validator isn't parsing the JSON-string arrays. The `strengths` and `findings` parameters must be passed as actual JSON arrays/objects, not strings. Let me restructure using proper nested parameter encoding.

73. user (2026-06-10T05:44:05.124Z)

Output does not match required schema: root: must have required property 'strengths', root: must have required property 'findings'

74. assistant (2026-06-10T05:44:05.828Z)

You've hit your session limit · resets 3:50pm (Asia/Calcutta)